Puberty and emotions study sensitivity model outputs

H1 Fully Adjusted Model output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.5888602 0.03723716 0.5062275 0.6503184
```

Summary of Posterior Distribution

Parameter	Mean	Median	CI	pd
b_Intercept	-0.39	-0.39	[-0.60, -0.18]	99.98%
b_pds_total_s	0.07	0.07	[-0.14, 0.28]	74.34%
b_adhd_statusY	0.61	0.61	[0.35, 0.88]	100.00%
b_age_cont_s	-0.07	-0.07	[-0.23, 0.09]	81.17%
b_sed_percentile_s	0.15	0.15	[0.03, 0.27]	99.06%
b_bmi_s	0.18	0.18	[0.05, 0.31]	99.66%
b_smfq_total_s	0.57	0.57	[0.45, 0.71]	100%
b_pds_total_s:adhd_statusY	-0.05	-0.05	[-0.30, 0.18]	67.02%

Parameter	% in ROPE	meaningful
b_Intercept	0.38%	TRUE
b_pds_total_s	55.96%	FALSE
b_adhd_statusY	0.01%	TRUE
b_age_cont_s	61.07%	FALSE
b_sed_percentile_s	22.15%	FALSE
b_bmi_s	11.43%	FALSE
b_smfq_total_s	0%	TRUE
b_pds_total_s:adhd_statusY	53.46%	FALSE

H1 Fully Adjusted Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

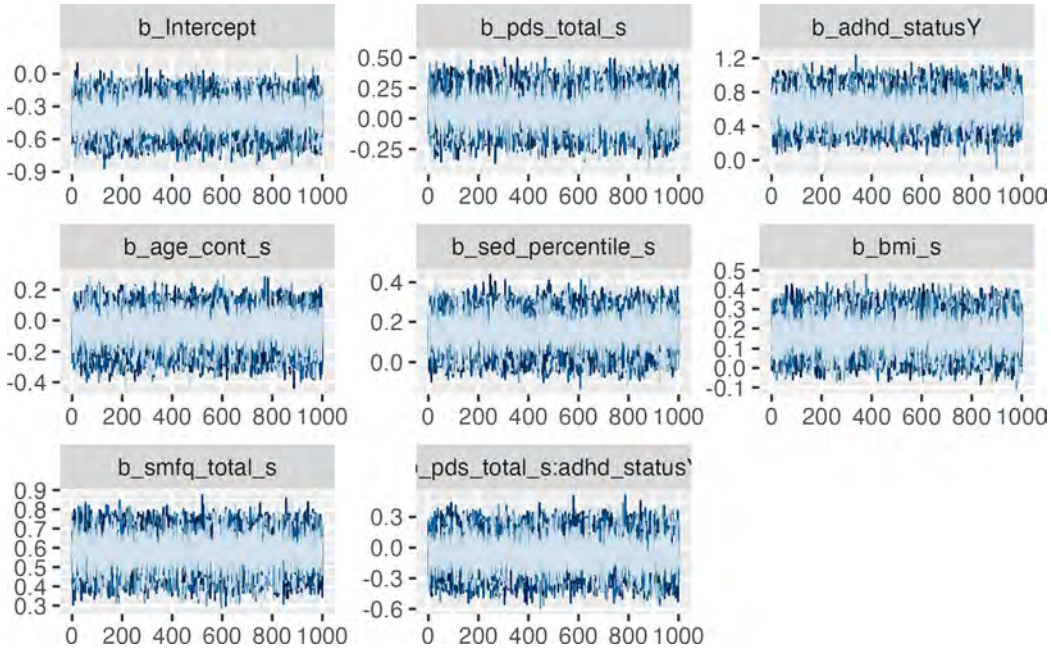
# A tibble: 12 x 4
   variable                    rhat ess_bulk ess_tail
   <chr>                      <dbl>    <dbl>    <dbl>
 1 b_Intercept                 1.00   25583.   92710.
 2 b_pds_total_s               1.00   52250.   86721.
 3 b_adhd_statusY              1.01   18617.   92049.
 4 b_age_cont_s                1.01    5421.   46852.
 5 b_sed_percentile_s          1.03    2741.   12269.
 6 b_bmi_s                     1.01   15183.   95018.
 7 b_smfq_total_s              1.01   14290.   91095.
 8 b_pds_total_s:adhd_statusY  1.01   15364.   89858.
 9 sigma                       1.01    7215.   86387.
10 Intercept                   1.00  144003.   91379.
11 lprior                      1.00  111680.   91588.
12 lp__                        1.16     491.    1218.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

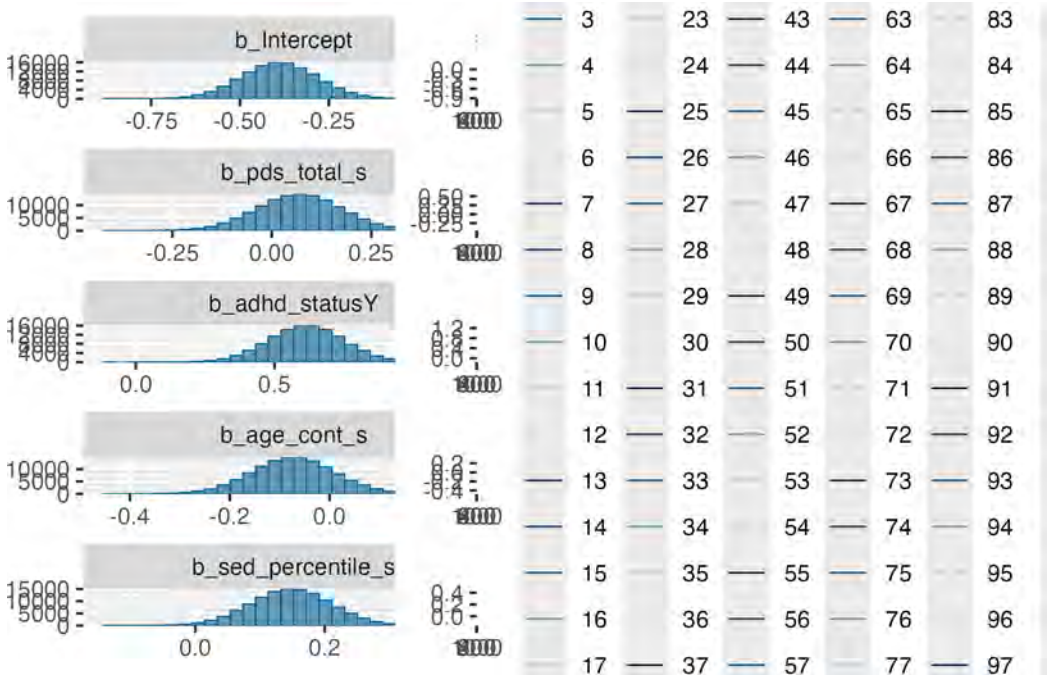
Priors specified for the model.

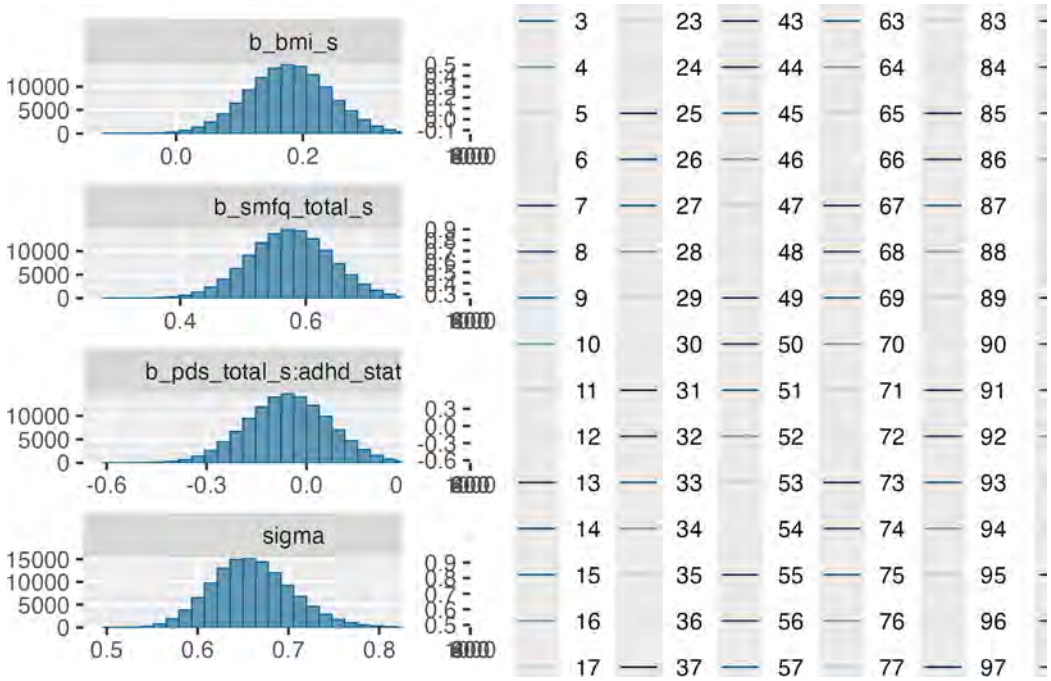
          prior     class                     coef group resp dpar nlpar lb ub
   normal(0, 1)         b
   normal(0, 1)         b             adhd_statusY
   normal(0, 1)         b               age_cont_s
   normal(0, 1)         b                    bmi_s
   normal(0, 1)         b              pds_total_s
   normal(0, 1)         b pds_total_s:adhd_statusY
   normal(0, 1)         b         sed_percentile_s
   normal(0, 1)         b              smfq_total_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                                    0
 tag       source
             user
     (vectorized)
     (vectorized)
     (vectorized)
     (vectorized)
     (vectorized)
```

```
      (vectorized)
      (vectorized)
              user
              user


=== Posterior Predictive Checks ===

Density plots for each parameter.
```

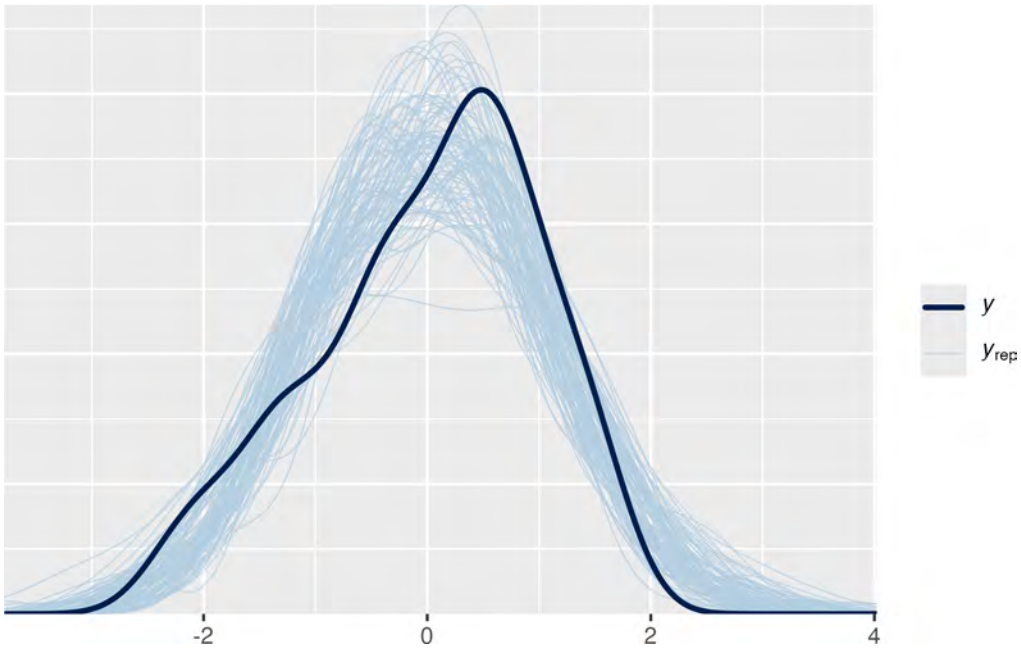




=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results with caution until a more principled approach has been implemented.



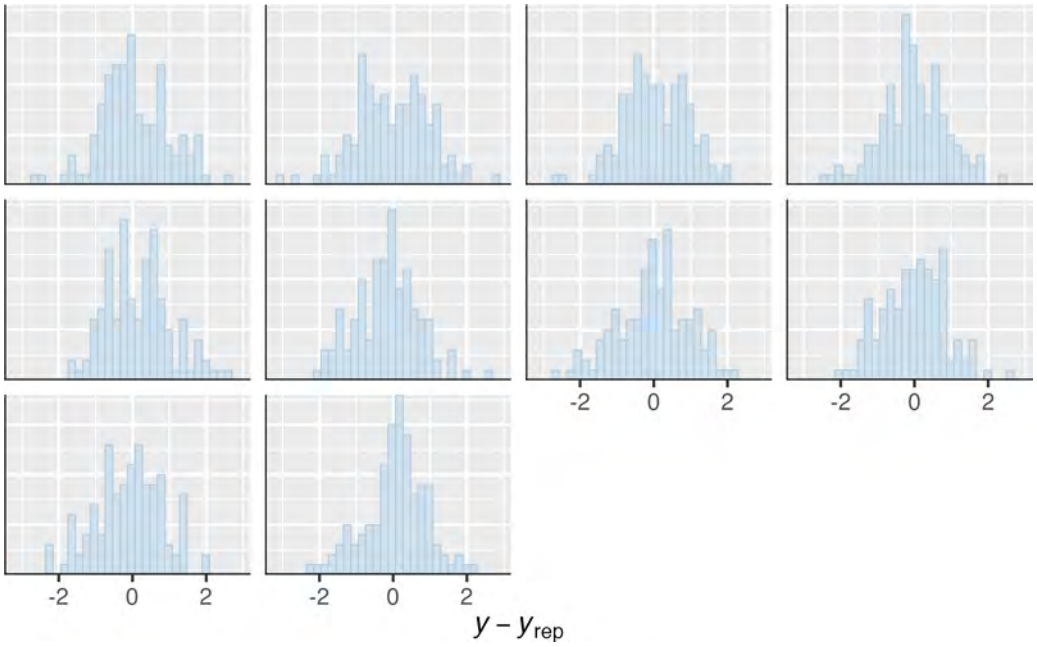

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

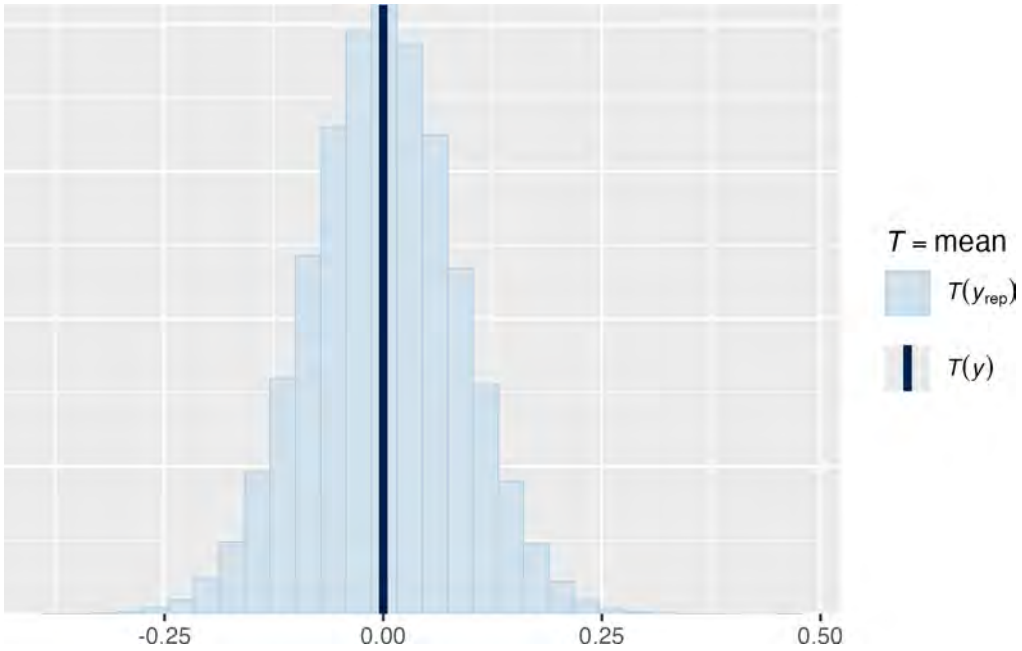
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



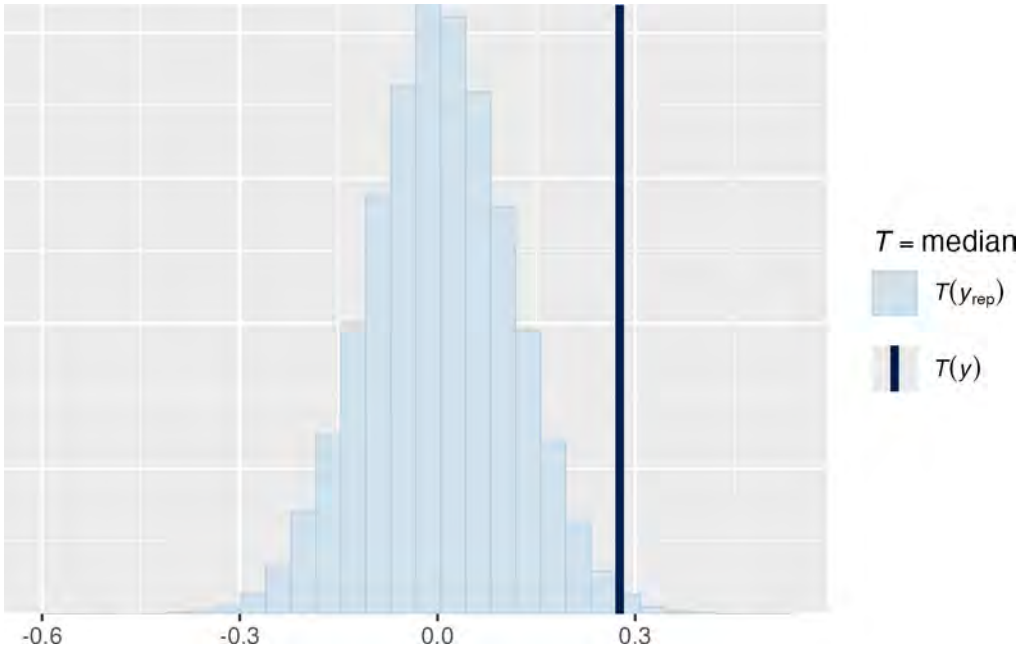

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



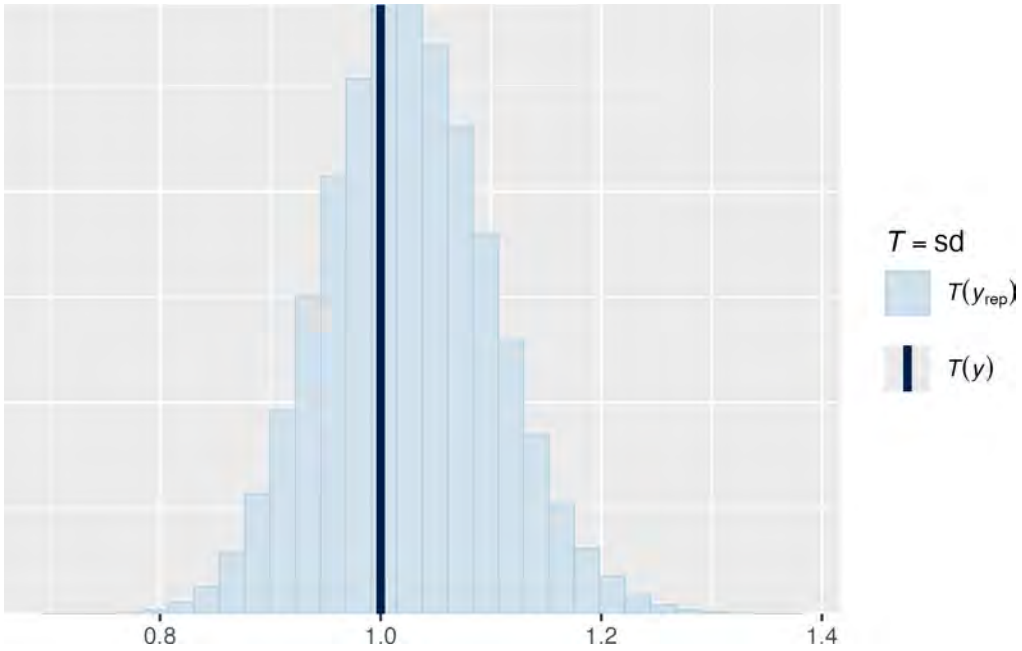

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

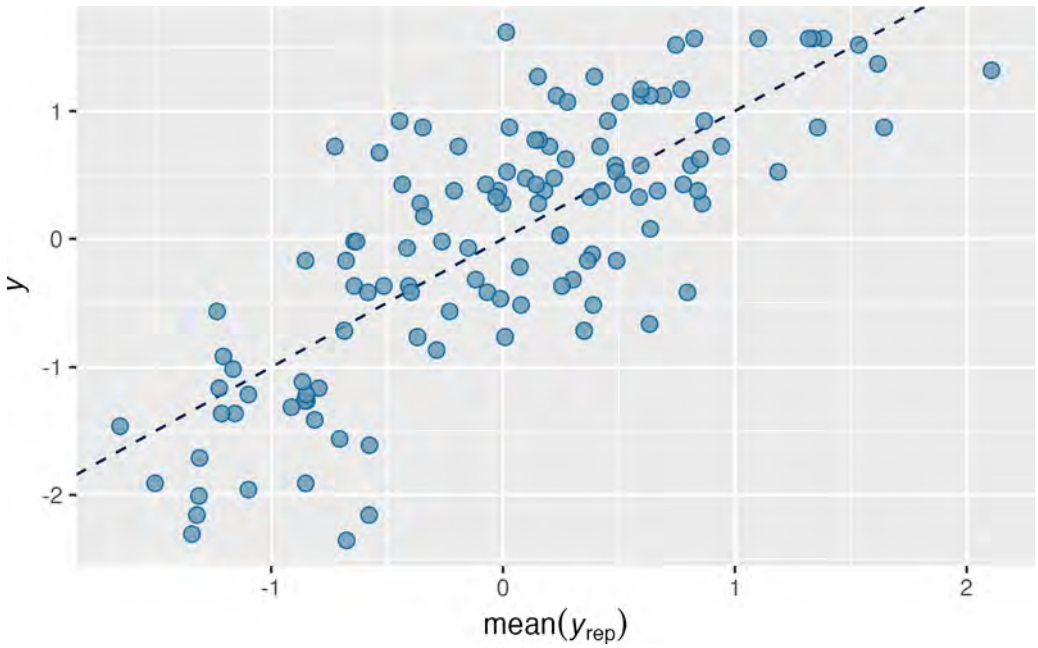



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

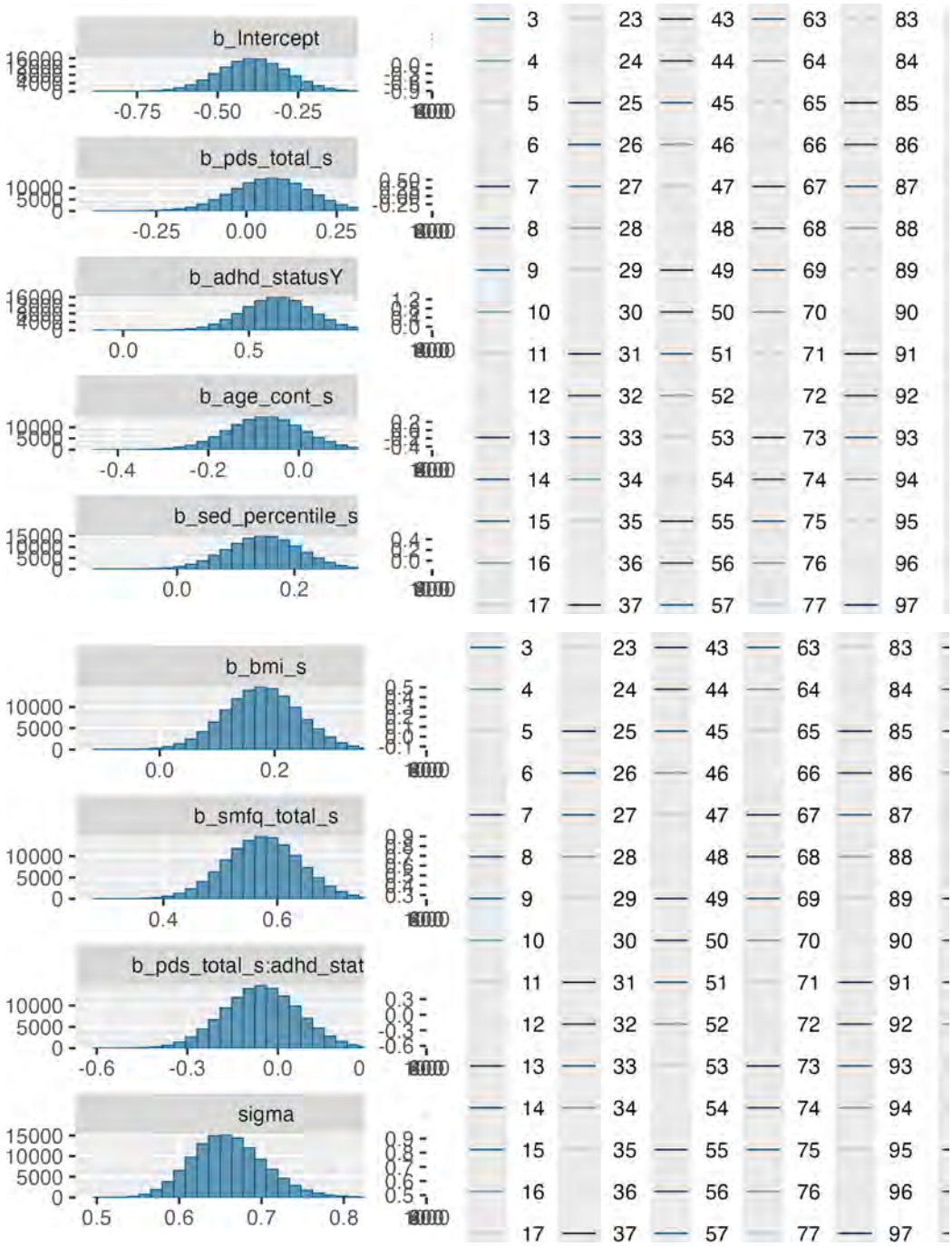
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Fully Adjusted Model output

```
Estimate  Est.Error      Q2.5     Q97.5
```

```
R2 0.5981698 0.03663128 0.5159221 0.6587268
```

```
Summary of Posterior Distribution
```

Parameter	Mean	Median	CI	pd
b_Intercept	-0.38	-0.38	[-0.58, -0.17]	99.98%
b_pubertal_timing_s	0.16	0.16	[-0.02, 0.34]	95.73%
b_adhd_statusY	0.61	0.61	[0.35, 0.87]	100%
b_sed_percentile_s	0.15	0.15	[0.03, 0.27]	99.30%
b_bmi_s	0.16	0.16	[0.04, 0.28]	99.52%
b_smfq_total_s	0.55	0.55	[0.43, 0.68]	100%
b_pubertal_timing_s:adhd_statusY	-0.22	-0.22	[-0.45, 0.03]	96.16%

Parameter	% in ROPE	meaningful
b_Intercept	0.46%	TRUE
b_pubertal_timing_s	26.39%	FALSE
b_adhd_statusY	8.33e-03%	TRUE
b_sed_percentile_s	19.93%	FALSE
b_bmi_s	16.20%	FALSE
b_smfq_total_s	0%	TRUE
b_pubertal_timing_s:adhd_statusY	16.53%	FALSE

H2 Fully Adjusted Model diagnostics and PP checks

```
=== Rhat and ESS ===
```

```
Rhat values close to 1 indicate convergence. ESS should ideally be > 400.
```

```
# A tibble: 11 x 4
   variable                          rhat ess_bulk ess_tail
   <chr>                            <dbl>    <dbl>    <dbl>
 1 b_Intercept                       1.00   93982.   93908.
 2 b_pubertal_timing_s               1.01   14111.   88806.
 3 b_adhd_statusY                    1.01   23470.   93195.
 4 b_sed_percentile_s                1.03    2694.   11807.
 5 b_bmi_s                           1.01   21273.   91553.
 6 b_smfq_total_s                    1.01   12413.   94214.
 7 b_pubertal_timing_s:adhd_statusY  1.01   19624.   92670.
 8 sigma                             1.01    6105.   79181.
```

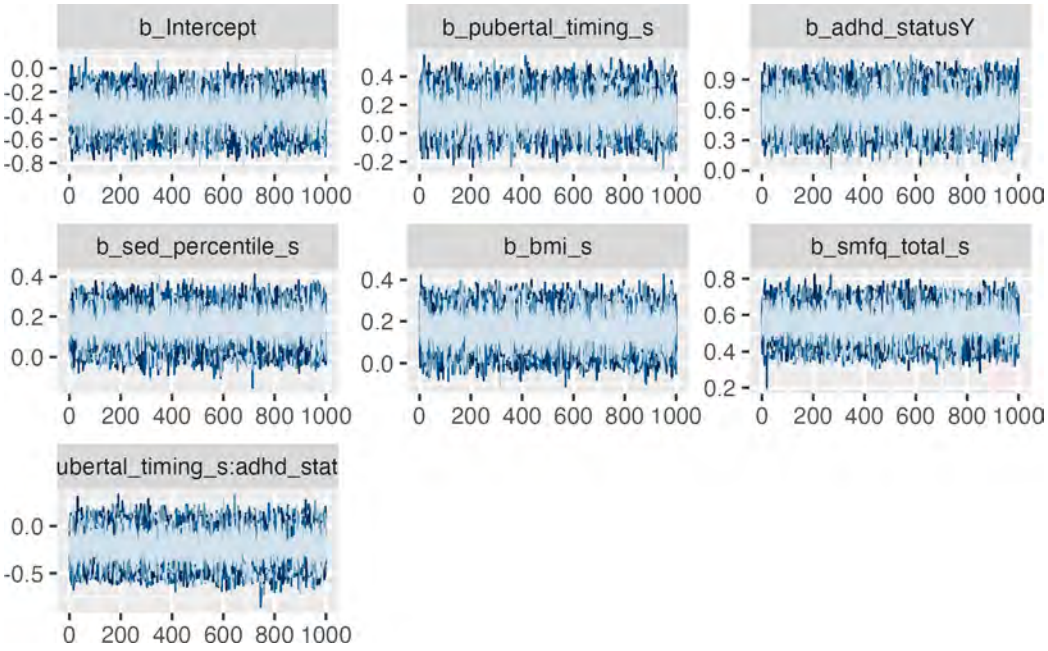
```
 9 Intercept                          1.00  168711.   95384.
10 lprior                             1.00   50337.   96821.
11 lp__                               1.19     417.     889.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



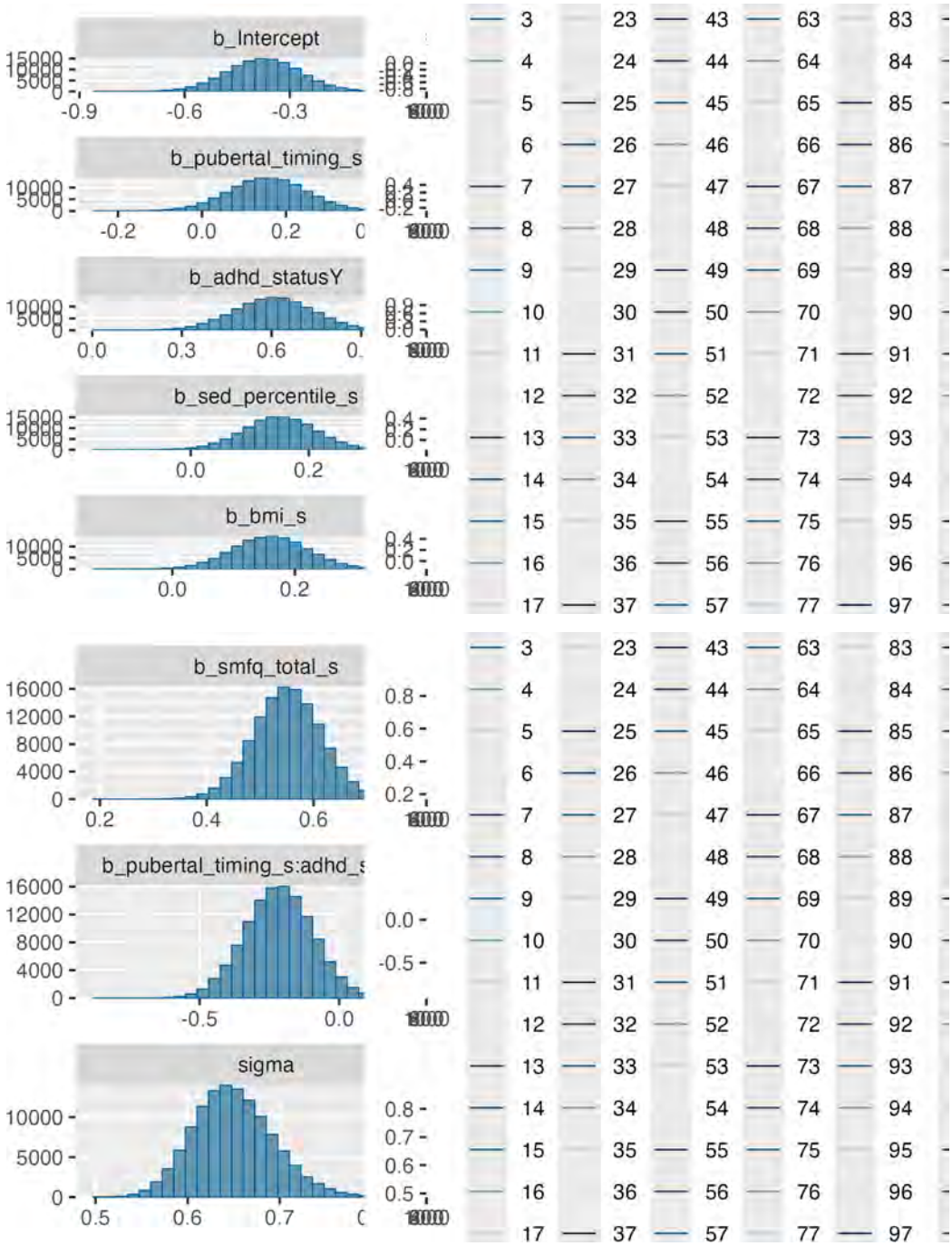

```
=== Prior Summary ===

Priors specified for the model.
```

```
          prior     class                           coef group resp dpar nlpar
   normal(0, 1)         b
   normal(0, 1)         b                     adhd_statusY
   normal(0, 1)         b                            bmi_s
   normal(0, 1)         b                pubertal_timing_s
   normal(0, 1)         b pubertal_timing_s:adhd_statusY
   normal(0, 1)         b                 sed_percentile_s
   normal(0, 1)         b                     smfq_total_s
   normal(0, 1) Intercept
 exponential(1)     sigma
 lb ub tag       source
                   user
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
                   user
  0                user


=== Posterior Predictive Checks ===

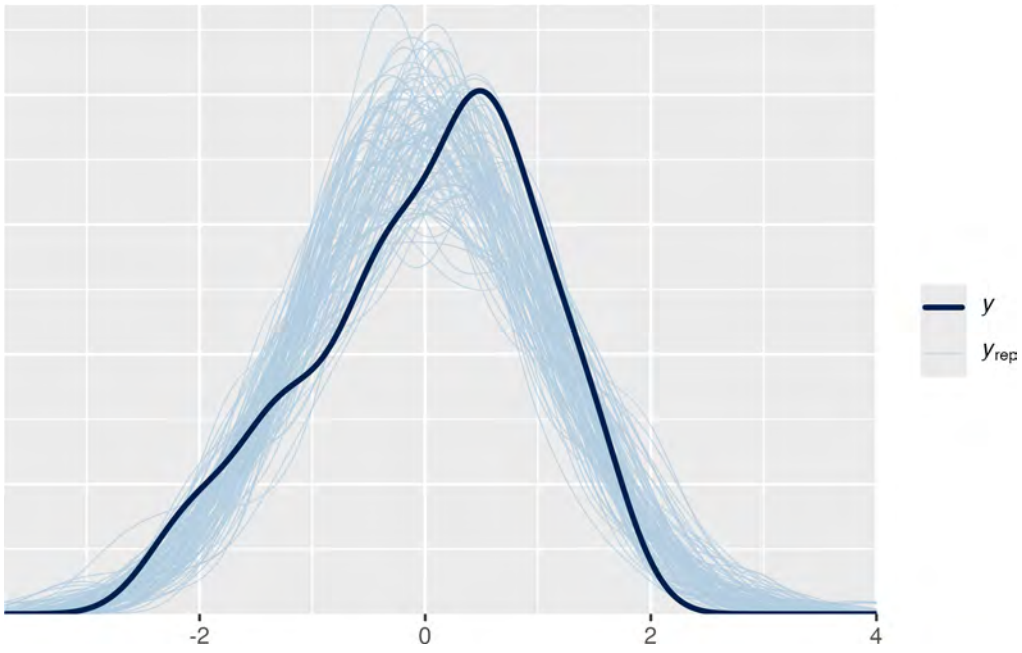
Density plots for each parameter.
```




```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.
```

```
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



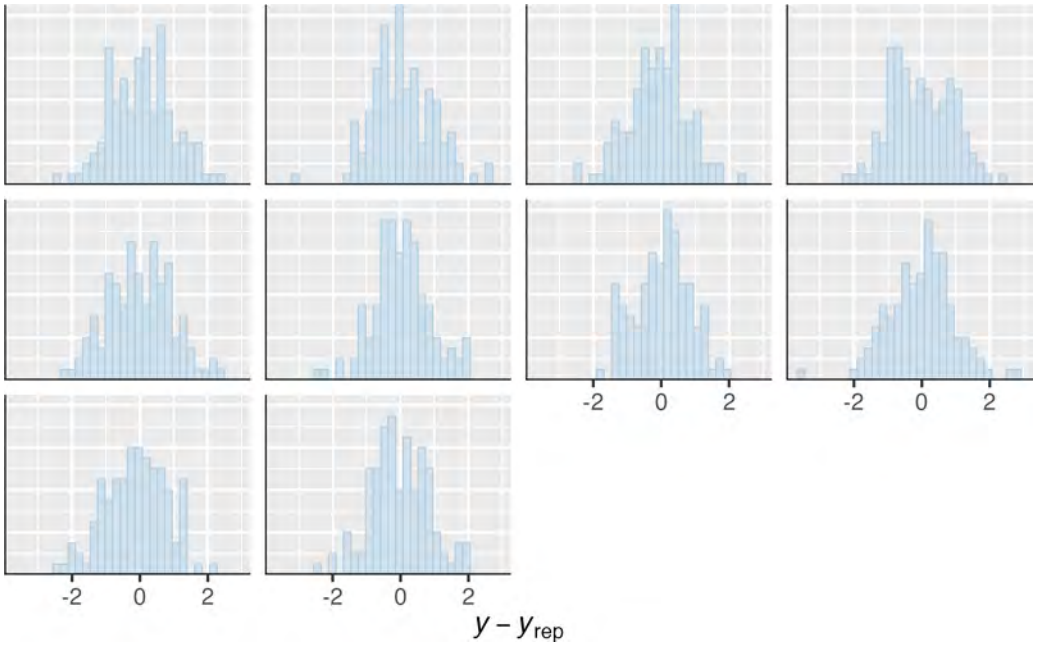

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

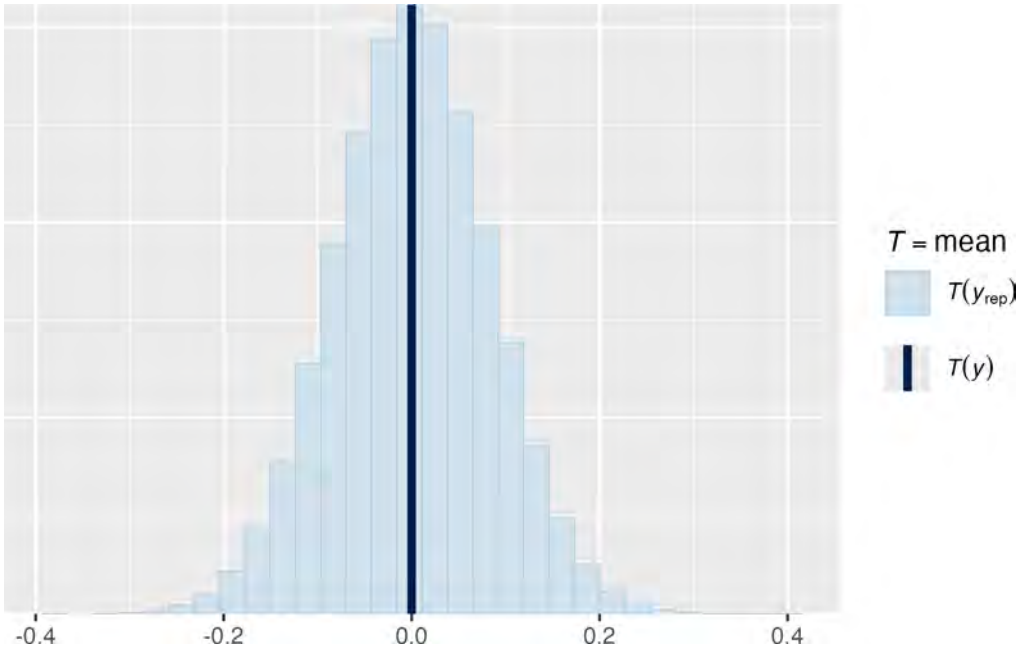
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



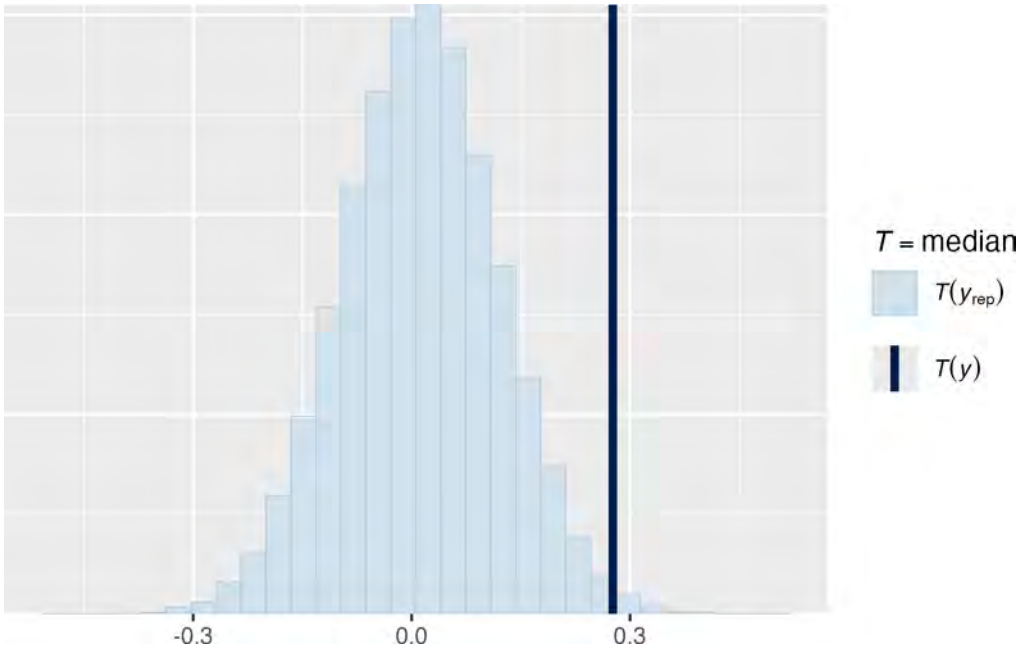

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



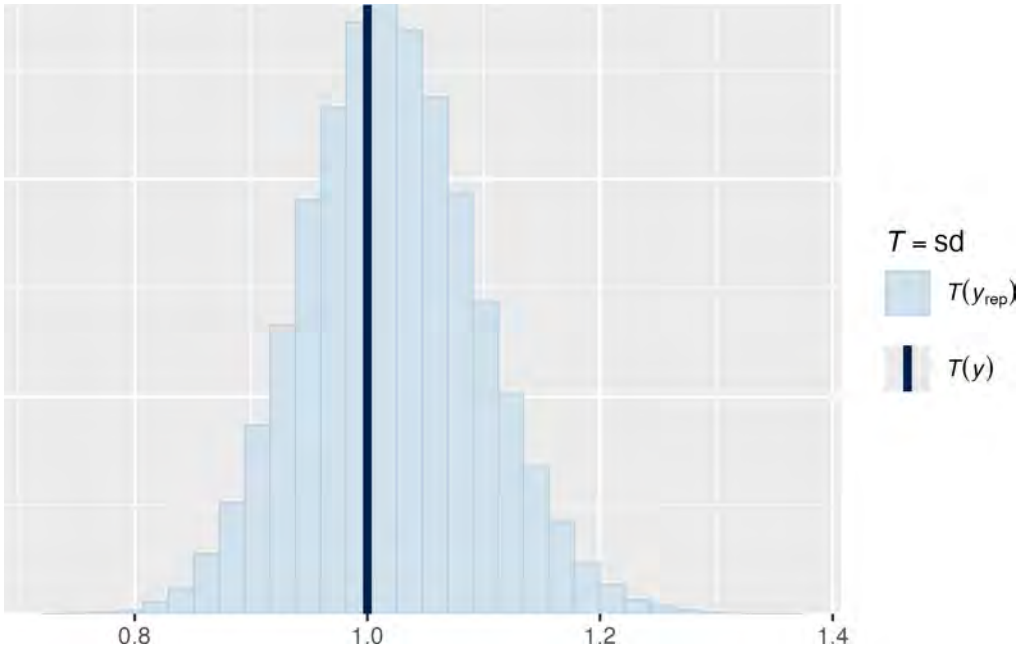

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

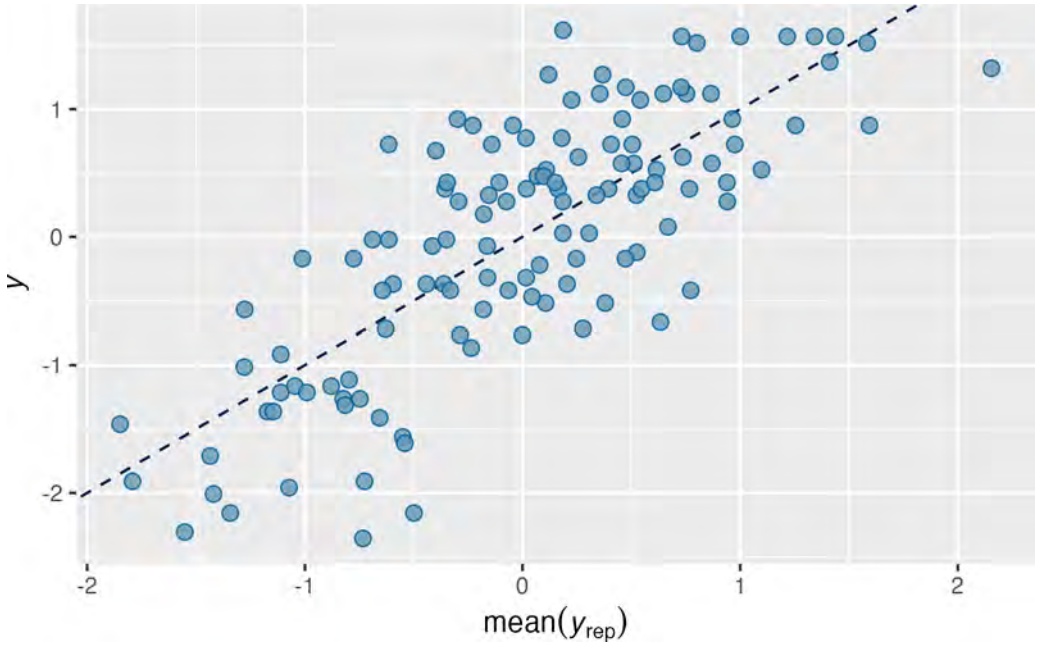



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

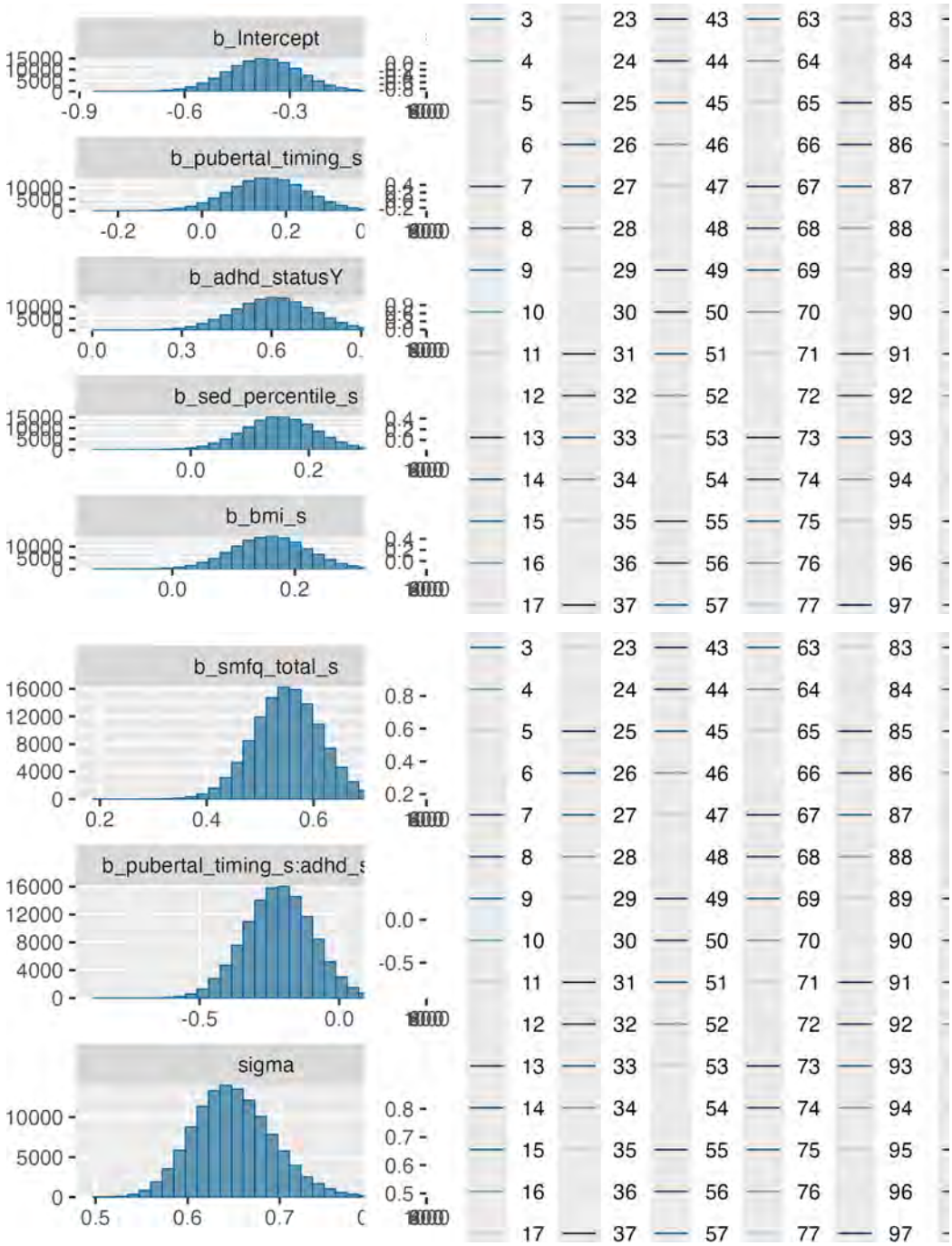
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H1 Observed (Non-Imputed) Model output

```
Estimate  Est.Error      Q2.5     Q97.5
```

```
R2 0.2985912 0.05758005 0.1801954 0.4061519
```

```
Summary of Posterior Distribution
```

Parameter	Mean	Median	CI	pd
b_Intercept	-0.60	-0.60	[-0.88, -0.34]	100%
b_pds_total_s	0.20	0.20	[-0.05, 0.47]	93.10%
b_adhd_statusY	0.95	0.95	[0.63, 1.30]	100%
b_age_cont_s	9.44e-05	-7.70e-04	[-0.23, 0.22]	50.28%
b_sed_percentile_s	0.09	0.09	[-0.07, 0.26]	85.97%
b_pds_total_s:adhd_statusY	-0.06	-0.06	[-0.40, 0.25]	64.08%

Parameter	% in ROPE	meaningful
b_Intercept	0.07%	TRUE
b_pds_total_s	21.35%	FALSE
b_adhd_statusY	0%	TRUE
b_age_cont_s	62.20%	FALSE
b_sed_percentile_s	52.35%	FALSE
b_pds_total_s:adhd_statusY	43.82%	FALSE

H1 Observed Model diagnostics and PP checks

```
=== Rhat and ESS ===
```

```
Rhat values close to 1 indicate convergence. ESS should ideally be > 400.
```

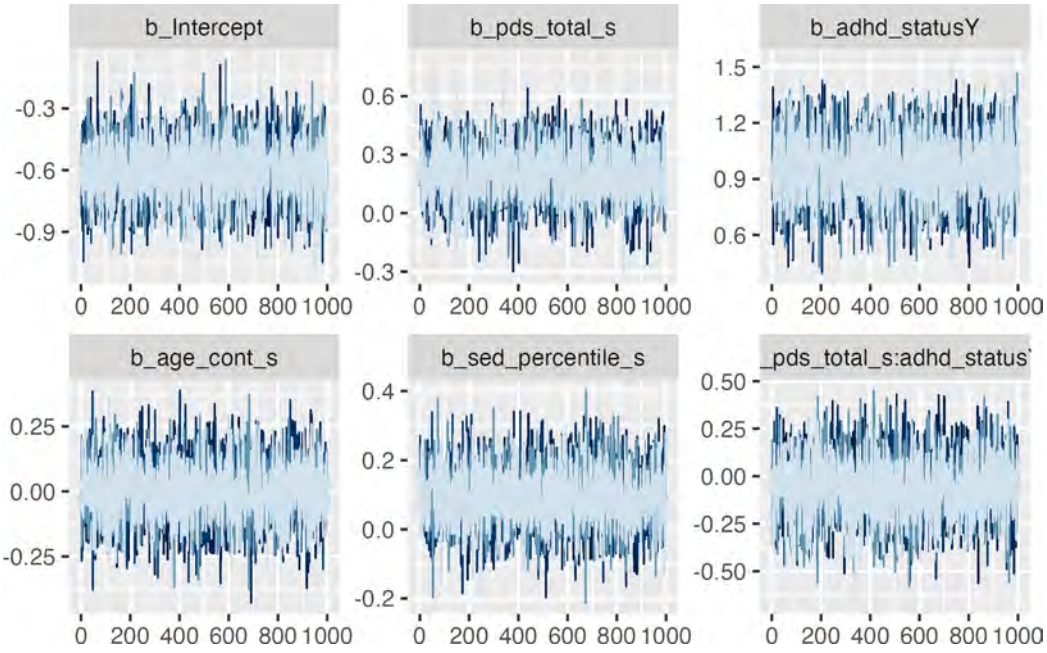
```
# A tibble: 10 x 4
   variable                    rhat ess_bulk ess_tail
   <chr>                      <dbl>    <dbl>    <dbl>
 1 b_Intercept                 1.00    4718.    3017.
 2 b_pds_total_s               1.00    2576.    2545.
 3 b_adhd_statusY              1.00    4535.    3251.
 4 b_age_cont_s                1.00    3543.    3008.
 5 b_sed_percentile_s          1.00    4418.    2709.
 6 b_pds_total_s:adhd_statusY  1.00    3252.    2726.
 7 sigma                       1.00    3650.    3217.
 8 Intercept                   1.00    5083.    3073.
 9 lprior                      1.00    4492.    3090.
10 lp__                        1.00    1895.    2573.
```

```
=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

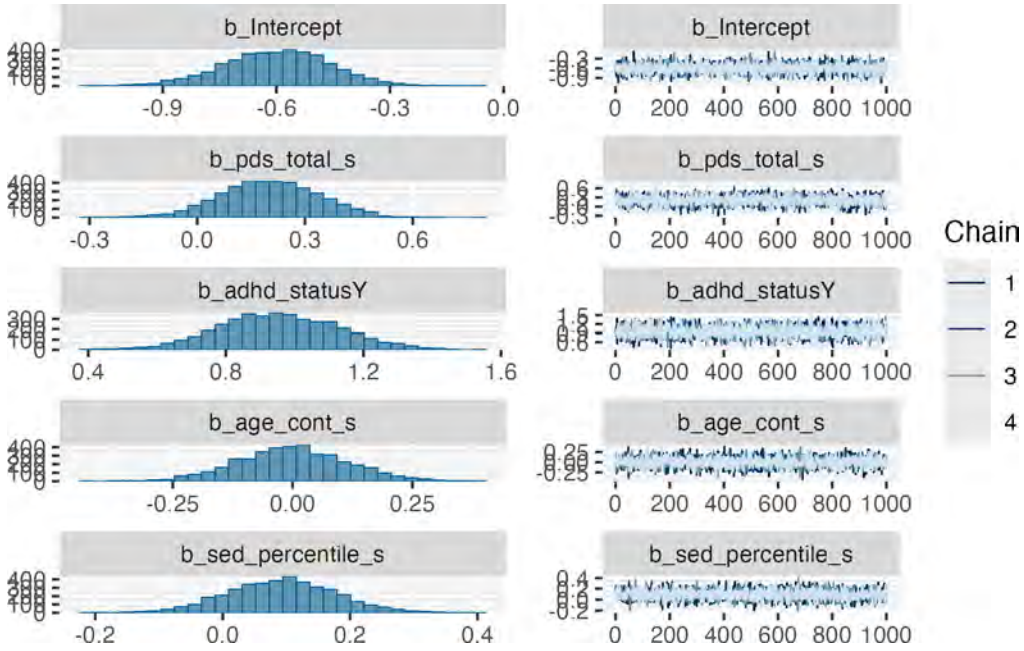
Priors specified for the model.

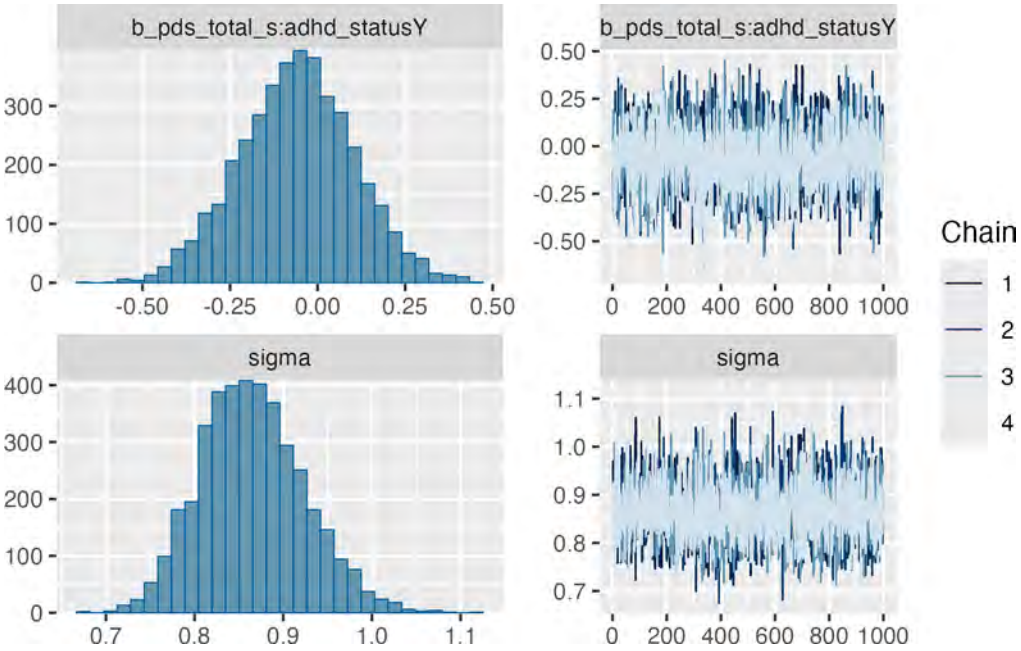
          prior     class                     coef group resp dpar nlpar lb ub
   normal(0, 1)         b
   normal(0, 1)         b             adhd_statusY
   normal(0, 1)         b               age_cont_s
   normal(0, 1)         b              pds_total_s
   normal(0, 1)         b pds_total_s:adhd_statusY
   normal(0, 1)         b         sed_percentile_s
```

```
  normal(0, 1) Intercept
exponential(1)     sigma                                                    0
tag       source
            user
    (vectorized)
    (vectorized)
    (vectorized)
    (vectorized)
    (vectorized)
            user
            user


=== Posterior Predictive Checks ===

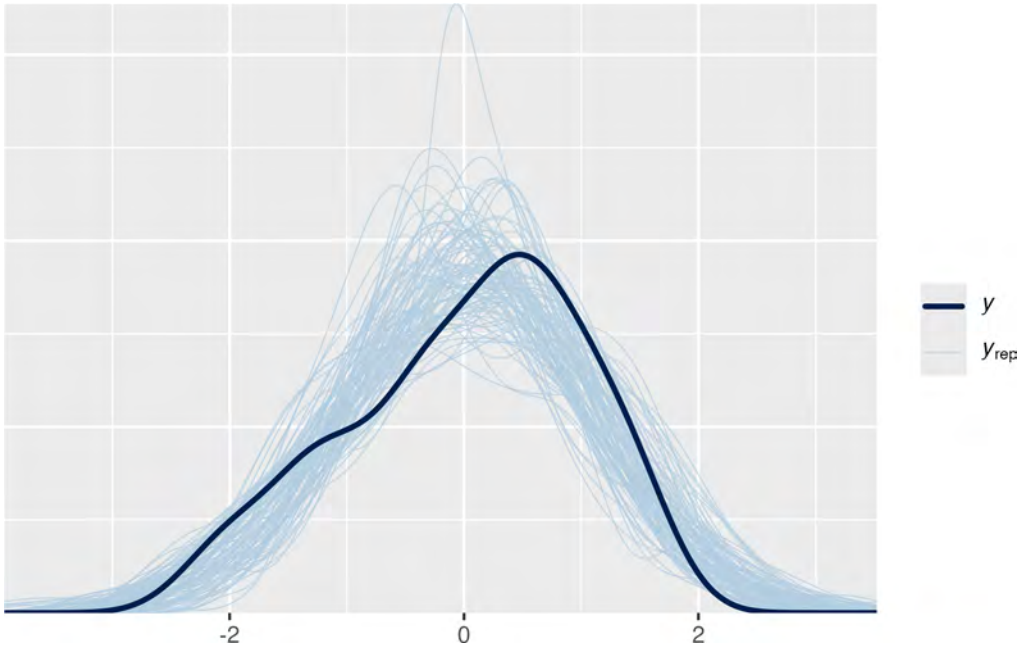
Density plots for each parameter.
```






```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.
```

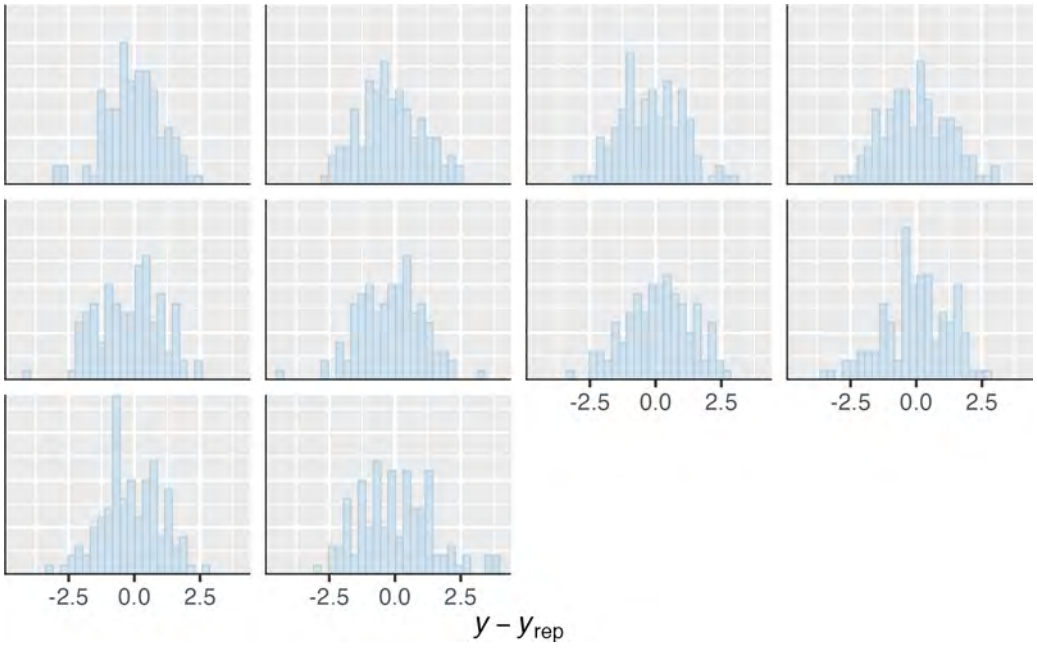


```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

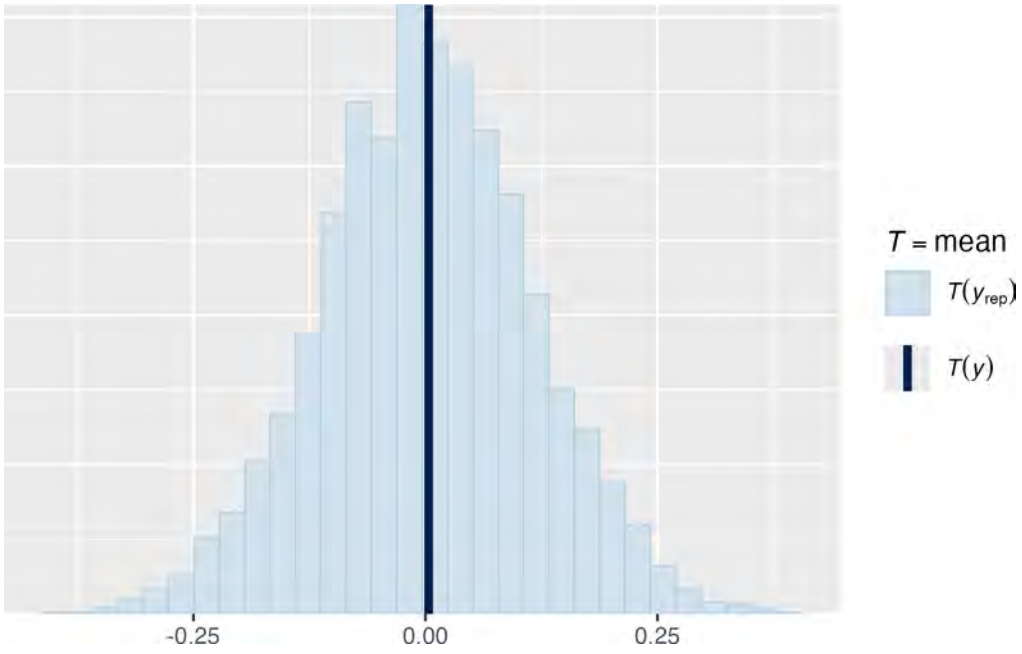
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

Observed mean (dark line) should fall within the distribution of replicated means.

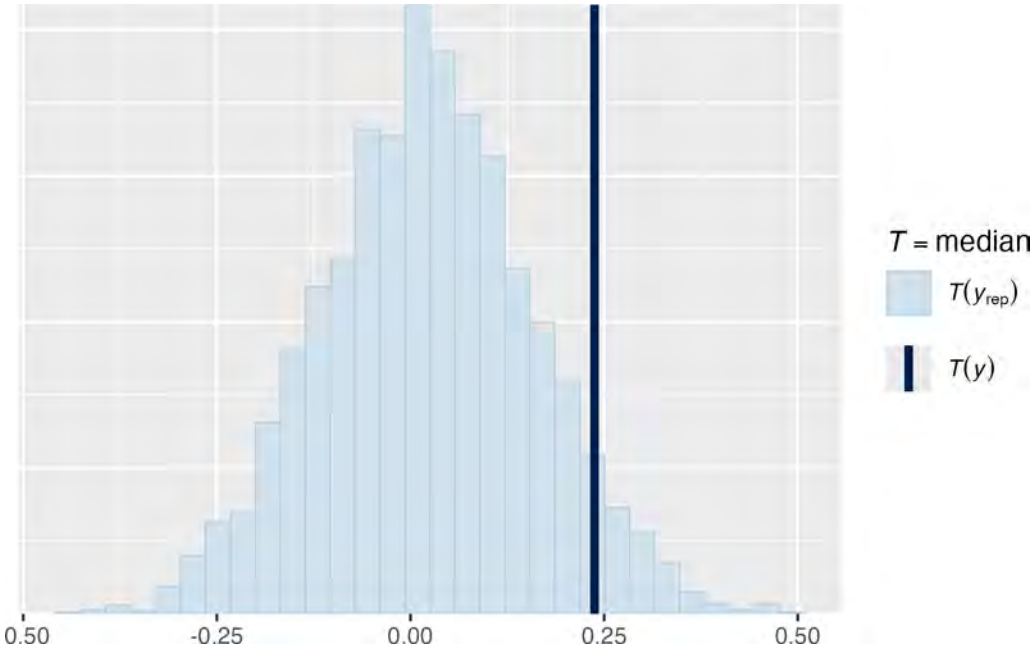
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

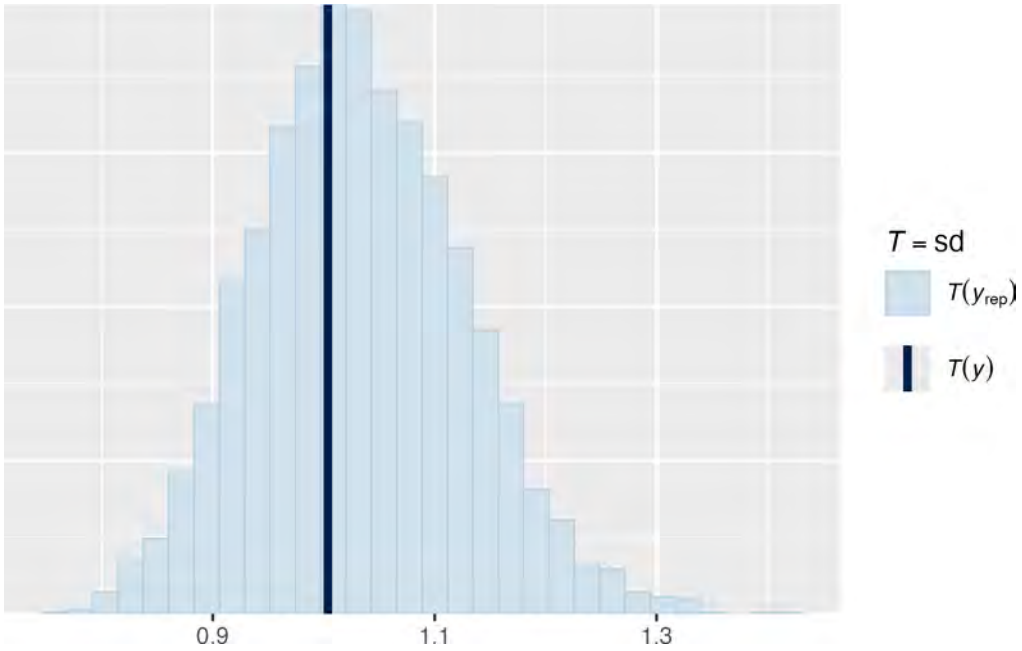
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

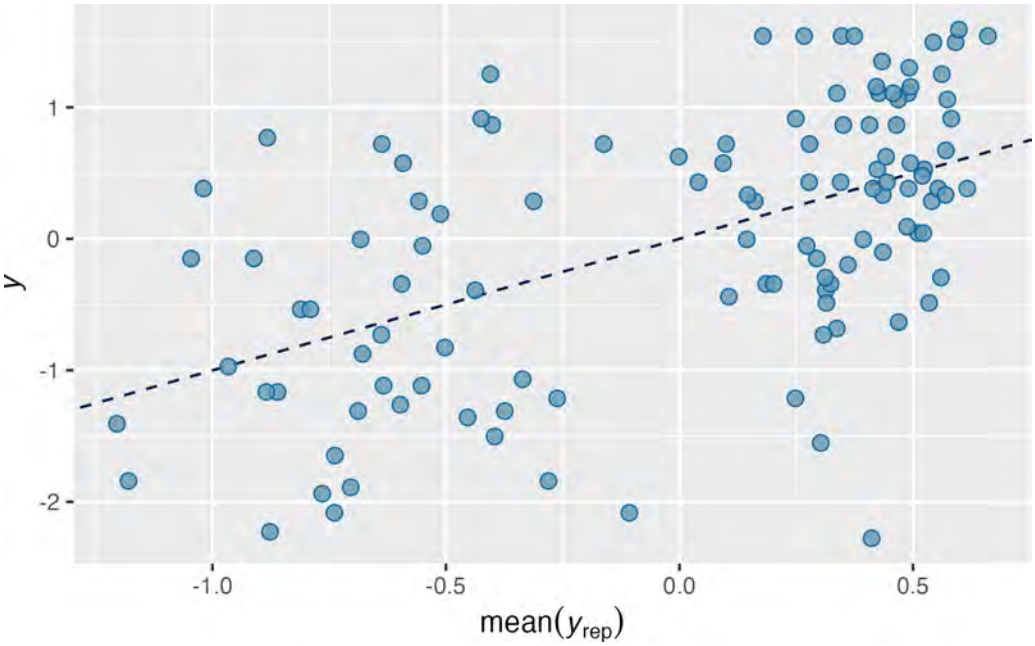
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Scatter Average ===

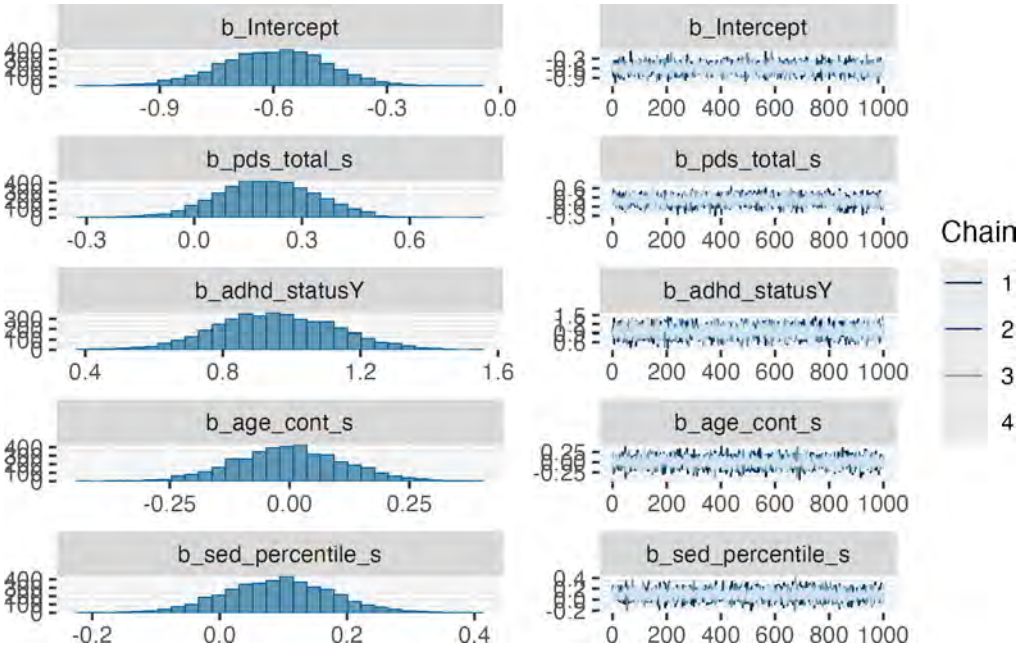
Points should cluster around the diagonal. Systematic patterns indicate misfit.

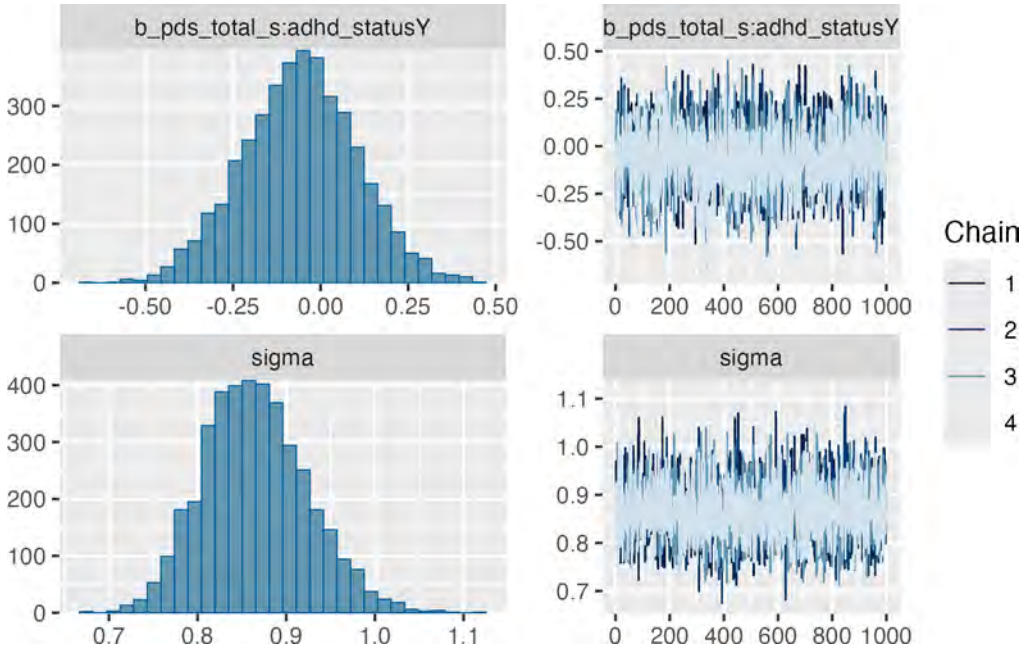
Using all posterior draws for ppc type 'scatter_avg' by default.
```




```
=== Pareto-k ===
```

```
All Pareto k estimates are good (k < 0.7).
```





H2 Observed (Non-Imputed) Model output

```
    Estimate  Est.Error     Q2.5     Q97.5
R2 0.3129592 0.05729467 0.195159 0.4188081
```

```
Summary of Posterior Distribution

Parameter                          |  Mean | Median |             CI |     pd
-------------------------------------------------------------------------------
b_Intercept                        | -0.61 |  -0.61 | [-0.85, -0.33] |   100%
b_pubertal_timing_s                |  0.32 |   0.32 | [ 0.10,  0.57] | 99.72%
b_adhd_statusY                     |  0.99 |   0.99 | [ 0.67,  1.30] |   100%
b_sed_percentile_s                 |  0.12 |   0.13 | [-0.03,  0.30] | 93.90%
b_pubertal_timing_s:adhd_statusY   | -0.35 |  -0.35 | [-0.65, -0.03] | 98.55%

Parameter                          | % in ROPE | meaningful
-----------------------------------------------------------
b_Intercept                        |        0% |       TRUE
b_pubertal_timing_s                |     3.17% |       TRUE
b_adhd_statusY                     |        0% |       TRUE
b_sed_percentile_s                 |    37.43% |      FALSE
b_pubertal_timing_s:adhd_statusY   |     5.88% |      FALSE
```

H2 Observed (Non-Imputed) Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

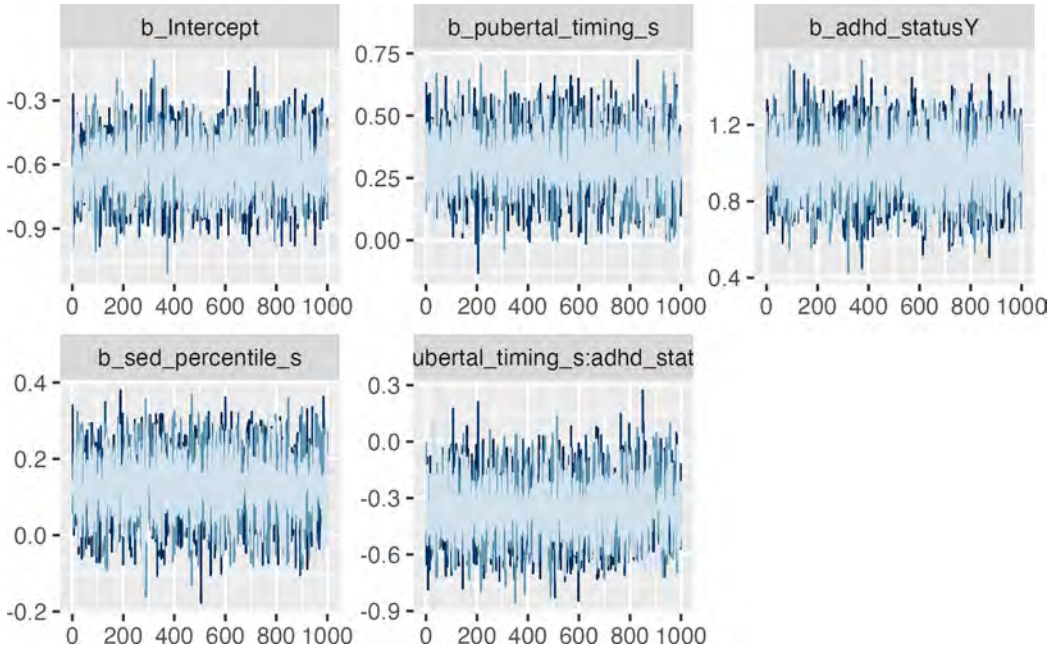
# A tibble: 9 x 4
  variable                         rhat ess_bulk ess_tail
  <chr>                           <dbl>    <dbl>    <dbl>
1 b_Intercept                      1.00    4588.    2716.
2 b_pubertal_timing_s              1.00    2603.    2635.
3 b_adhd_statusY                   1.00    4824.    2685.
4 b_sed_percentile_s               1.00    4693.    2917.
5 b_pubertal_timing_s:adhd_statusY 1.00    2584.    2497.
6 sigma                            1.00    5001.    2806.
7 Intercept                        1.00    4245.    2630.
8 lprior                           1.00    4355.    2835.
9 lp__                             1.00    1792.    2328.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



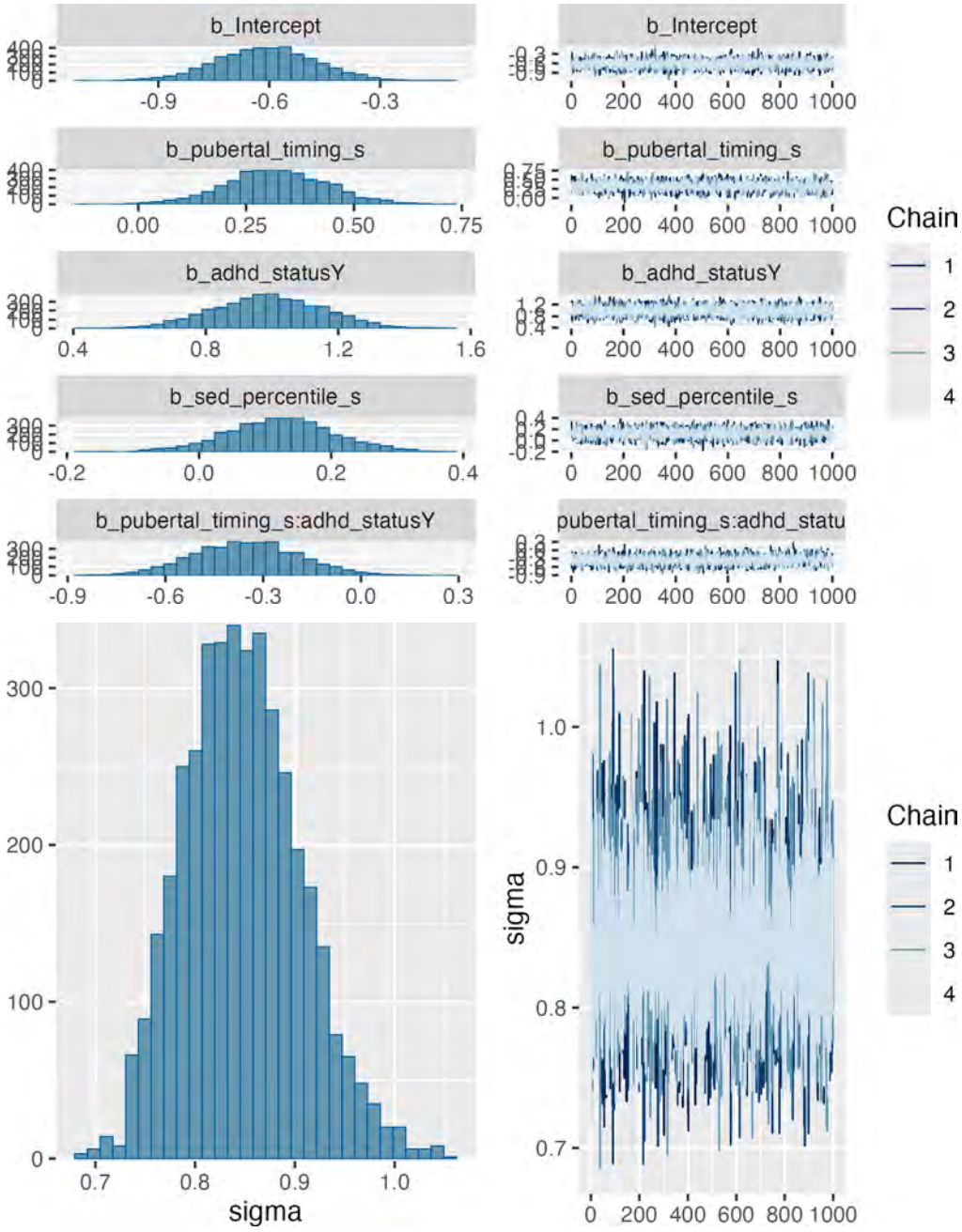

```
=== Prior Summary ===

Priors specified for the model.

          prior     class                          coef group resp dpar nlpar
   normal(0, 1)         b
   normal(0, 1)         b                   adhd_statusY
   normal(0, 1)         b              pubertal_timing_s
   normal(0, 1)         b pubertal_timing_s:adhd_statusY
   normal(0, 1)         b               sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma
 lb ub tag       source
                   user
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
                   user
  0                user

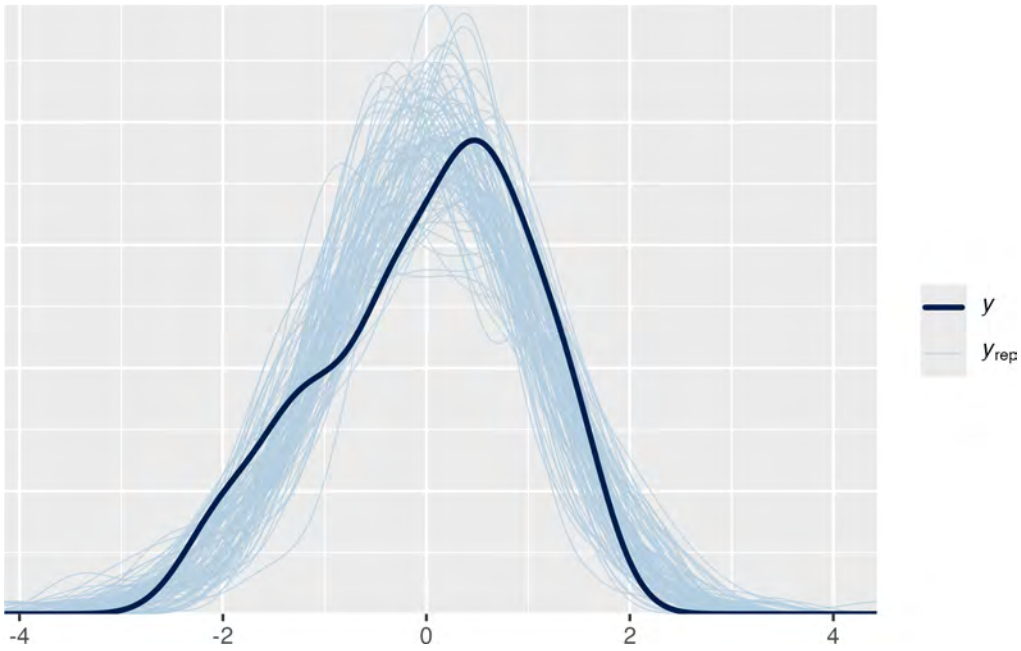

=== Posterior Predictive Checks ===
```

Density plots for each parameter.



=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

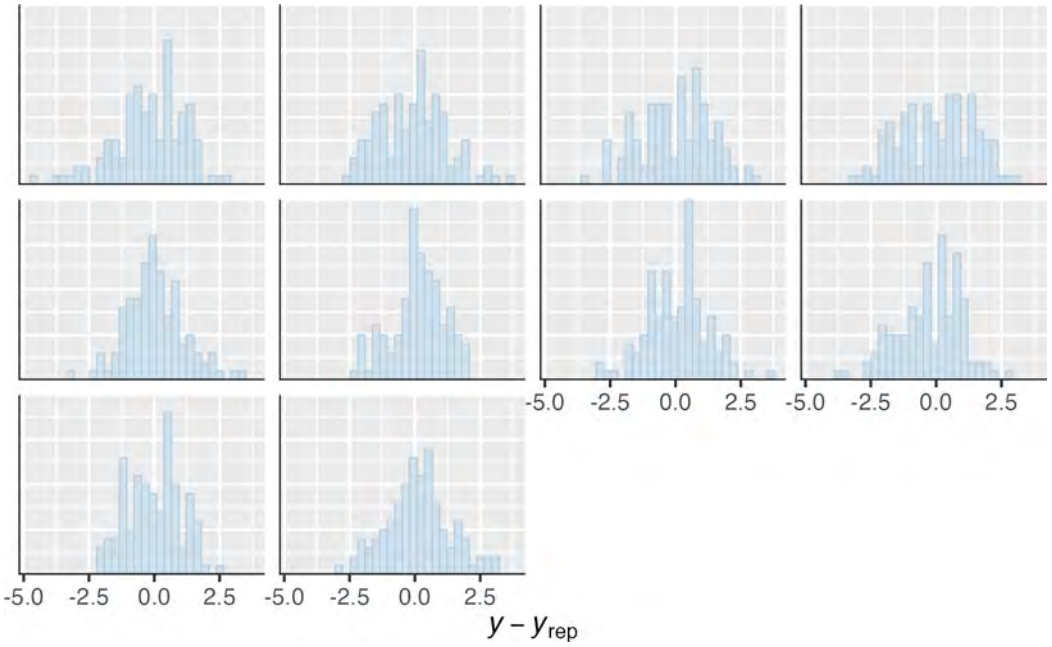


=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

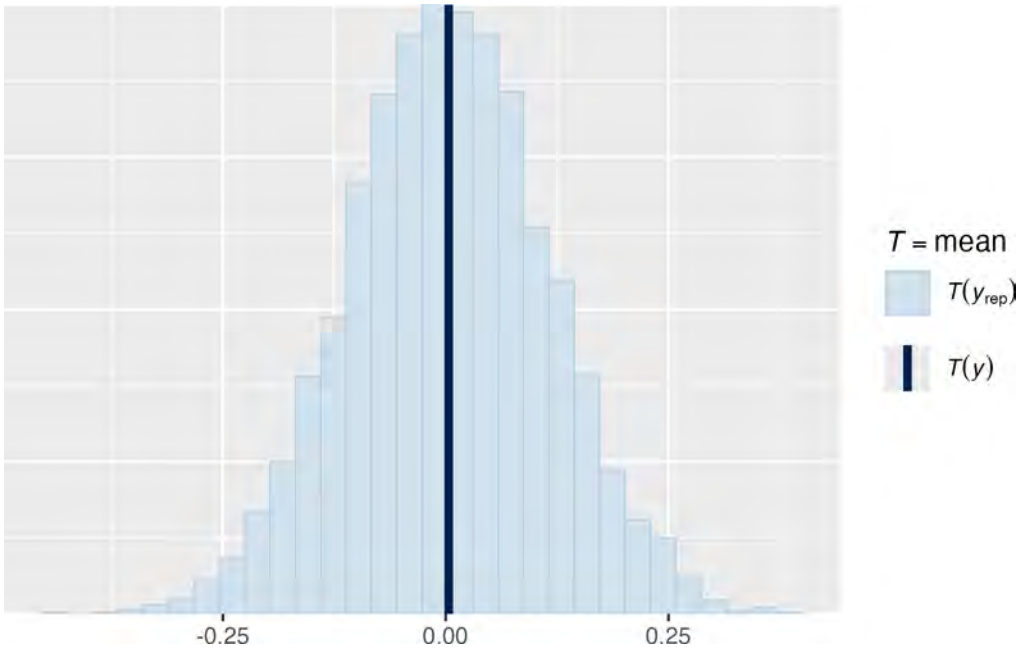
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.




```
=== PPC: Mean ===

Observed mean (dark line) should fall within the distribution of replicated means.

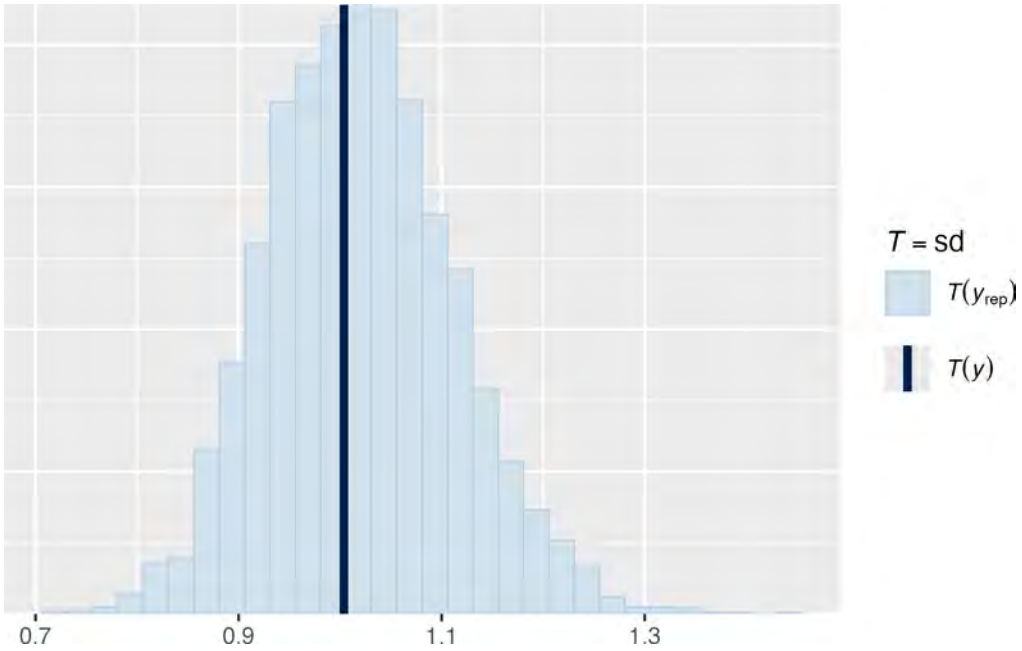
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

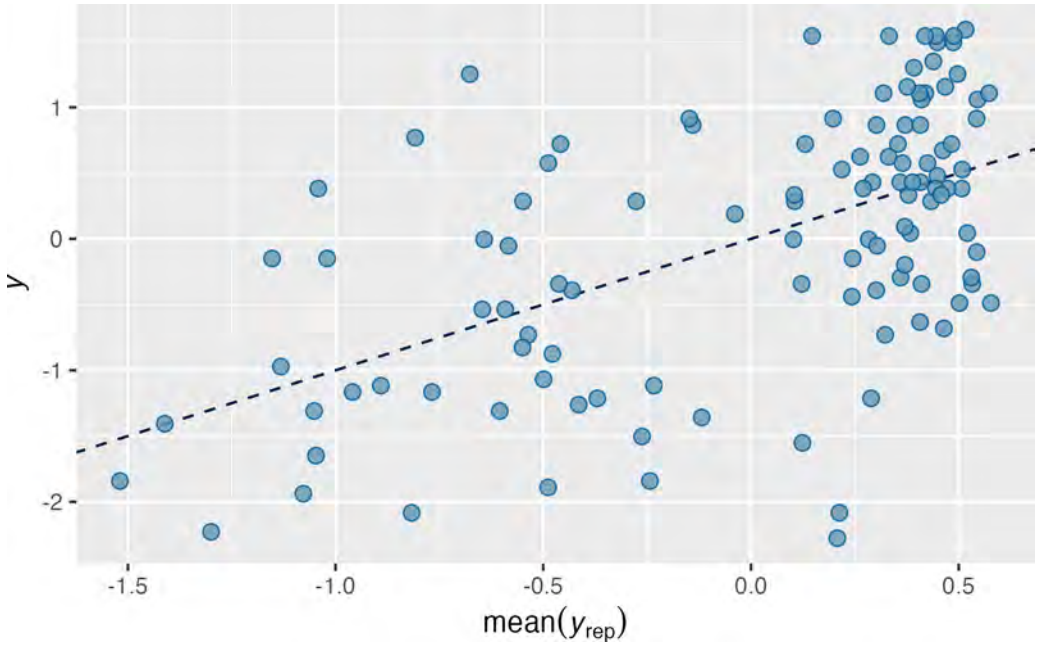
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Scatter Average ===

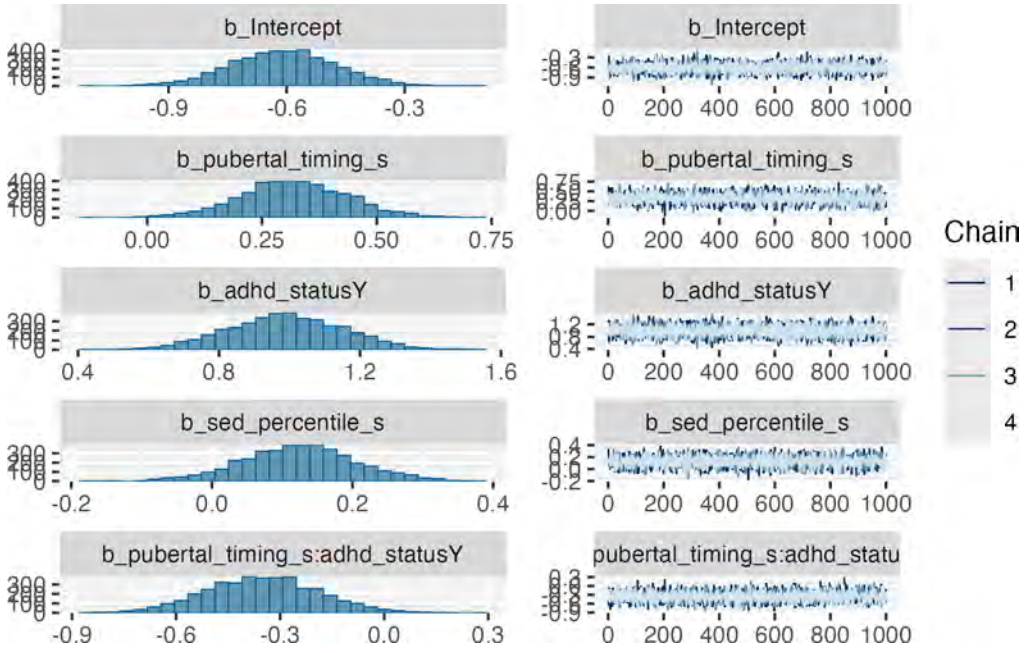
Points should cluster around the diagonal. Systematic patterns indicate misfit.

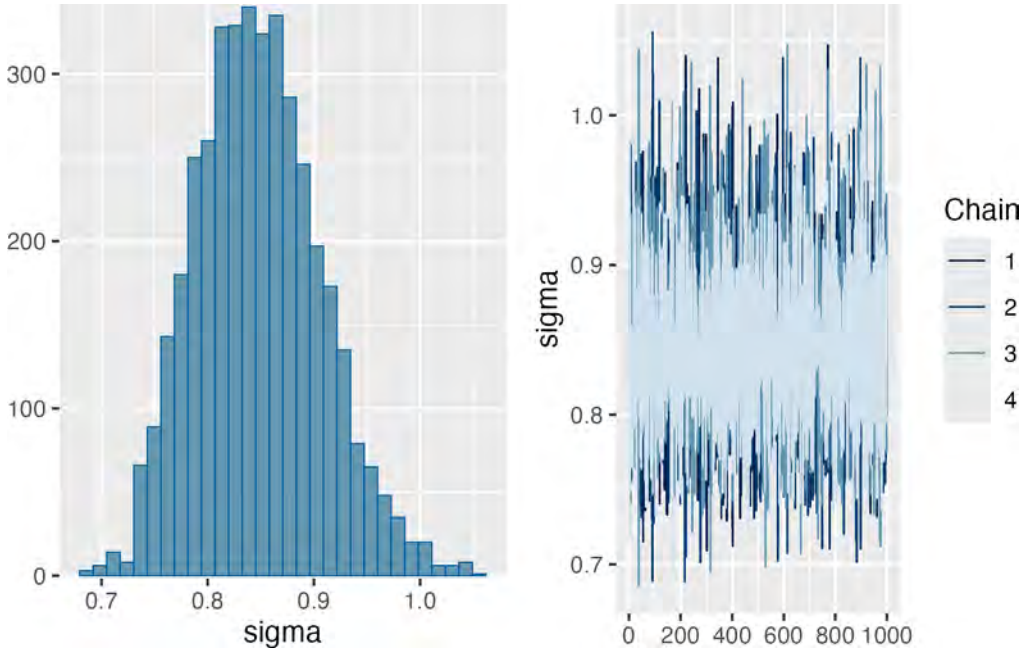
Using all posterior draws for ppc type 'scatter_avg' by default.
```




```
=== Pareto-k ===
```

```
All Pareto k estimates are good (k < 0.7).
```





H1 Prior 1 Model output

```
    Estimate Est.Error      Q2.5     Q97.5
R2 0.2995884 0.0554383 0.1853446 0.4019177
```

```
Summary of Posterior Distribution

Parameter                   |      Mean |    Median |             CI |     pd
------------------------------------------------------------------------------
b_Intercept                 |     -0.61 |     -0.61 | [-0.88, -0.36] |   100%
b_pds_total_s               |      0.21 |      0.21 | [-0.05,  0.47] | 94.06%
b_adhd_statusY              |      0.96 |      0.96 | [ 0.63,  1.27] |   100%
b_age_cont_s                | -6.07e-05 | -2.88e-04 | [-0.20,  0.20] | 50.12%
b_sed_percentile_s          |      0.09 |      0.09 | [-0.07,  0.24] | 86.67%
b_pds_total_s:adhd_statusY  |     -0.06 |     -0.06 | [-0.36,  0.25] | 64.62%

Parameter                   | % in ROPE | meaningful
--------------------------------------------------
b_Intercept                 | 9.17e-03% |       TRUE
b_pds_total_s               |    20.12% |      FALSE
b_adhd_statusY              |        0% |       TRUE
b_age_cont_s                |    66.60% |      FALSE
b_sed_percentile_s          |    55.93% |      FALSE
```

```
b_pds_total_s:adhd_statusY |    45.02% |      FALSE
```

H1 Prior 1 Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

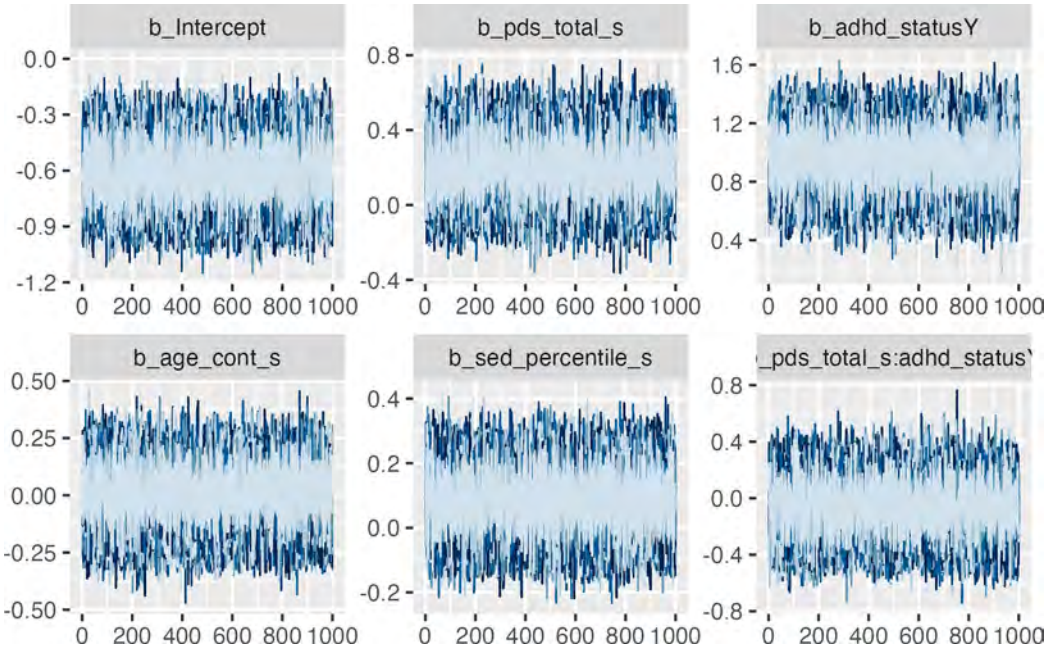
# A tibble: 10 x 4
   variable                   rhat ess_bulk ess_tail
   <chr>                     <dbl>    <dbl>    <dbl>
 1 b_Intercept                1.00  129257.   93913.
 2 b_pds_total_s              1.00   80948.   87089.
 3 b_adhd_statusY             1.00  134751.   93657.
 4 b_age_cont_s               1.00   33224.   90691.
 5 b_sed_percentile_s         1.00   32793.   94043.
 6 b_pds_total_s:adhd_statusY 1.00   94848.   91131.
 7 sigma                      1.00  126868.   91958.
 8 Intercept                  1.00  145705.   91509.
 9 lprior                     1.00  137507.   93132.
10 lp__                       1.03    2559.    3686.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



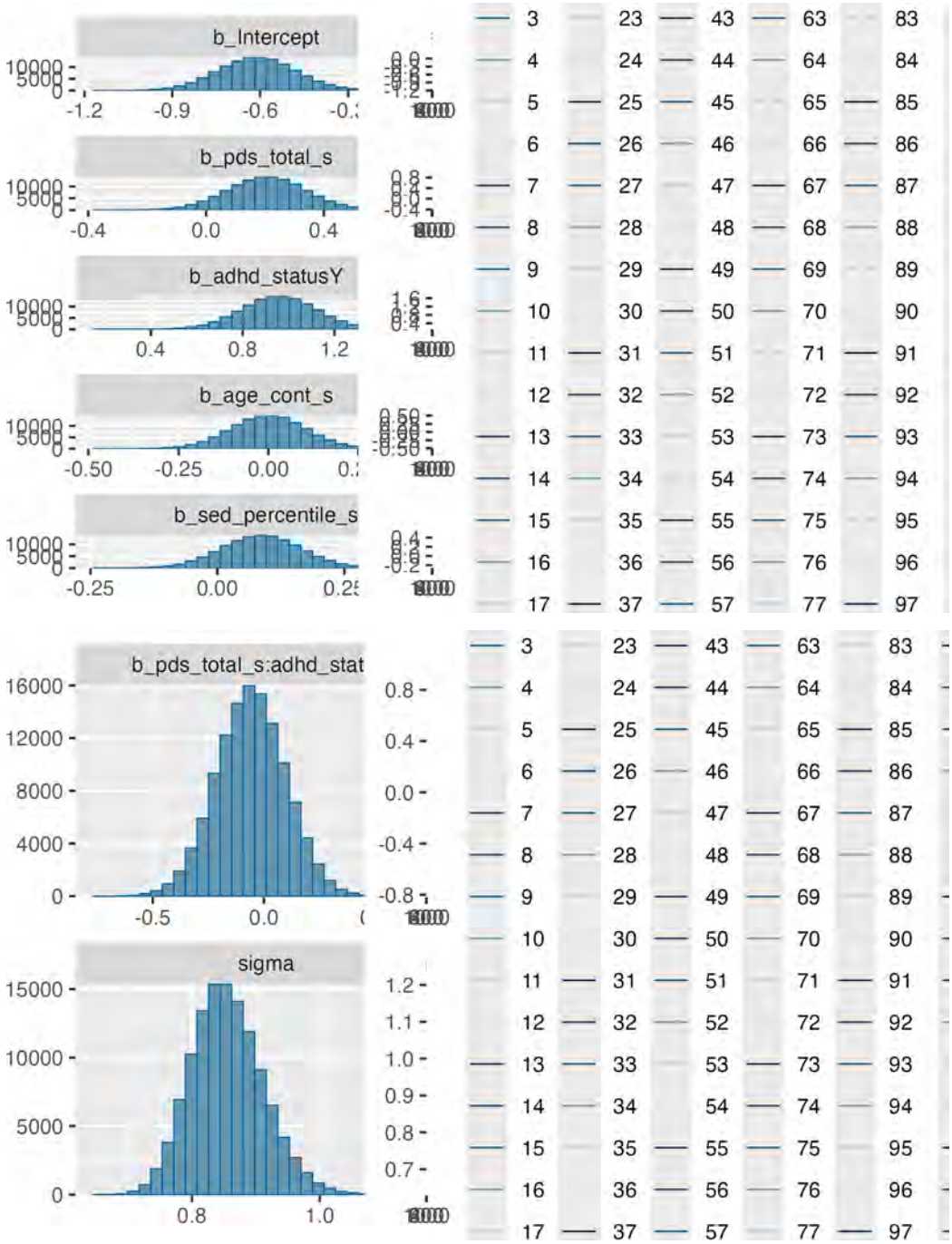

```
=== Prior Summary ===

Priors specified for the model.

          prior     class                     coef group resp dpar nlpar lb ub
 normal(0, 0.75)         b
 normal(0, 0.75)         b             adhd_statusY
 normal(0, 0.75)         b               age_cont_s
 normal(0, 0.75)         b              pds_total_s
 normal(0, 0.75)         b pds_total_s:adhd_statusY
 normal(0, 0.75)         b         sed_percentile_s
    normal(0, 1) Intercept
  exponential(1)     sigma                                                   0
 tag       source
             user
     (vectorized)
     (vectorized)
     (vectorized)
     (vectorized)
     (vectorized)
             user
             user
```

```
=== Posterior Predictive Checks ===

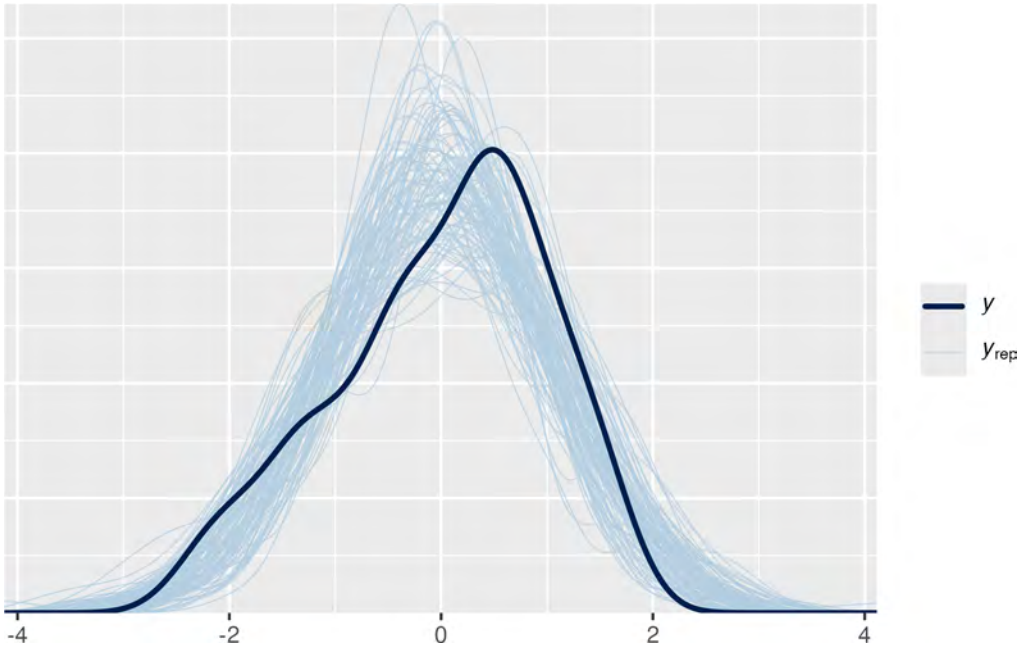
Density plots for each parameter.
```



```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



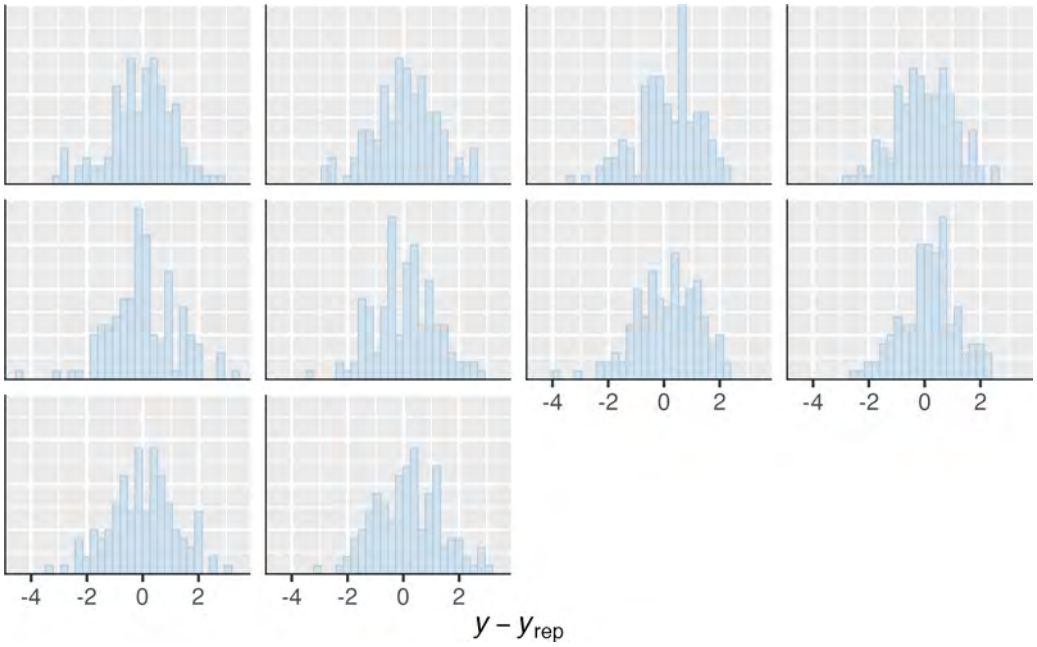

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

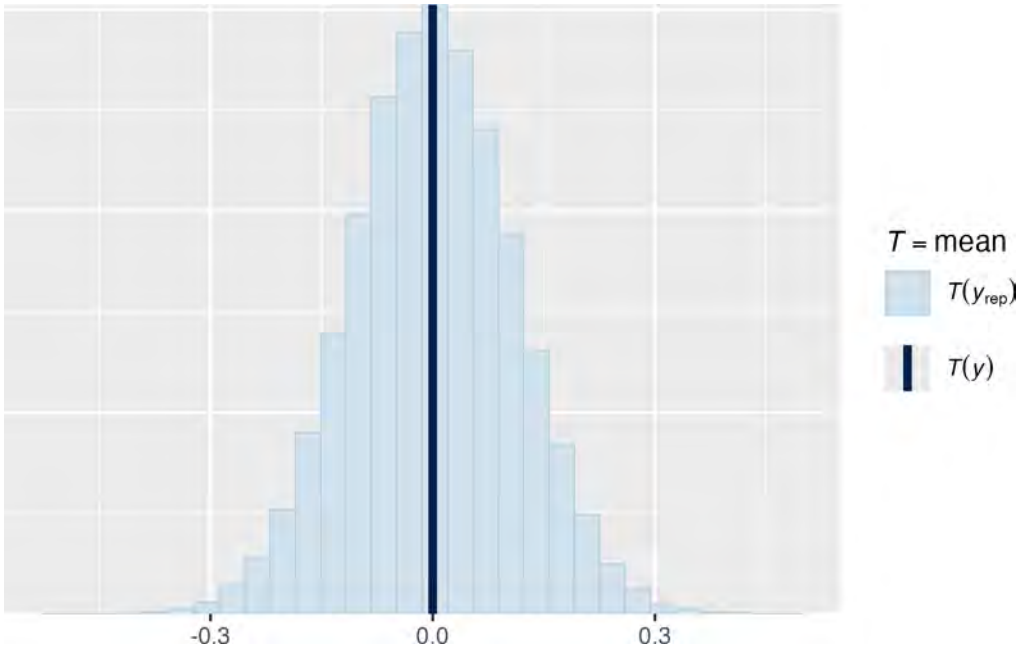
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



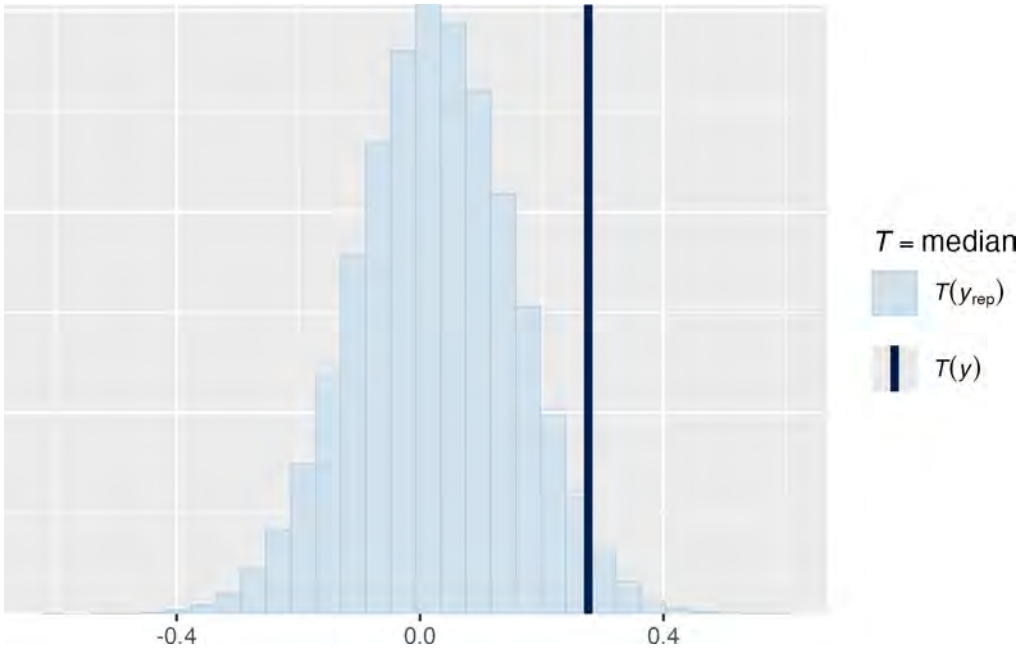

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



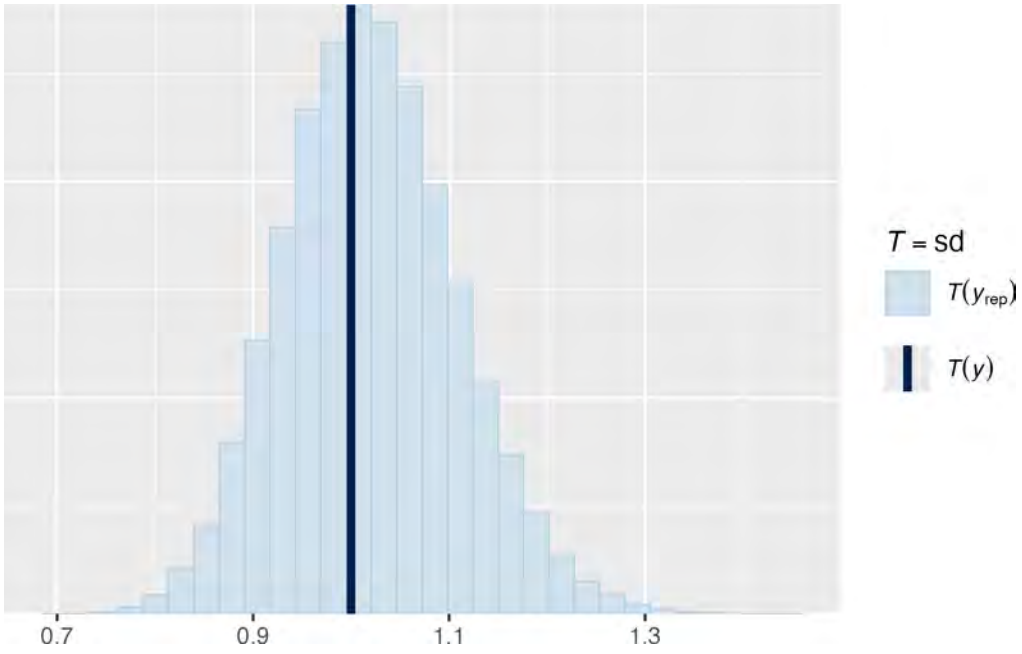

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

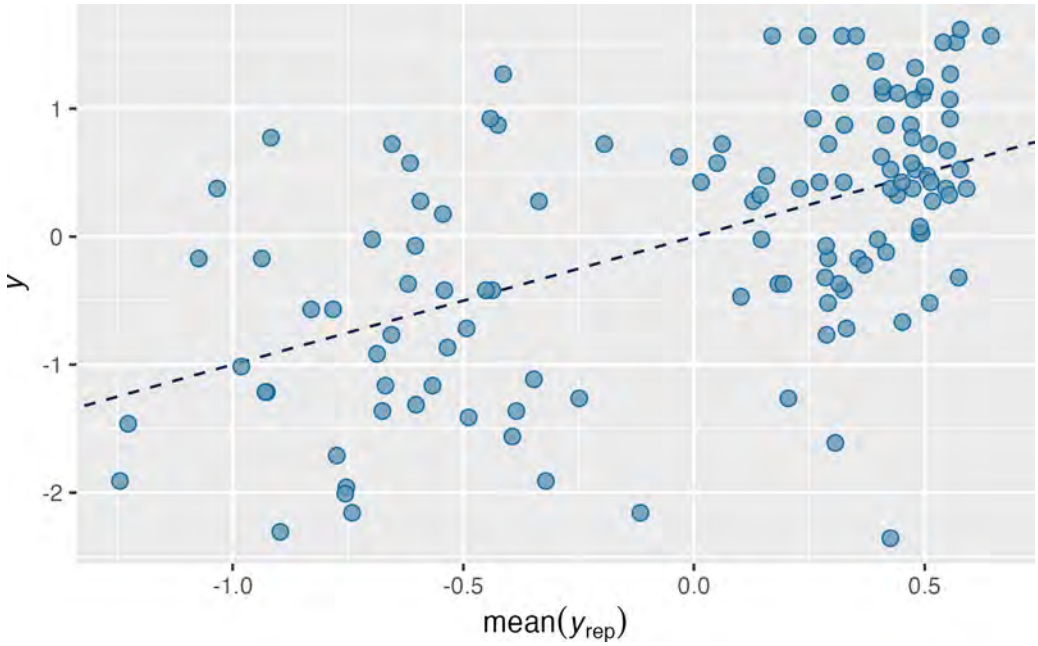



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

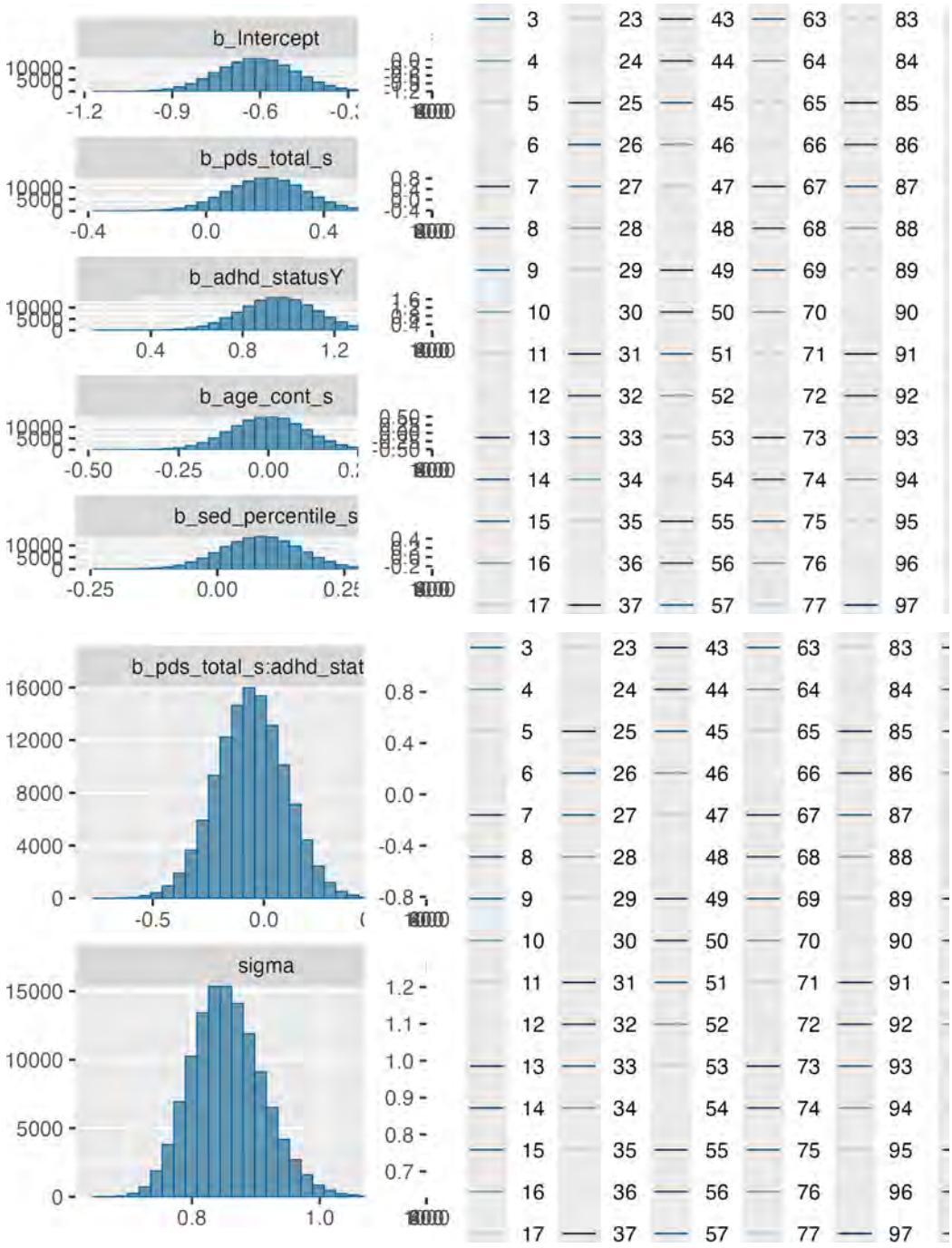
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Prior 1 Model output

```
Estimate  Est.Error      Q2.5     Q97.5
```

```
R2 0.309593 0.05603917 0.1942682 0.4123329

Summary of Posterior Distribution

Parameter                         |  Mean | Median |             CI |     pd
-------------------------------------------------------------------------------
b_Intercept                       | -0.61 |  -0.61 | [-0.86, -0.36] |   100%
b_pubertal_timing_s               |  0.31 |   0.31 | [ 0.09,  0.53] | 99.62%
b_adhd_statusY                    |  0.99 |   0.99 | [ 0.67,  1.30] |   100%
b_sed_percentile_s                |  0.12 |   0.12 | [-0.04,  0.26] | 93.34%
b_pubertal_timing_s:adhd_statusY  | -0.33 |  -0.34 | [-0.63, -0.03] | 98.56%

Parameter                         | % in ROPE | meaningful
-------------------------------------------------------------
b_Intercept                       | 3.33e-03% |       TRUE
b_pubertal_timing_s               |     3.26% |       TRUE
b_adhd_statusY                    |        0% |       TRUE
b_sed_percentile_s                |    41.83% |      FALSE
b_pubertal_timing_s:adhd_statusY  |     6.03% |      FALSE
```

H2 Prior 1 Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

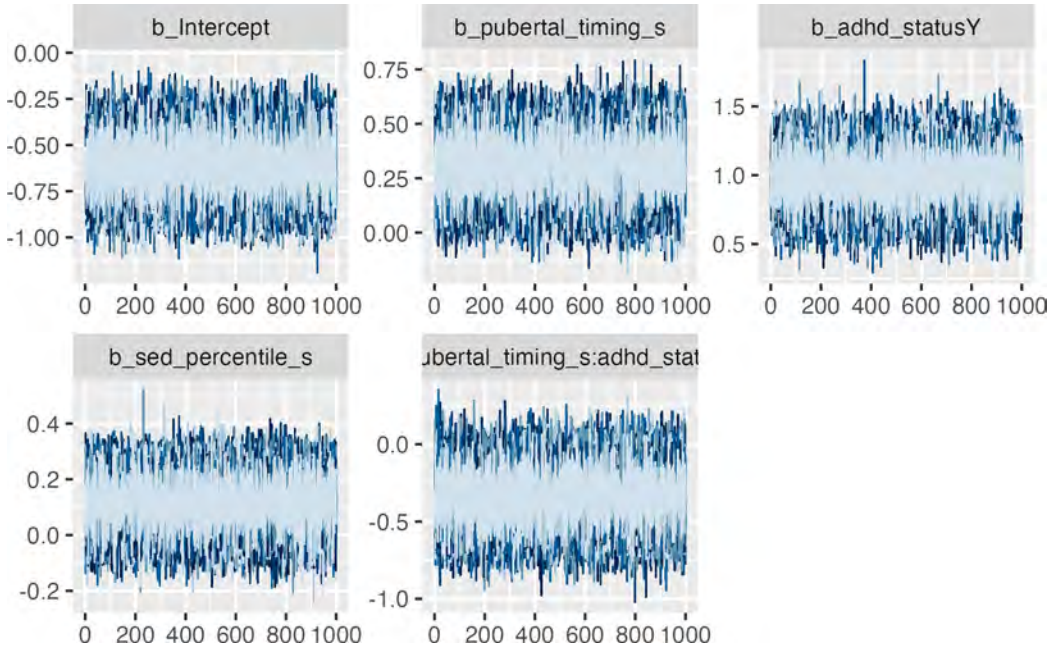
# A tibble: 9 x 4
  variable                         rhat ess_bulk ess_tail
  <chr>                           <dbl>    <dbl>    <dbl>
1 b_Intercept                      1.00  143207.   90246.
2 b_pubertal_timing_s              1.00   63300.   91441.
3 b_adhd_statusY                   1.00  146476.   89514.
4 b_sed_percentile_s               1.01   21244.   90795.
5 b_pubertal_timing_s:adhd_statusY 1.00   78741.   90279.
6 sigma                            1.00  137116.   92662.
7 Intercept                        1.00  155006.   90583.
8 lprior                           1.00  141629.   88348.
9 lp__                             1.05    1586.    1953.


=== Divergent Transitions ===
```

```
Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

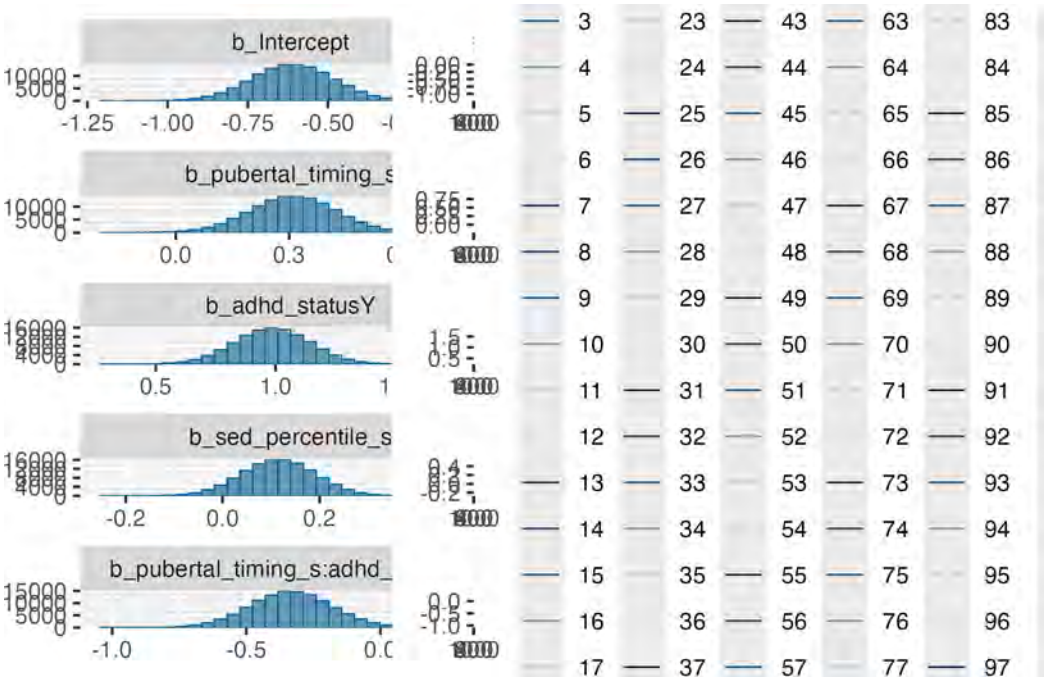
Priors specified for the model.

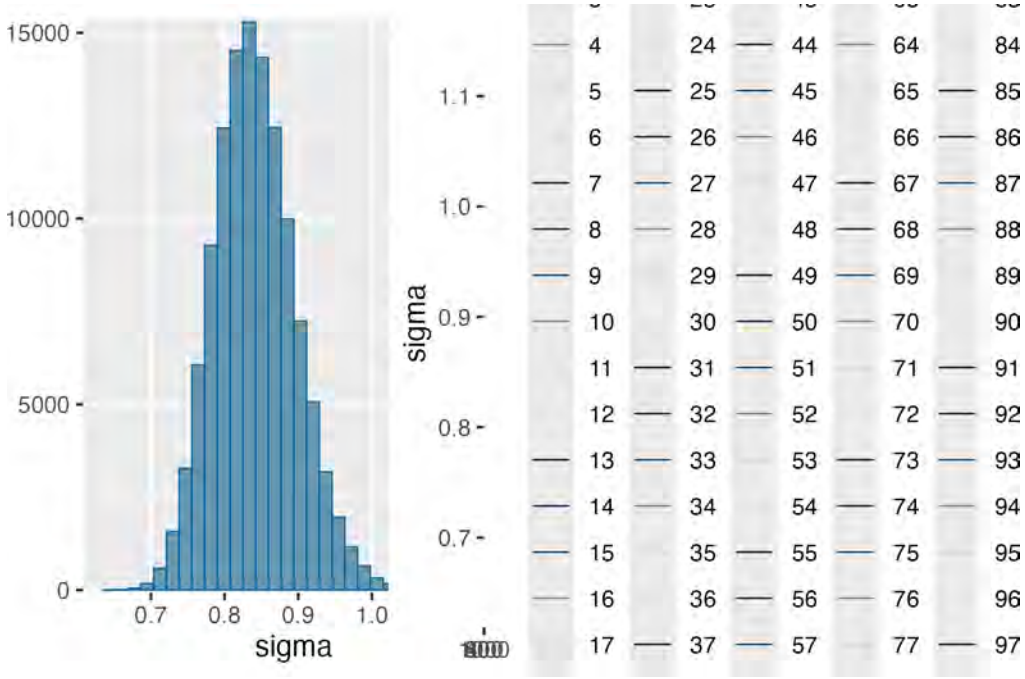
           prior     class                          coef group resp dpar nlpar
 normal(0, 0.75)         b
 normal(0, 0.75)         b                 adhd_statusY
 normal(0, 0.75)         b             pubertal_timing_s
 normal(0, 0.75)         b pubertal_timing_s:adhd_statusY
 normal(0, 0.75)         b              sed_percentile_s
    normal(0, 1) Intercept
  exponential(1)     sigma
 lb ub tag       source
                   user
```

```
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
                   user
  0                user
```

```
=== Posterior Predictive Checks ===
```

Density plots for each parameter.

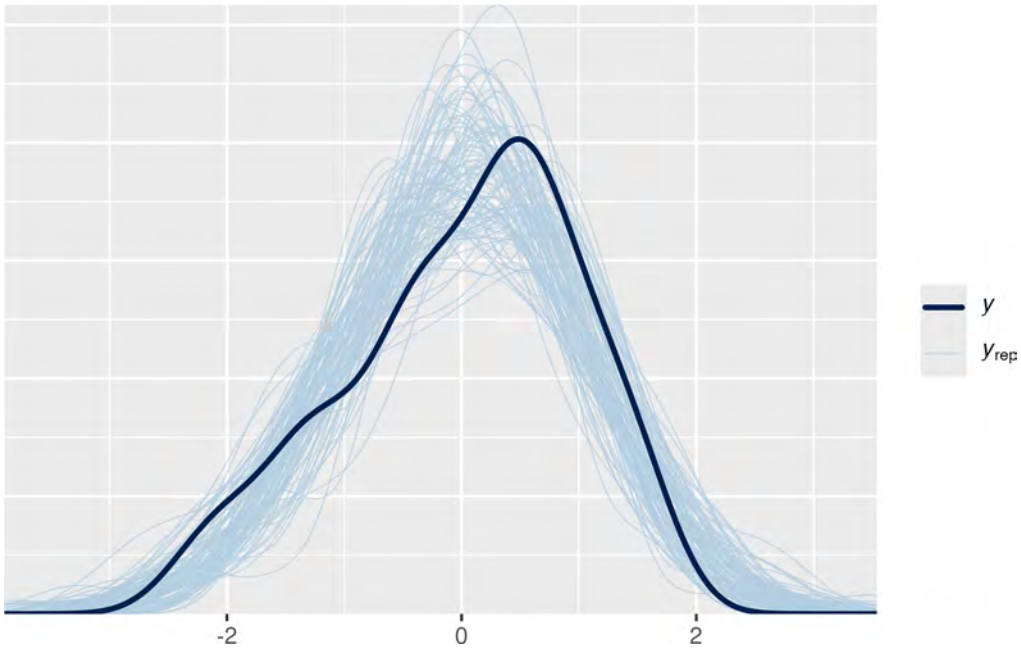





```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



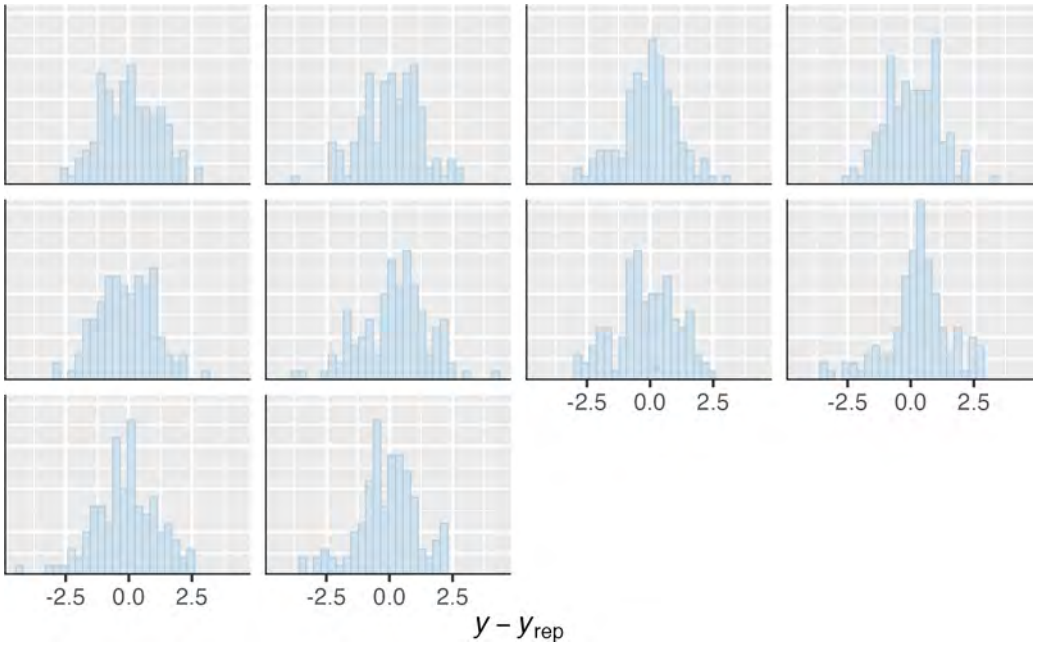

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

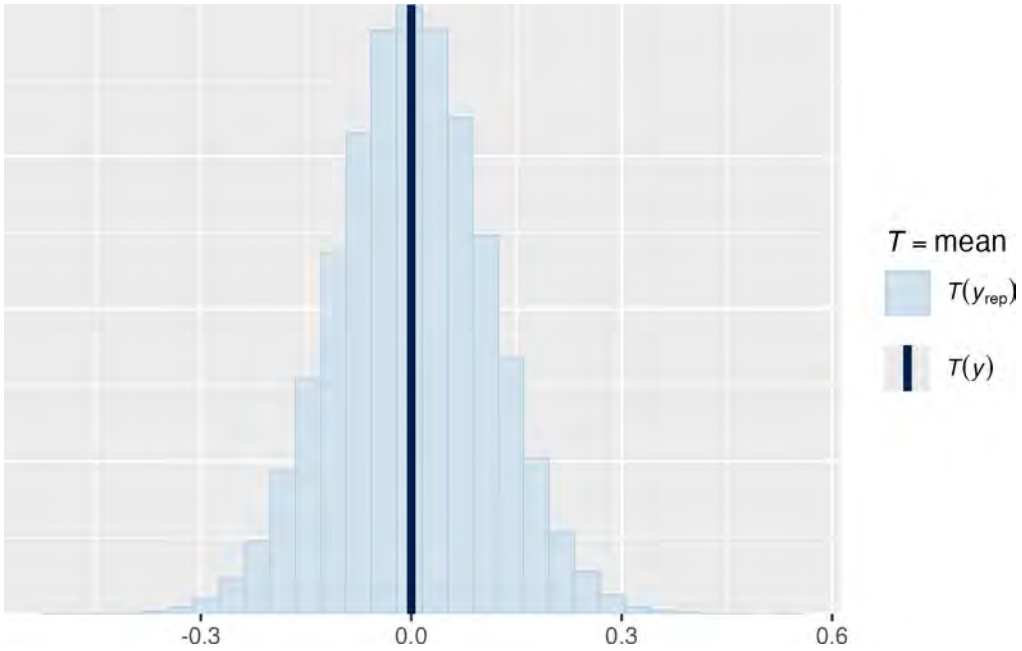
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



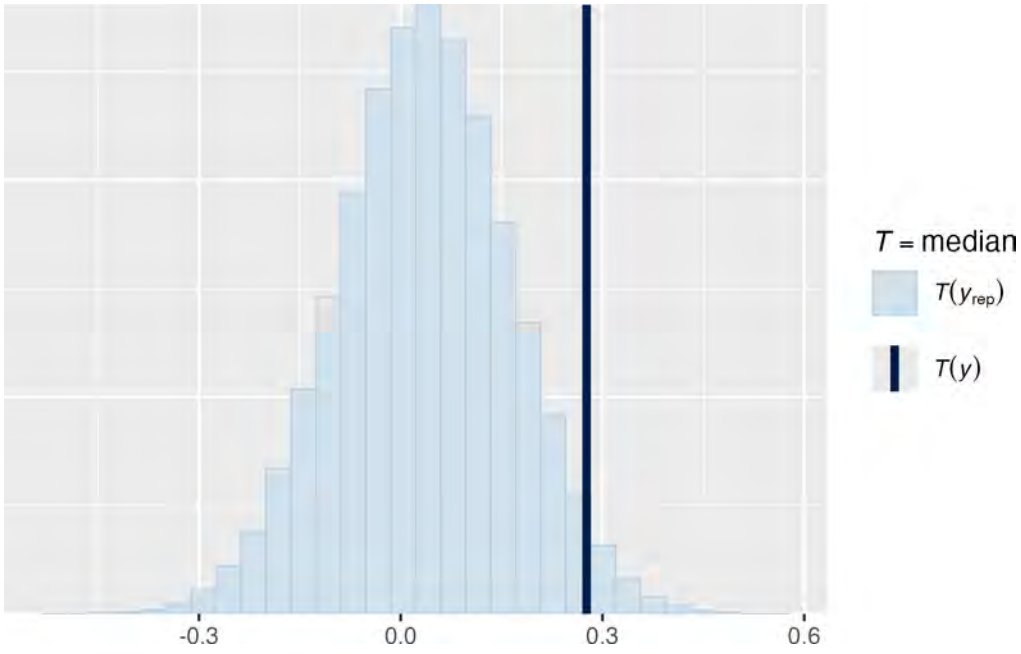

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

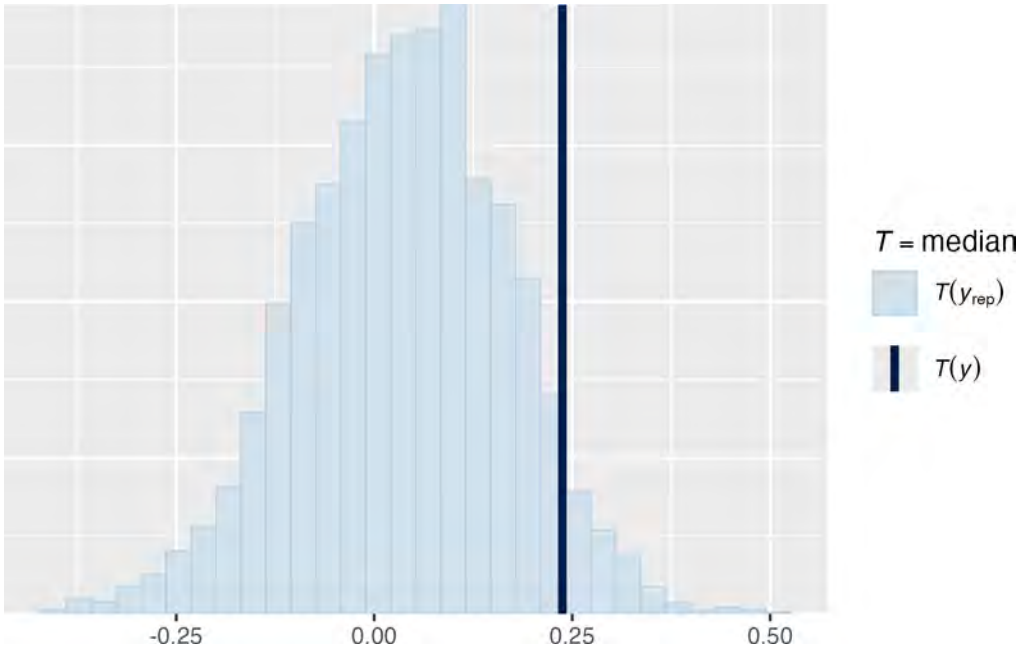
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



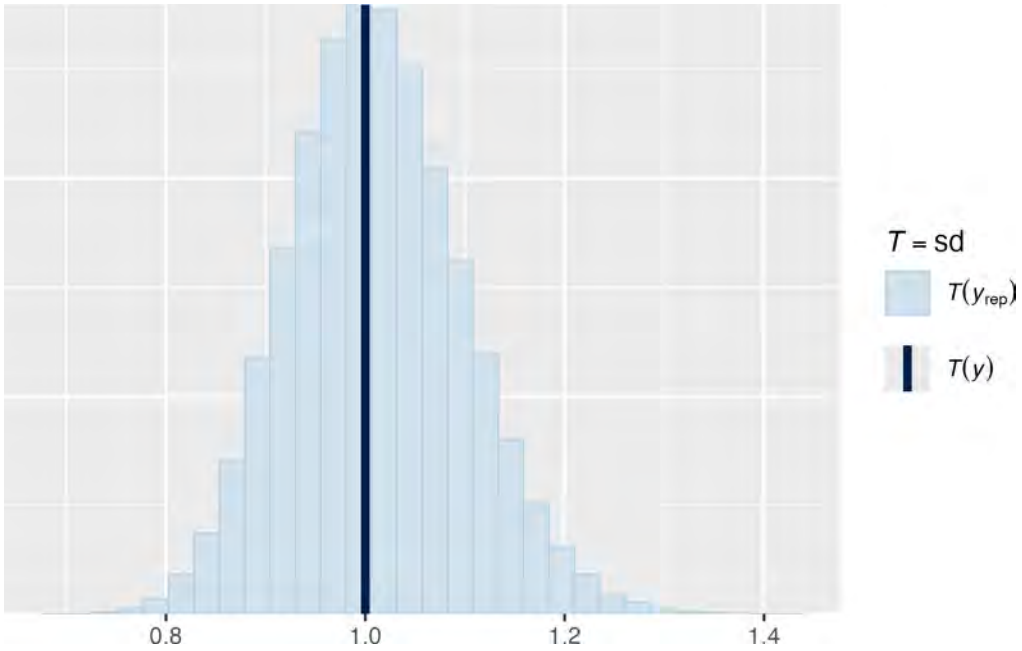

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

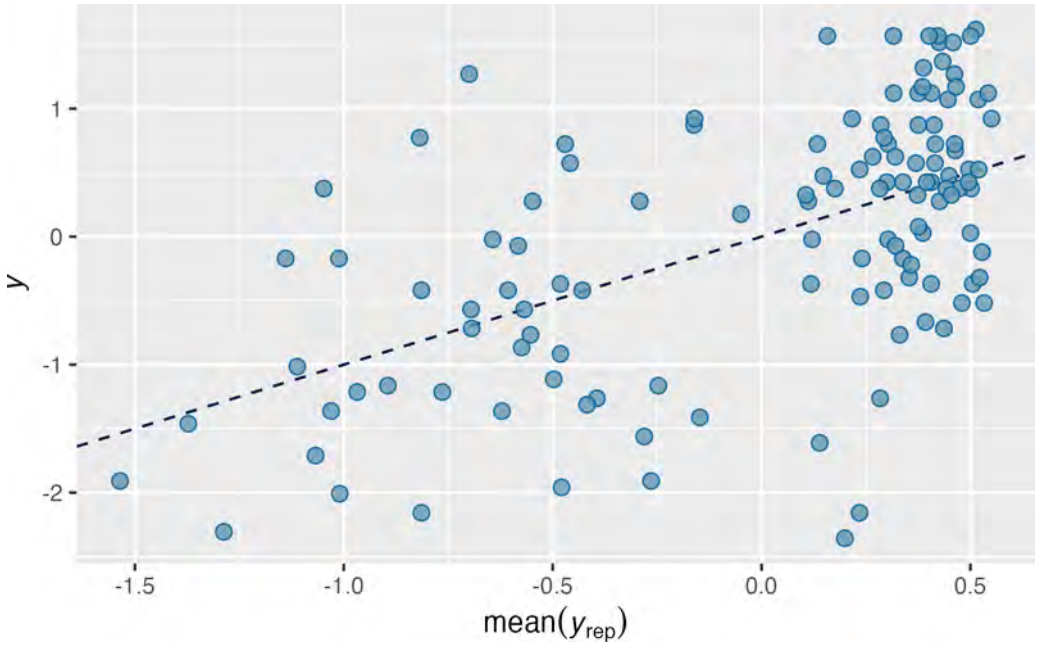



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

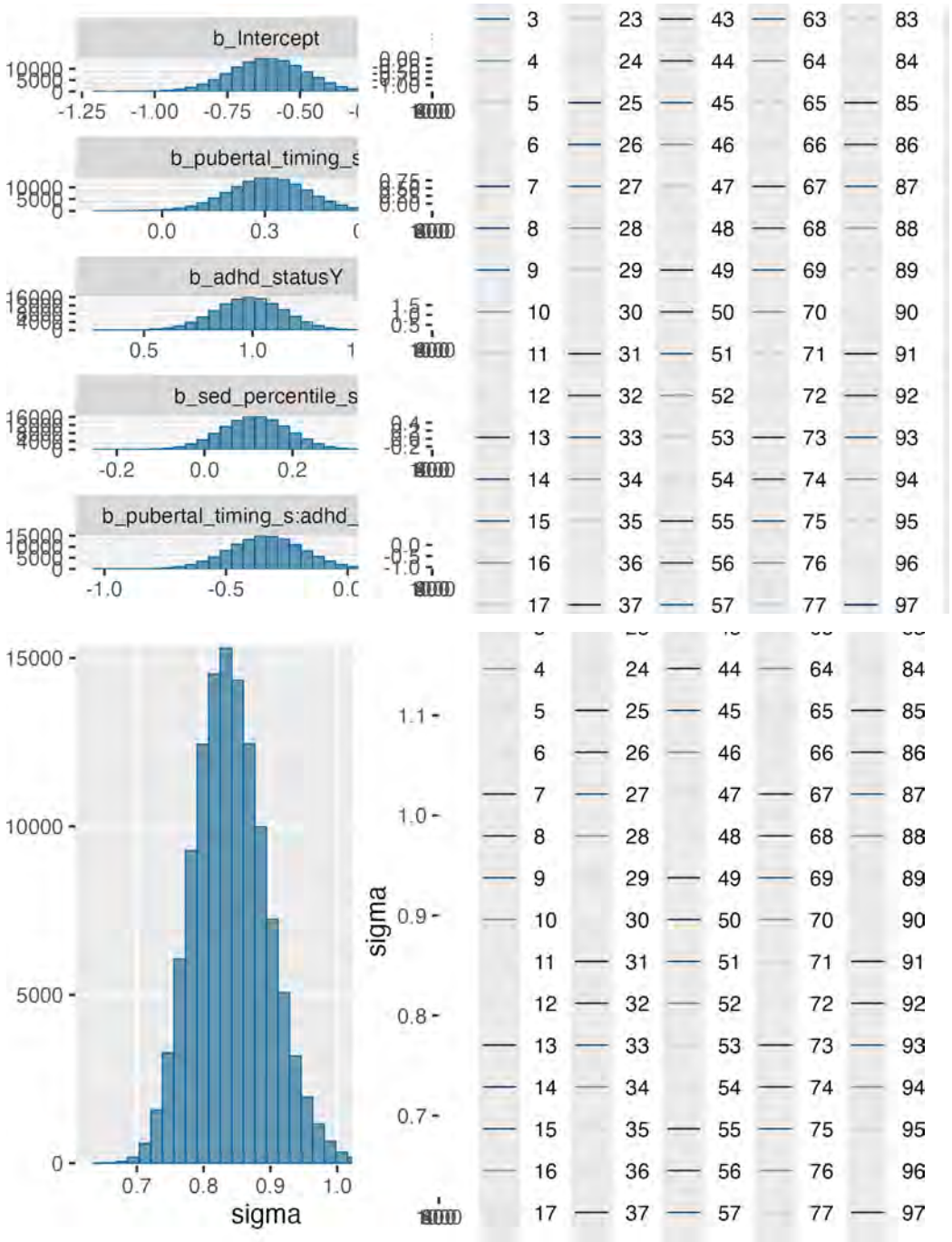
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H1 Prior 2 Model output

```
Estimate  Est.Error      Q2.5     Q97.5
```

```
R2 0.3123788 0.05531762 0.1987447 0.4135095
```

```
Summary of Posterior Distribution
```

Parameter	Mean	Median	CI	pd
b_Intercept	-0.64	-0.64	[-0.90, -0.37]	100%
b_pds_total_s	0.21	0.21	[-0.06, 0.47]	93.81%
b_adhd_statusY	1.00	1.00	[0.67, 1.33]	100%
b_age_cont_s	-3.36e-03	-3.12e-03	[-0.21, 0.21]	51.18%
b_sed_percentile_s	0.09	0.09	[-0.07, 0.24]	87.02%
b_pds_total_s:adhd_statusY	-0.06	-0.06	[-0.37, 0.26]	64.94%

Parameter	% in ROPE	meaningful
b_Intercept	5.00e-03%	TRUE
b_pds_total_s	20.08%	FALSE
b_adhd_statusY	0%	TRUE
b_age_cont_s	65.88%	FALSE
b_sed_percentile_s	54.54%	FALSE
b_pds_total_s:adhd_statusY	43.84%	FALSE

H1 Prior 2 Model diagnostics and PP checks

```
=== Rhat and ESS ===
```

```
Rhat values close to 1 indicate convergence. ESS should ideally be > 400.
```

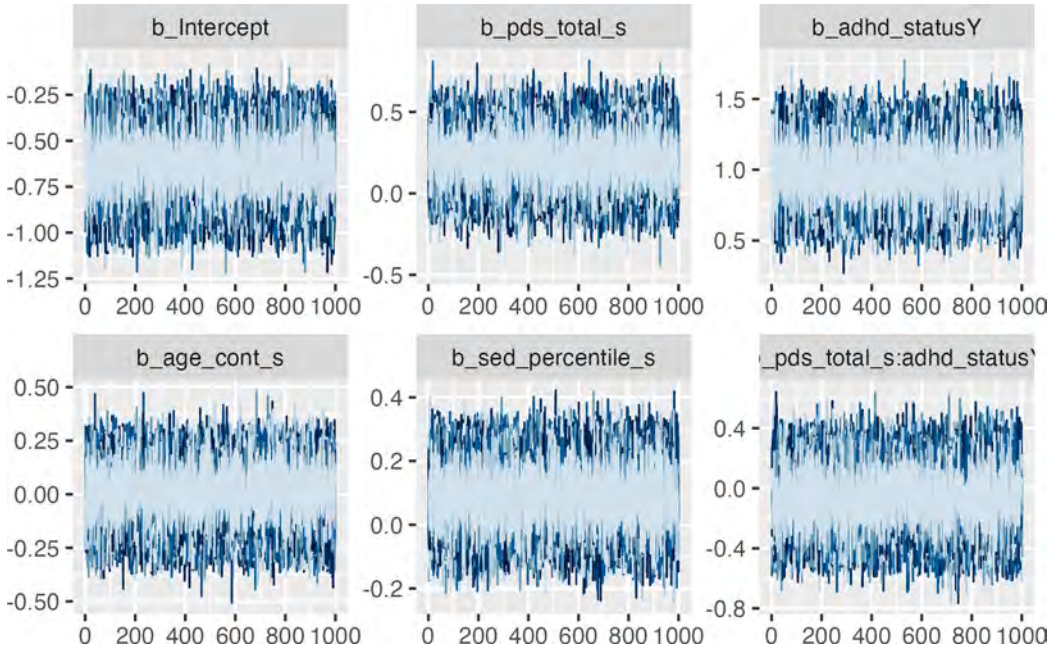
```
# A tibble: 10 x 4
   variable                    rhat ess_bulk ess_tail
   <chr>                      <dbl>    <dbl>    <dbl>
 1 b_Intercept                 1.00  126049.   92146.
 2 b_pds_total_s               1.00   82251.   83241.
 3 b_adhd_statusY              1.00  125215.   91761.
 4 b_age_cont_s                1.00   45974.   89228.
 5 b_sed_percentile_s          1.00   75599.   90697.
 6 b_pds_total_s:adhd_statusY  1.00   97668.   90413.
 7 sigma                       1.00  122192.   90274.
 8 Intercept                   1.00  140465.   90567.
 9 lprior                      1.00  117542.   90675.
10 lp__                        1.03    2256.    3201.
```

```
=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

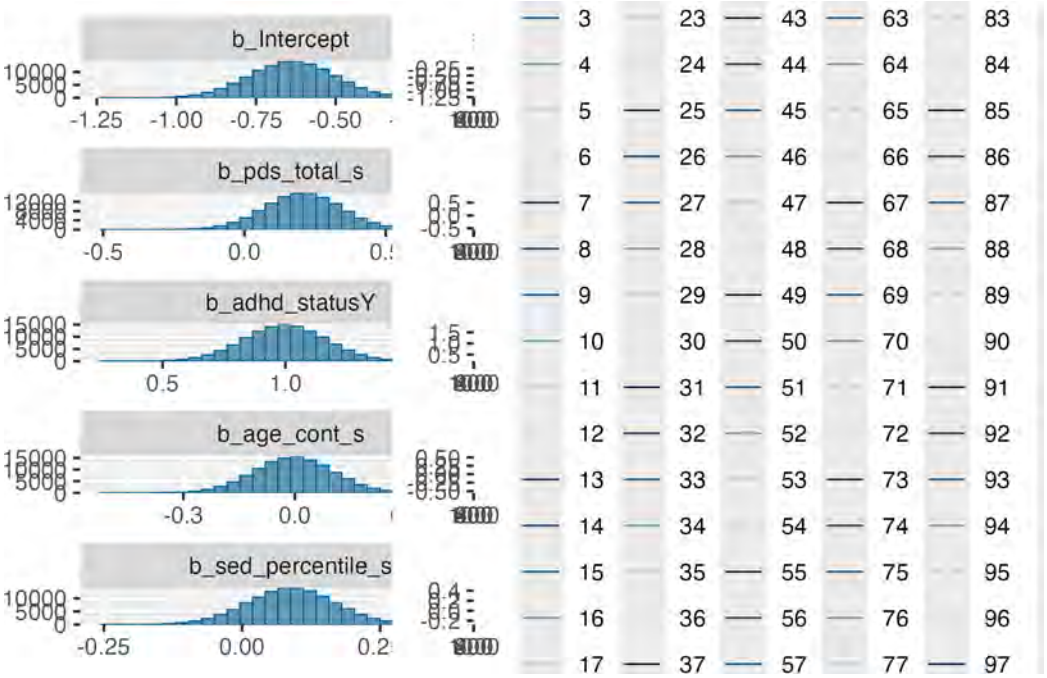
Priors specified for the model.

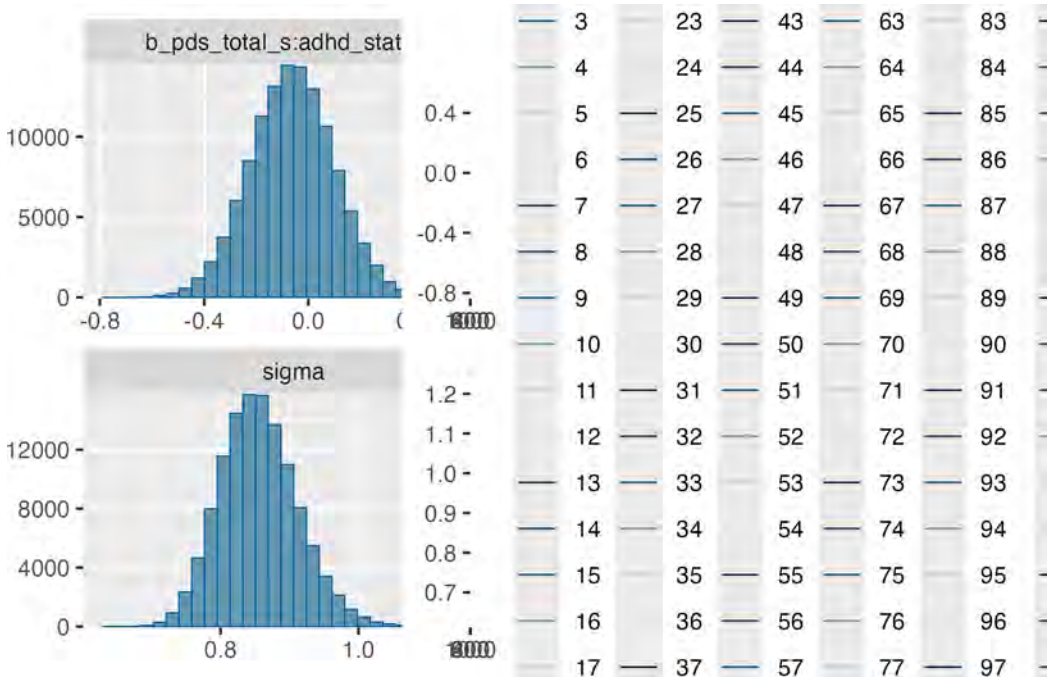
          prior     class                     coef group resp dpar nlpar lb ub
 normal(0, 2.5)         b
 normal(0, 2.5)         b             adhd_statusY
 normal(0, 2.5)         b               age_cont_s
 normal(0, 2.5)         b              pds_total_s
 normal(0, 2.5)         b pds_total_s:adhd_statusY
 normal(0, 2.5)         b         sed_percentile_s
```

```
  normal(0, 1) Intercept
exponential(1)     sigma                                                    0
tag       source
            user
    (vectorized)
    (vectorized)
    (vectorized)
    (vectorized)
    (vectorized)
            user
            user


=== Posterior Predictive Checks ===

Density plots for each parameter.
```

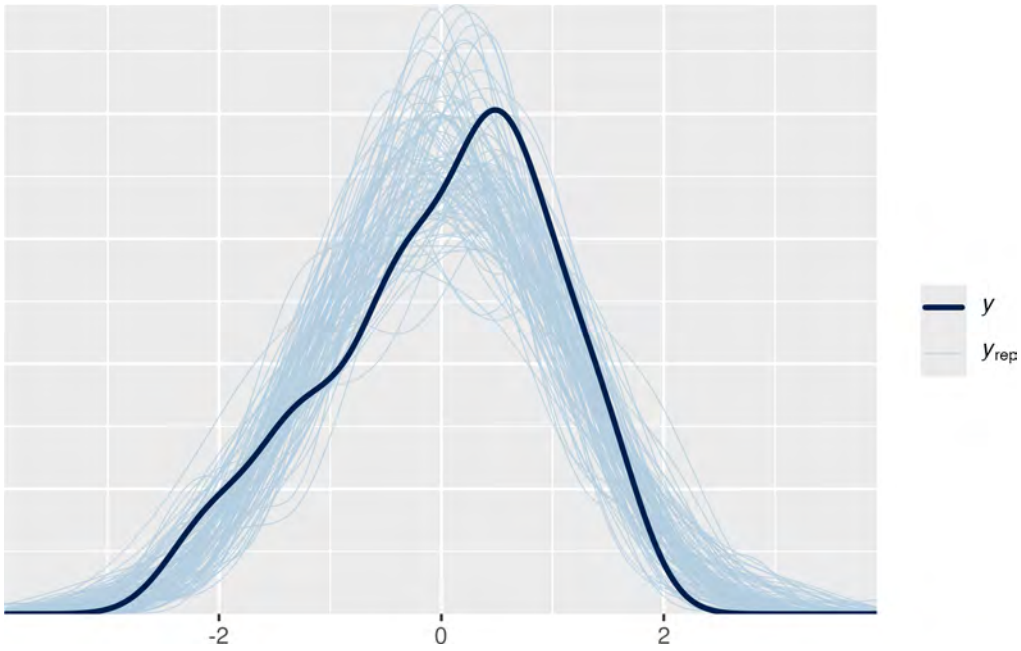





```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



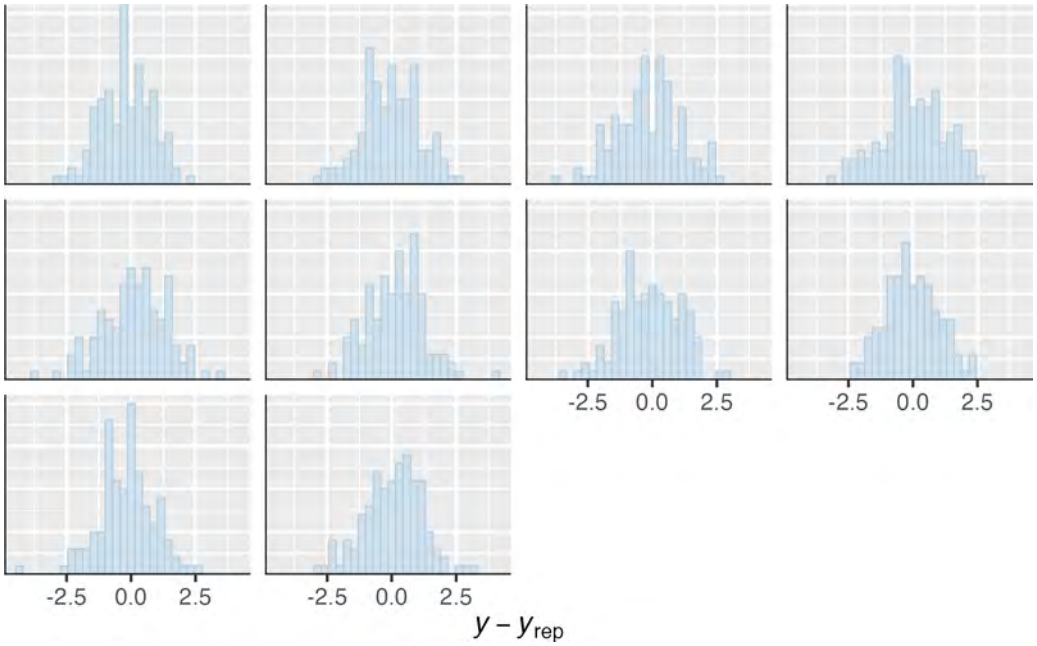

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

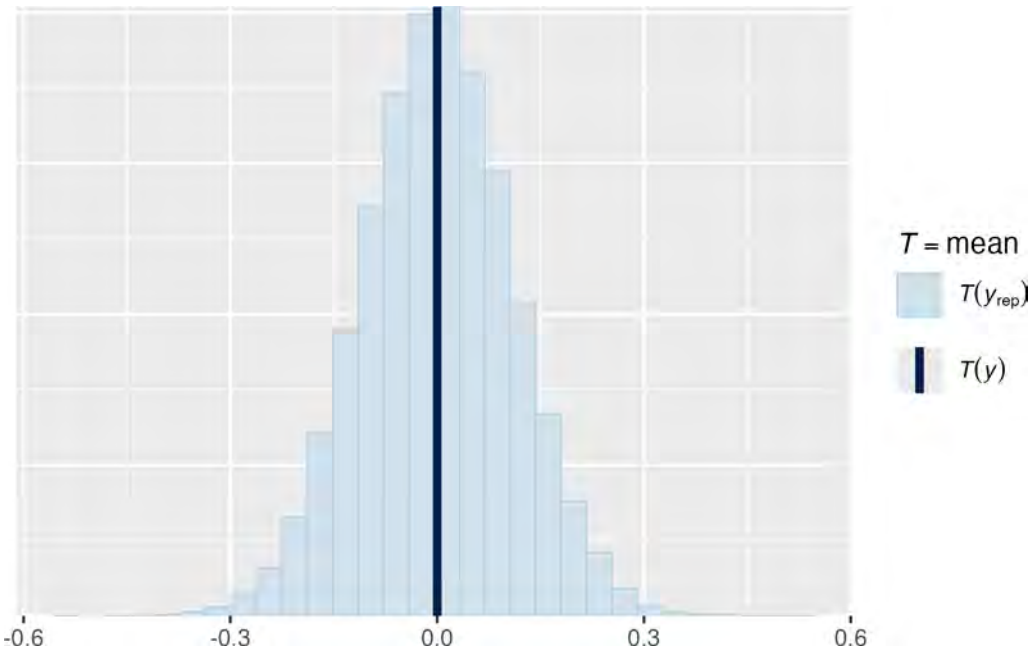
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



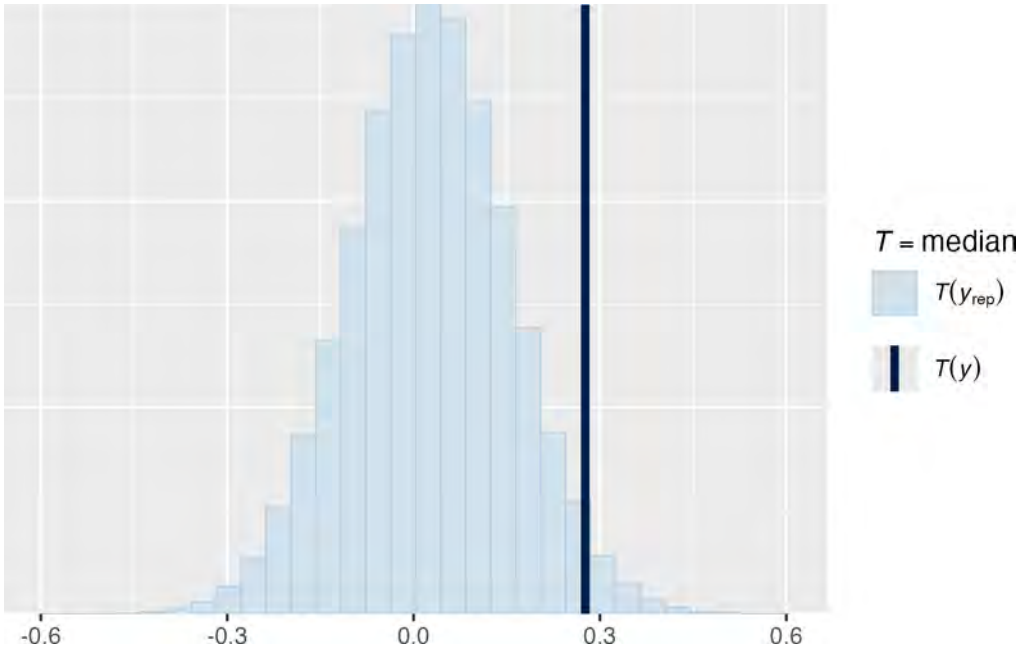

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



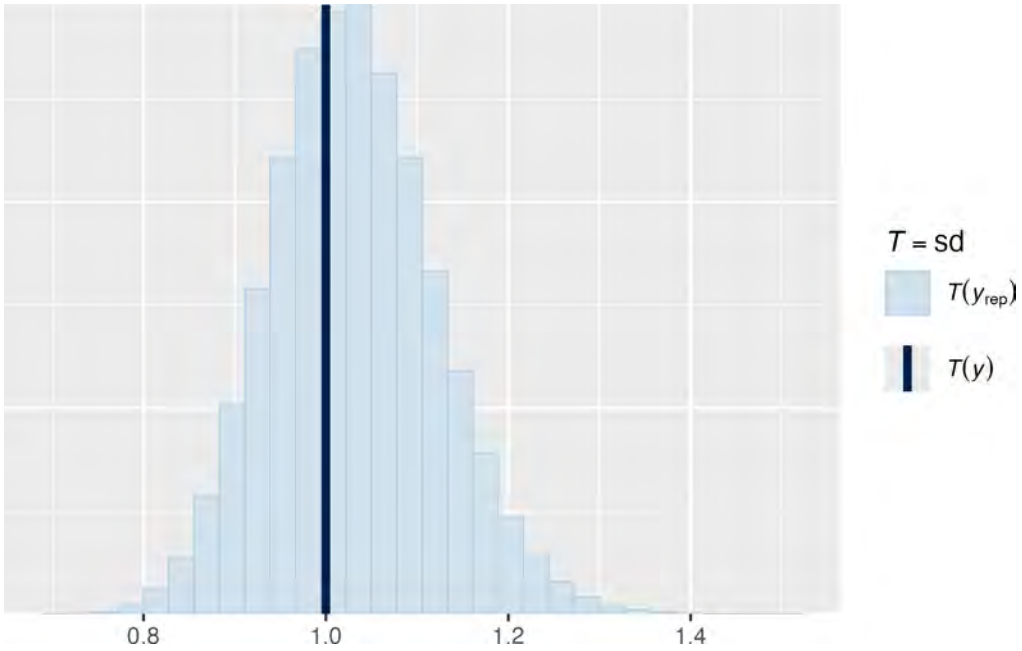

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

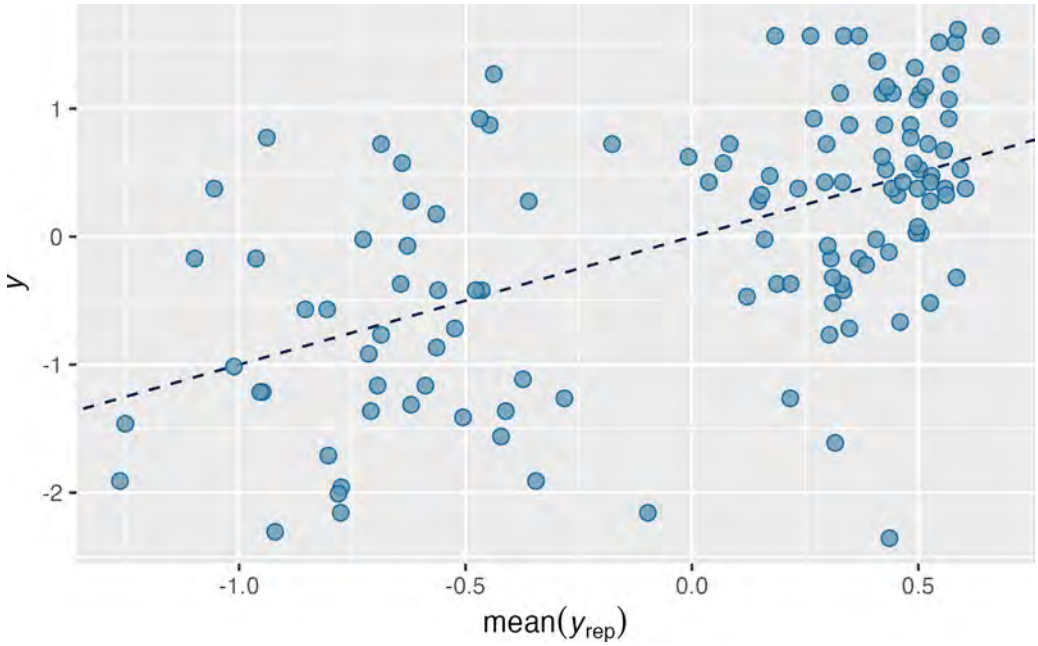



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

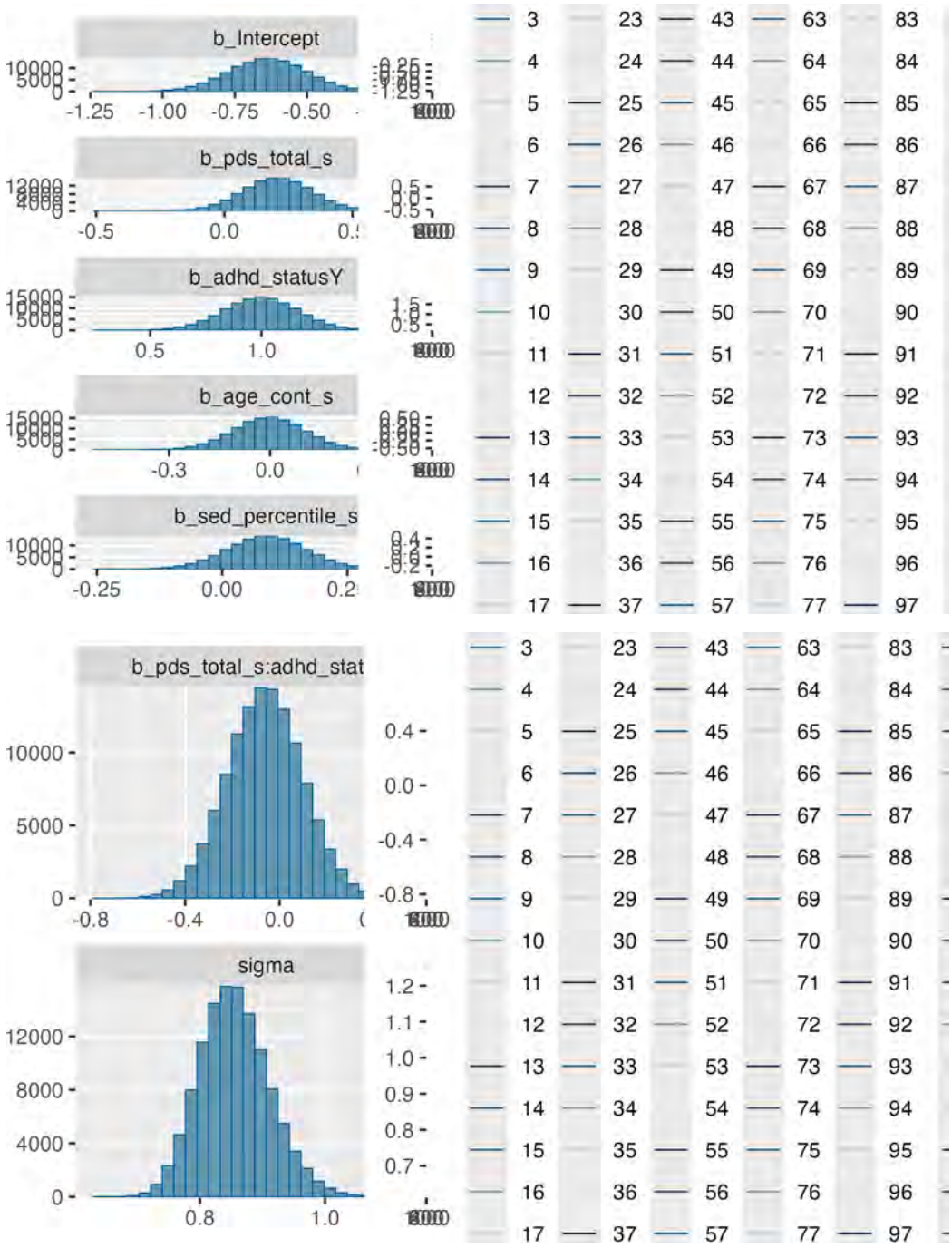
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Prior 2 Model output

```
Estimate  Est.Error      Q2.5     Q97.5
```

```
R2 0.3244019 0.05559742 0.209052 0.4252536
```

```
Summary of Posterior Distribution

Parameter                         |  Mean | Median |             CI |     pd
-------------------------------------------------------------------------------
b_Intercept                       | -0.63 |  -0.63 | [-0.89, -0.38] |   100%
b_pubertal_timing_s               |  0.32 |   0.32 | [ 0.09,  0.54] | 99.71%
b_adhd_statusY                    |  1.02 |   1.03 | [ 0.71,  1.34] |   100%
b_sed_percentile_s                |  0.12 |   0.12 | [-0.03,  0.27] | 93.82%
b_pubertal_timing_s:adhd_statusY  | -0.35 |  -0.35 | [-0.66, -0.04] | 98.75%

Parameter                         | % in ROPE | meaningful
-------------------------------------------------------------
b_Intercept                       | 8.33e-04% |       TRUE
b_pubertal_timing_s               |     2.91% |       TRUE
b_adhd_statusY                    |        0% |       TRUE
b_sed_percentile_s                |    40.28% |      FALSE
b_pubertal_timing_s:adhd_statusY  |     5.10% |      FALSE
```

H2 Prior 2 Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

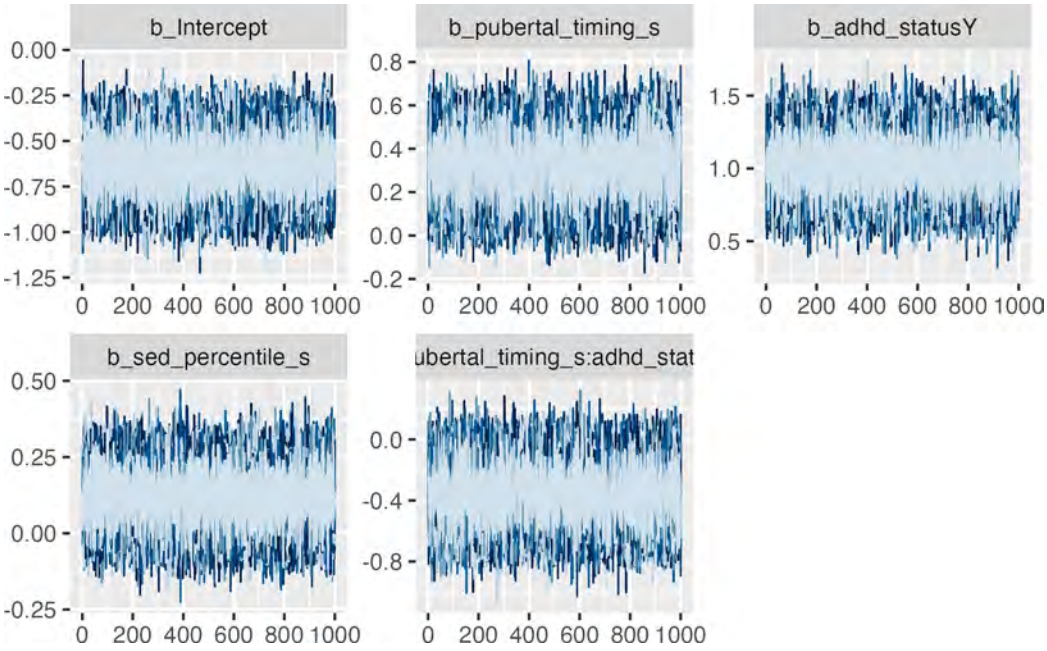
# A tibble: 9 x 4
  variable                         rhat ess_bulk ess_tail
  <chr>                           <dbl>    <dbl>    <dbl>
1 b_Intercept                      1.00  145384.   92360.
2 b_pubertal_timing_s              1.00   55104.   88042.
3 b_adhd_statusY                   1.00  149389.   91833.
4 b_sed_percentile_s               1.01   18669.   88550.
5 b_pubertal_timing_s:adhd_statusY 1.00   45659.   90519.
6 sigma                            1.00  138307.   88954.
7 Intercept                        1.00  154726.   90114.
8 lprior                           1.00  136197.   90509.
9 lp__                             1.05    1396.    1859.


=== Divergent Transitions ===
```

```
Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

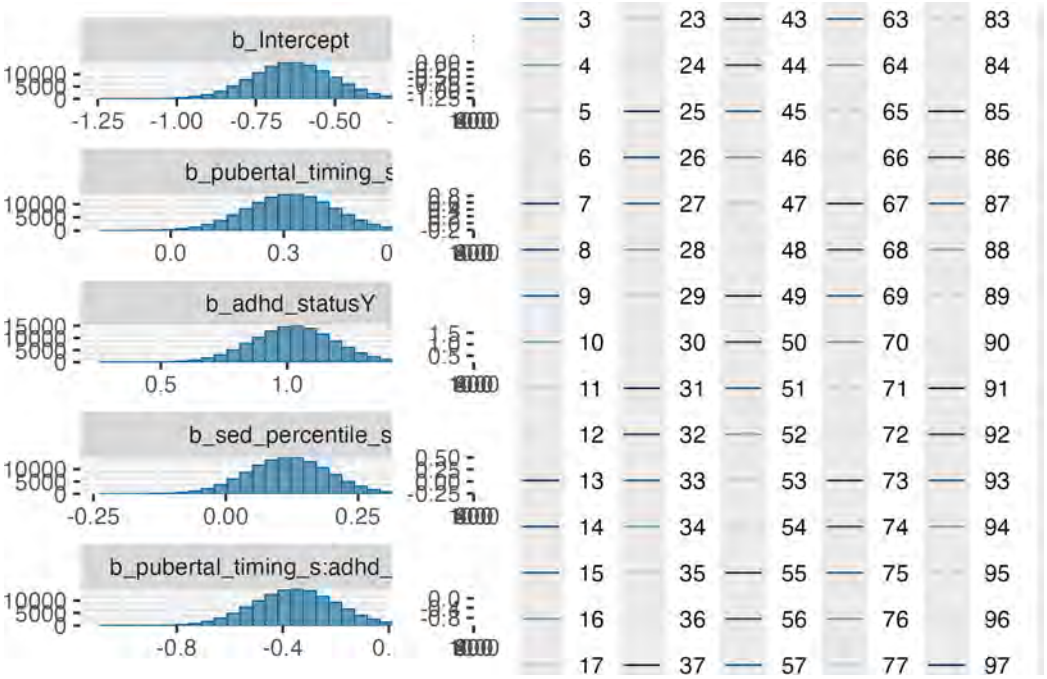
Priors specified for the model.

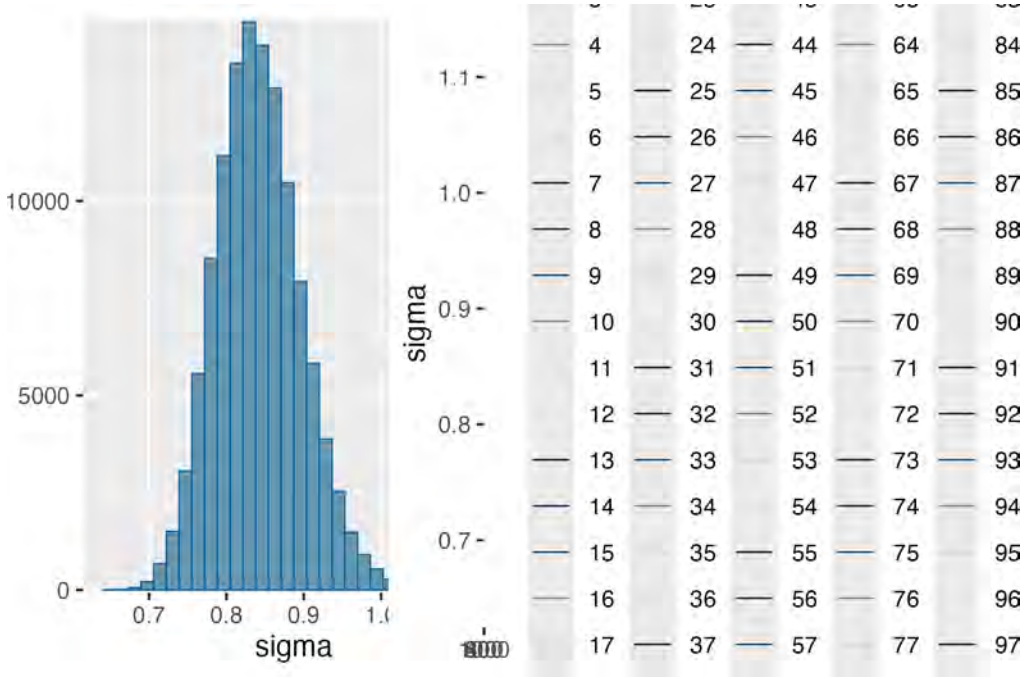
          prior     class                          coef group resp dpar nlpar
 normal(0, 2.5)         b
 normal(0, 2.5)         b                 adhd_statusY
 normal(0, 2.5)         b              pubertal_timing_s
 normal(0, 2.5)         b pubertal_timing_s:adhd_statusY
 normal(0, 2.5)         b               sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma
 lb ub tag       source
                   user
```

```
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
                   user
  0                user
```

```
=== Posterior Predictive Checks ===
```

Density plots for each parameter.

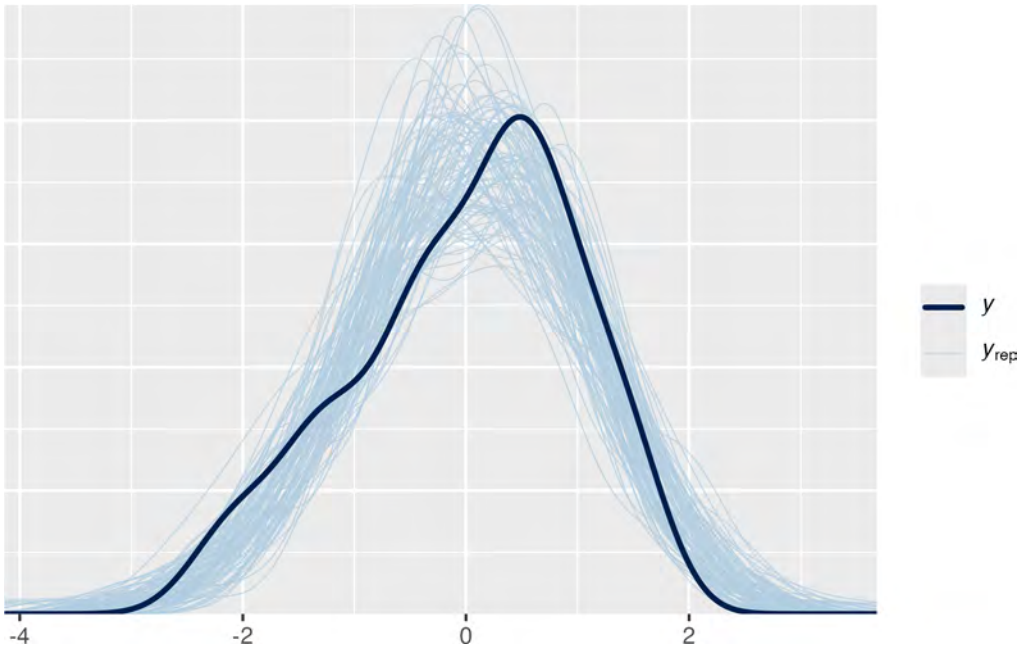





```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



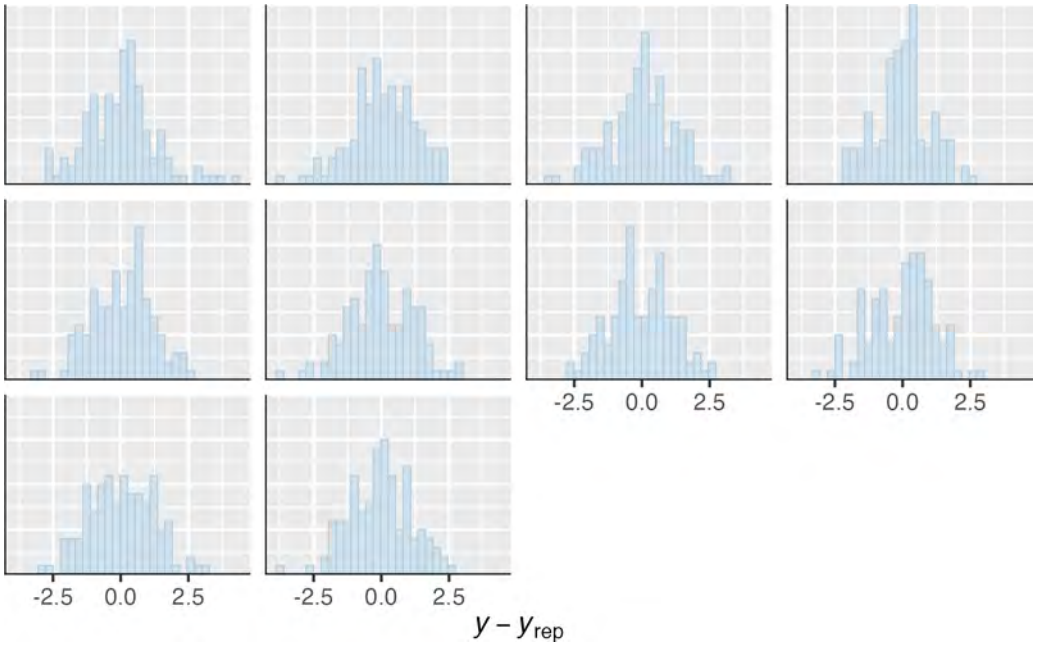

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

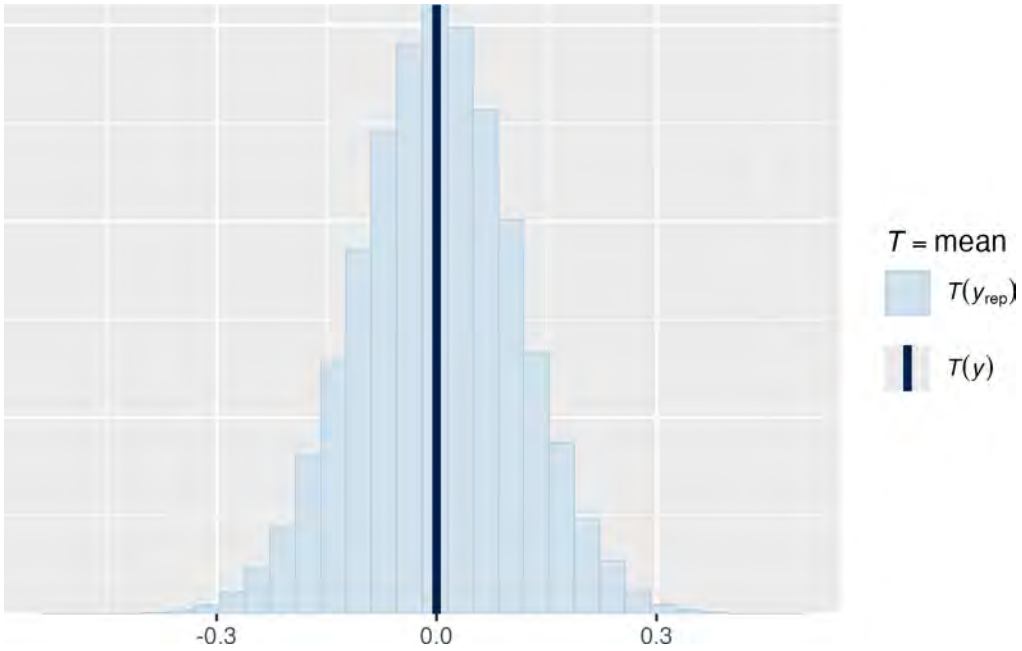
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



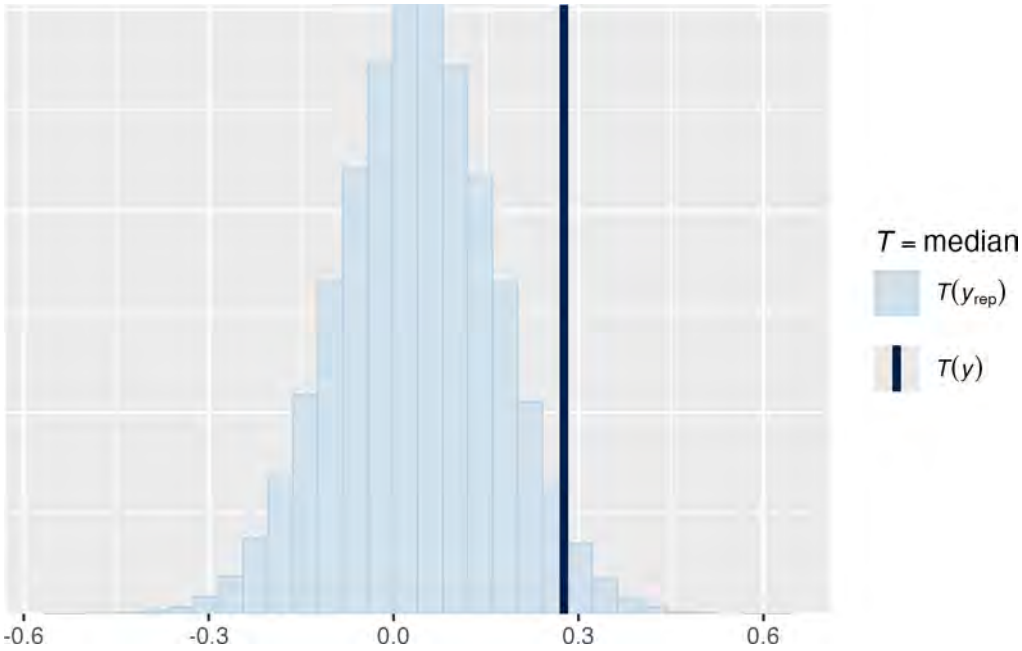

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



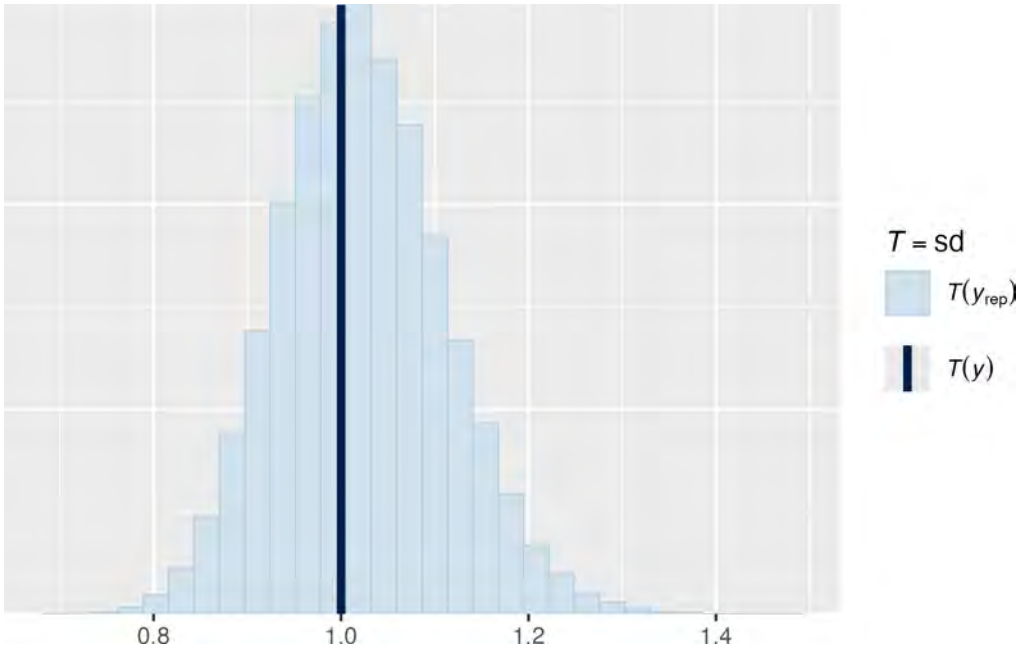

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

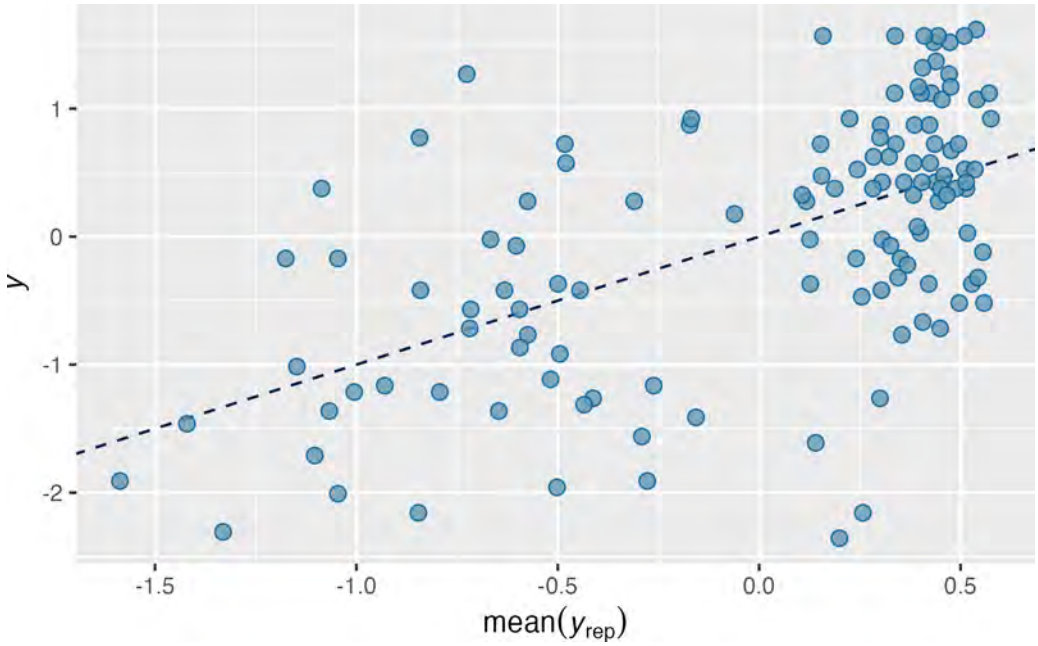



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

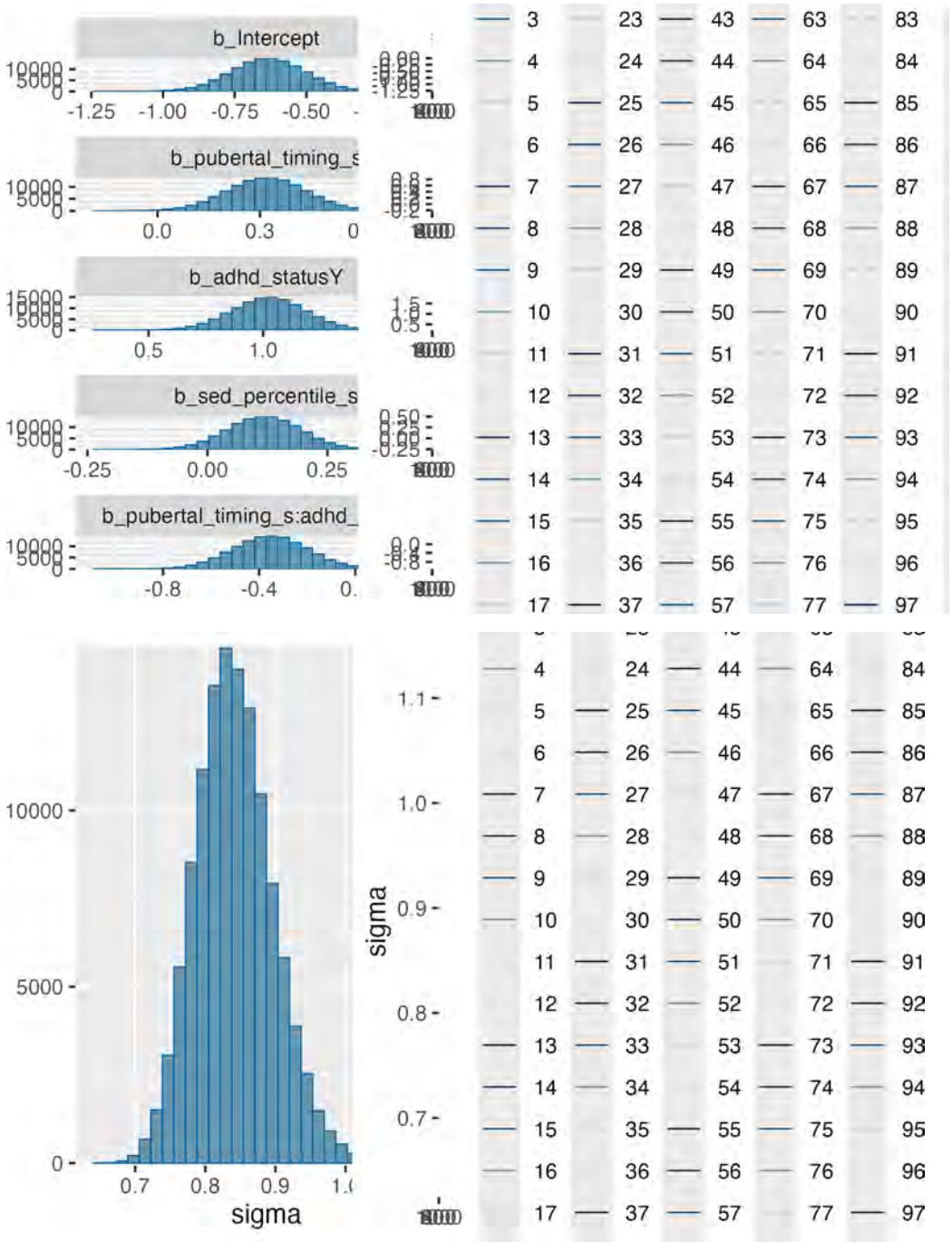
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H1 Prior 3 Model output

```
Estimate  Est.Error      Q2.5     Q97.5
```

```
R2 0.2832838 0.05533899 0.1708941 0.3865316
```

```
Summary of Posterior Distribution
```

Parameter	Mean	Median	CI	pd
b_Intercept	-0.58	-0.58	[-0.83, -0.32]	100.00%
b_pds_total_s	0.20	0.20	[-0.05, 0.46]	94.61%
b_adhd_statusY	0.91	0.91	[0.59, 1.21]	100%
b_age_cont_s	3.37e-03	3.29e-03	[-0.20, 0.20]	51.29%
b_sed_percentile_s	0.08	0.08	[-0.07, 0.24]	85.85%
b_pds_total_s:adhd_statusY	-0.05	-0.05	[-0.35, 0.25]	63.98%

Parameter	% in ROPE	meaningful
b_Intercept	0.02%	TRUE
b_pds_total_s	19.76%	FALSE
b_adhd_statusY	0%	TRUE
b_age_cont_s	67.38%	FALSE
b_sed_percentile_s	57.96%	FALSE
b_pds_total_s:adhd_statusY	46.83%	FALSE

H1 Prior 3 Model diagnostics and PP checks

```
=== Rhat and ESS ===
```

```
Rhat values close to 1 indicate convergence. ESS should ideally be > 400.
```

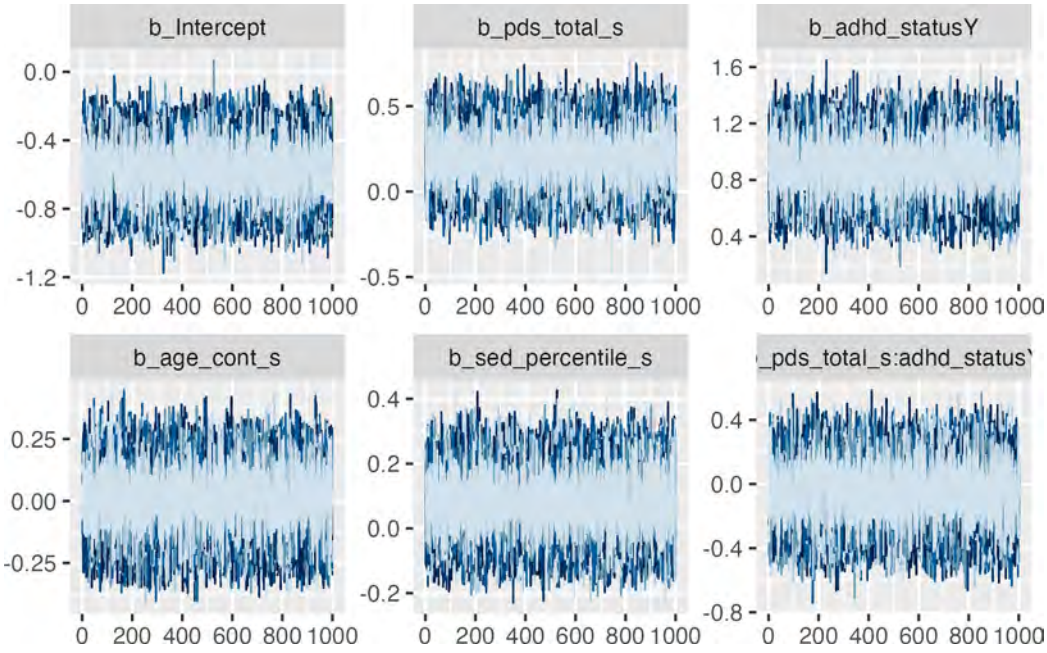
```
# A tibble: 10 x 4
   variable                    rhat ess_bulk ess_tail
   <chr>                      <dbl>    <dbl>    <dbl>
 1 b_Intercept                 1.00  127396.   92285.
 2 b_pds_total_s               1.00   82067.   87330.
 3 b_adhd_statusY              1.00  130659.   91408.
 4 b_age_cont_s                1.00   60911.   91881.
 5 b_sed_percentile_s          1.00   38084.   92960.
 6 b_pds_total_s:adhd_statusY  1.00  101009.   92193.
 7 sigma                       1.00  131579.   92916.
 8 Intercept                   1.00  141714.   89713.
 9 lprior                      1.00  136723.   91653.
10 lp__                        1.03    2927.    3880.
```

```
=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

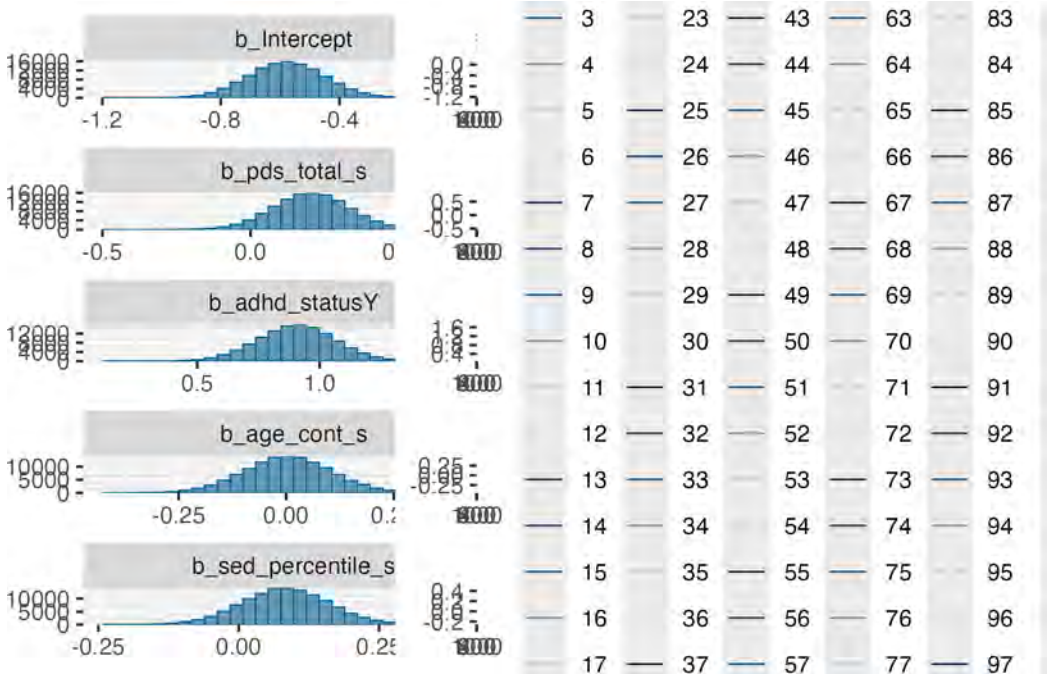
Priors specified for the model.

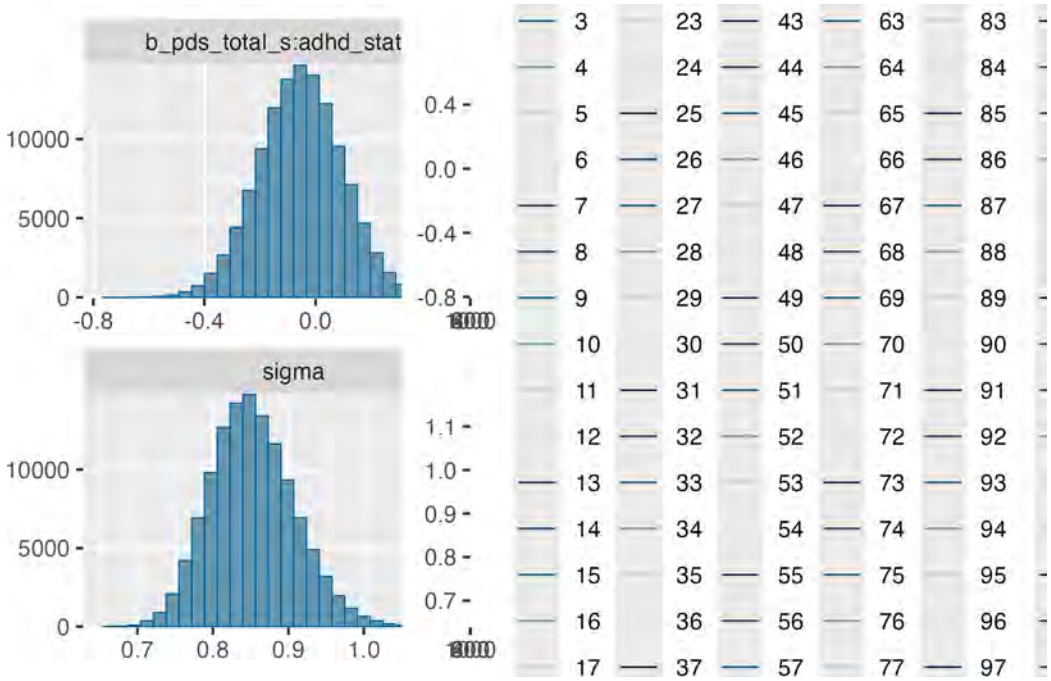
          prior     class                     coef group resp dpar nlpar lb ub
 normal(0, 0.5)         b
 normal(0, 0.5)         b             adhd_statusY
 normal(0, 0.5)         b               age_cont_s
 normal(0, 0.5)         b              pds_total_s
 normal(0, 0.5)         b pds_total_s:adhd_statusY
 normal(0, 0.5)         b         sed_percentile_s
```

```
 normal(0, 0.5) Intercept
 exponential(2)     sigma                                                    0
 tag       source
             user
     (vectorized)
     (vectorized)
     (vectorized)
     (vectorized)
     (vectorized)
             user
             user
```

```
=== Posterior Predictive Checks ===
```

Density plots for each parameter.

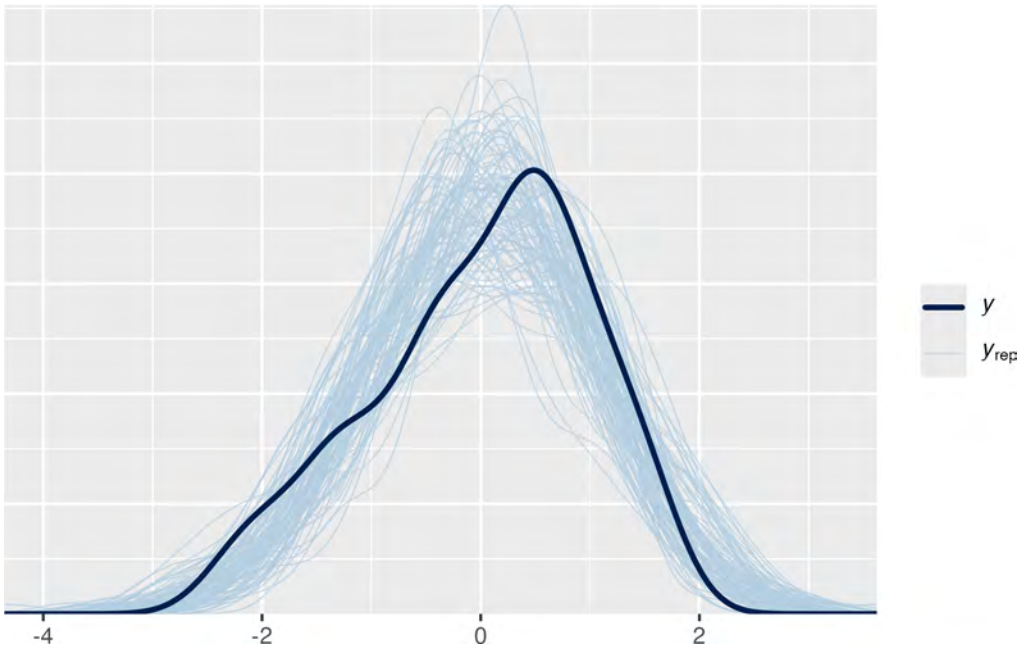





```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



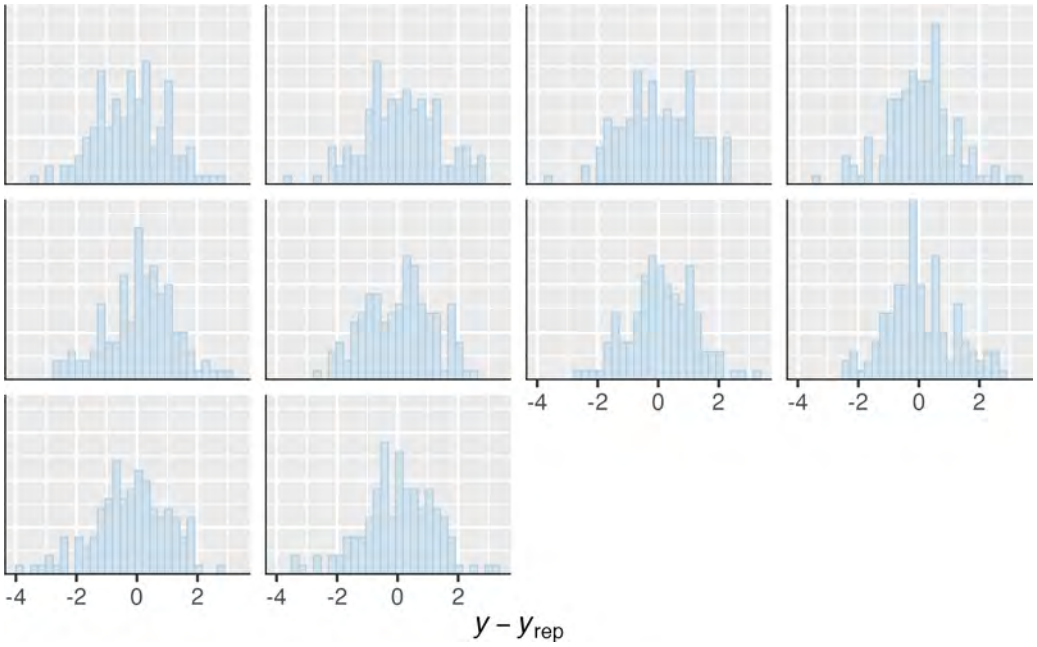

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

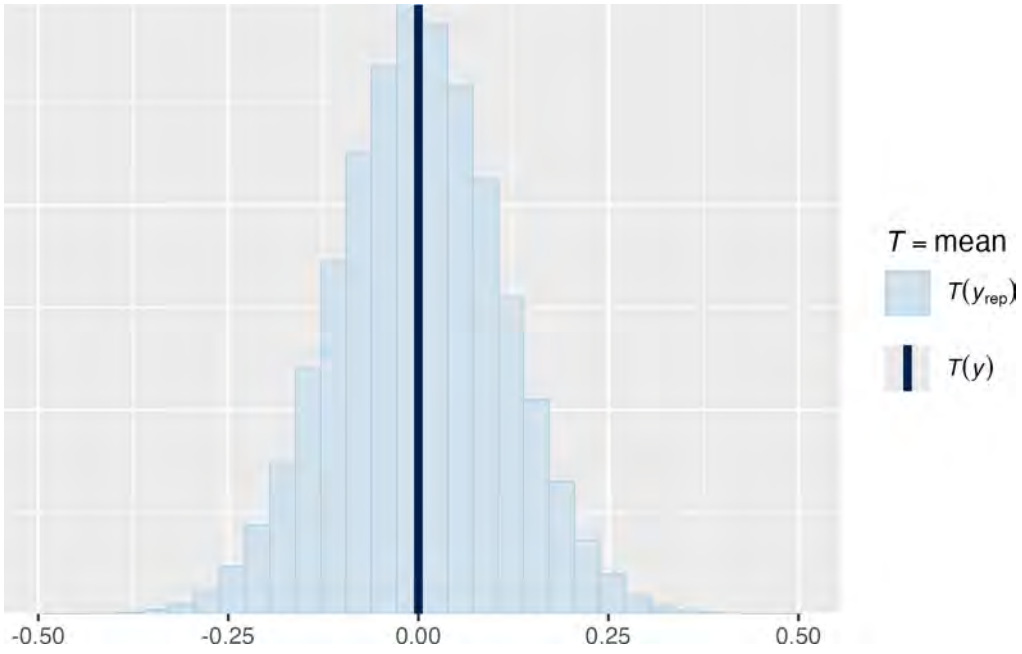
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



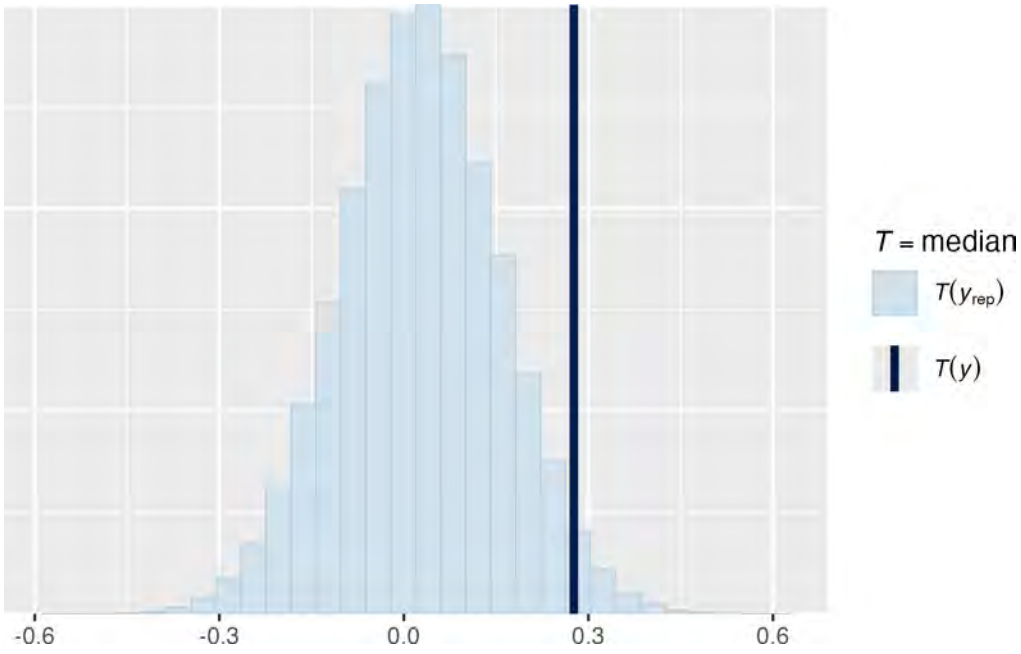

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



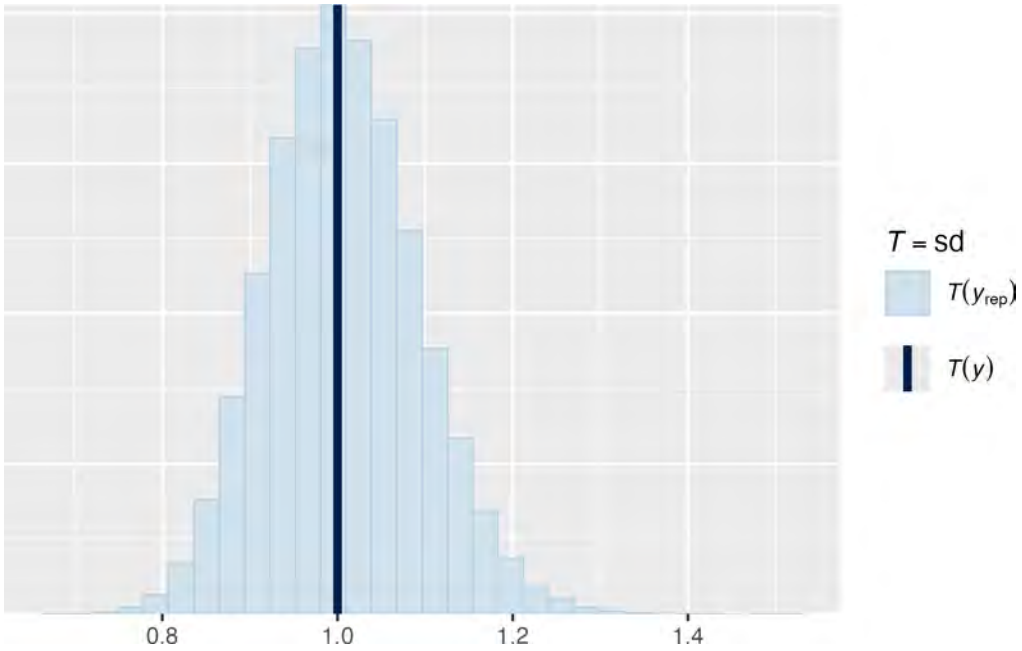

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

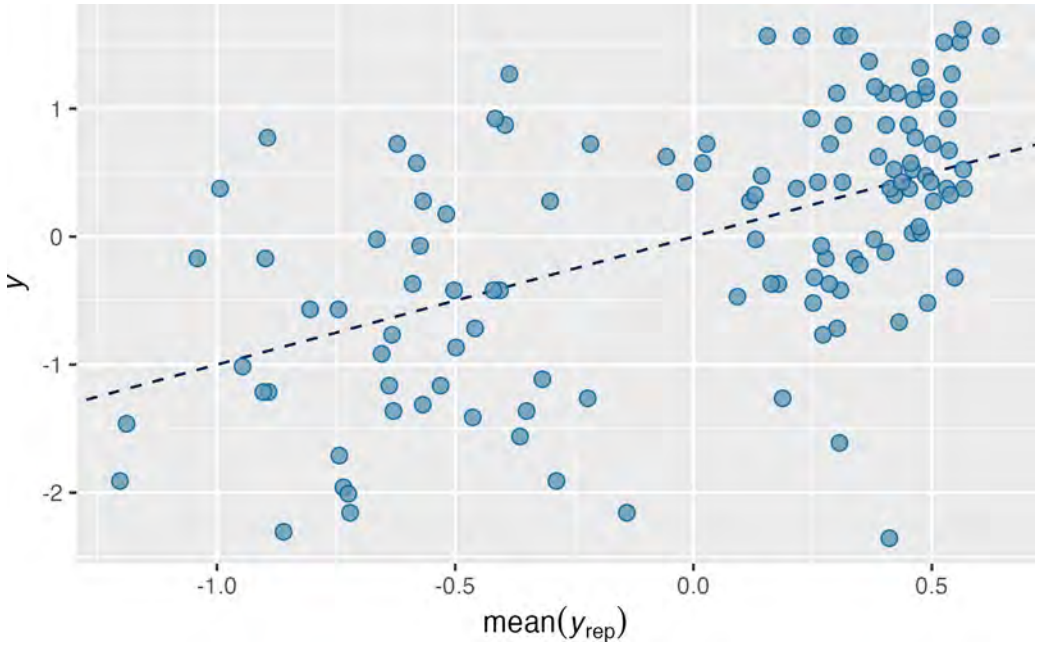



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

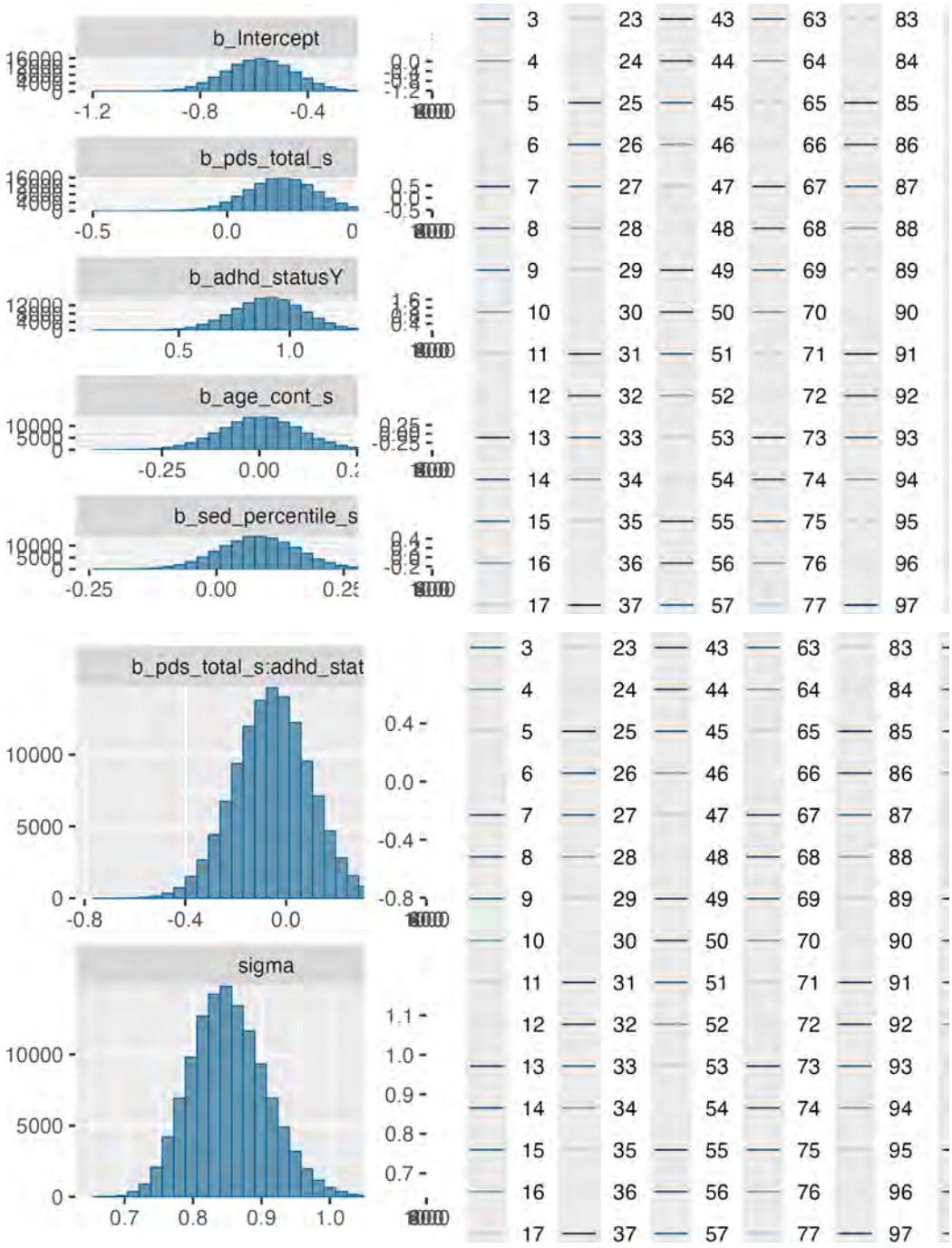
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Prior 3 Model output

```
Estimate  Est.Error      Q2.5      Q97.5
```

```
R2 0.2905126 0.05560292 0.177015 0.3940575

Summary of Posterior Distribution

Parameter                         |  Mean | Median |             CI |      pd
-------------------------------------------------------------------------------
b_Intercept                       | -0.58 |  -0.58 | [-0.82, -0.33] | 100.00%
b_pubertal_timing_s               |  0.30 |   0.30 | [ 0.08,  0.51] |  99.62%
b_adhd_statusY                    |  0.94 |   0.94 | [ 0.64,  1.23] |    100%
b_sed_percentile_s                |  0.11 |   0.11 | [-0.04,  0.26] |  92.67%
b_pubertal_timing_s:adhd_statusY  | -0.31 |  -0.31 | [-0.60, -0.02] |  98.26%

Parameter                         | % in ROPE | meaningful
-------------------------------------------------------------
b_Intercept                       |     0.01% |       TRUE
b_pubertal_timing_s               |     3.62% |       TRUE
b_adhd_statusY                    |        0% |       TRUE
b_sed_percentile_s                |    44.15% |      FALSE
b_pubertal_timing_s:adhd_statusY  |     7.14% |      FALSE
```

H2 Prior 3 Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

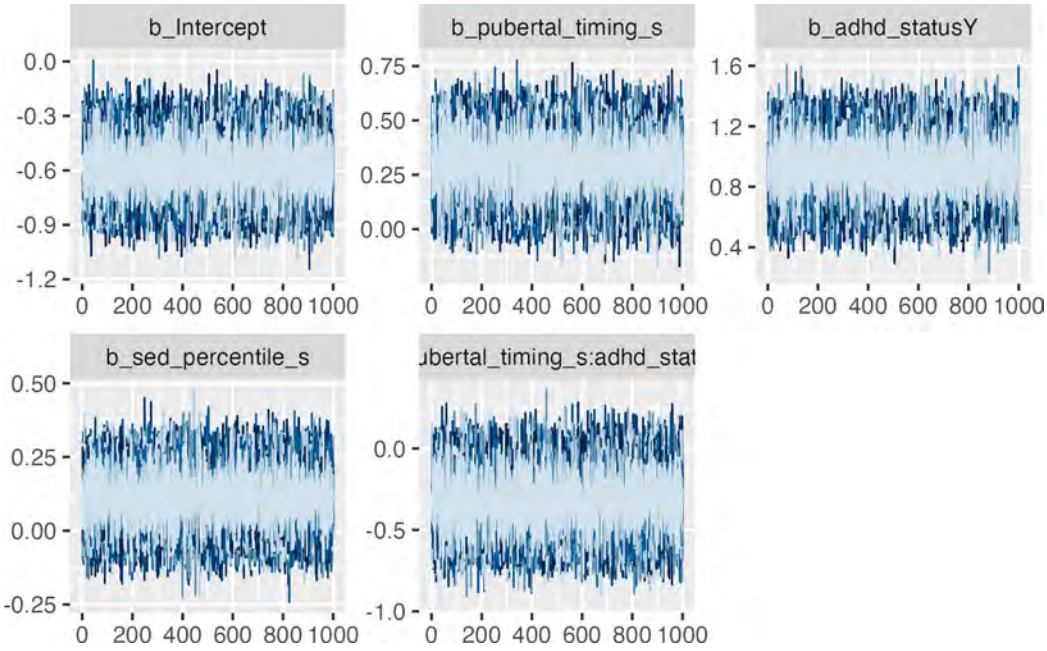
# A tibble: 9 x 4
  variable                         rhat ess_bulk ess_tail
  <chr>                           <dbl>    <dbl>    <dbl>
1 b_Intercept                      1.00  150497.   91311.
2 b_pubertal_timing_s              1.00   85682.   90560.
3 b_adhd_statusY                   1.00  151127.   92555.
4 b_sed_percentile_s               1.00   24663.   91639.
5 b_pubertal_timing_s:adhd_statusY 1.00   90862.   90729.
6 sigma                            1.00  143930.   91468.
7 Intercept                        1.00  159185.   91837.
8 lprior                           1.00  145627.   92436.
9 lp__                             1.04    1725.    2082.


=== Divergent Transitions ===
```

```
Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

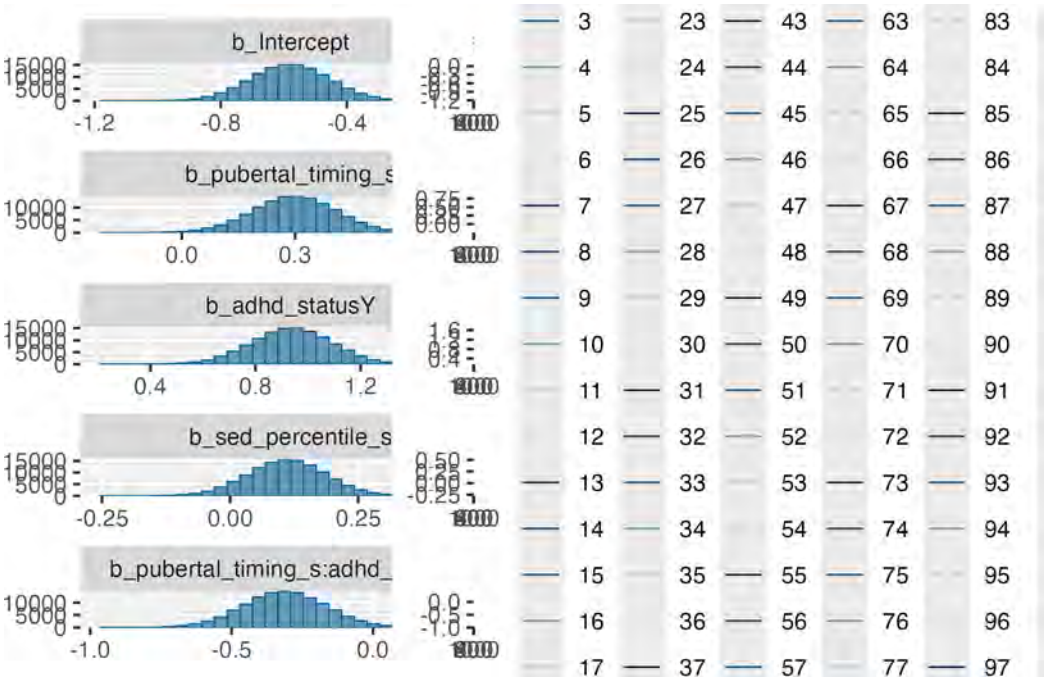
Priors specified for the model.

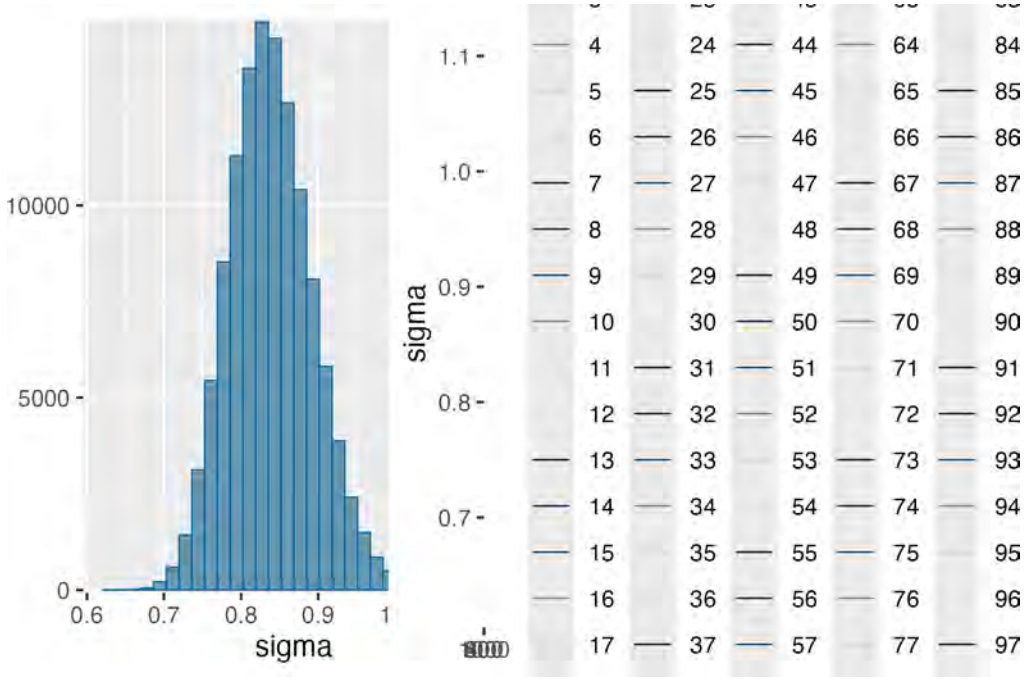
          prior     class                          coef group resp dpar nlpar
 normal(0, 0.5)         b
 normal(0, 0.5)         b                   adhd_statusY
 normal(0, 0.5)         b               pubertal_timing_s
 normal(0, 0.5)         b pubertal_timing_s:adhd_statusY
 normal(0, 0.5)         b                sed_percentile_s
 normal(0, 0.5) Intercept
 exponential(2)     sigma
 lb ub tag       source
                   user
```

```
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
                   user
  0                user
```

```
=== Posterior Predictive Checks ===
```

```
Density plots for each parameter.
```

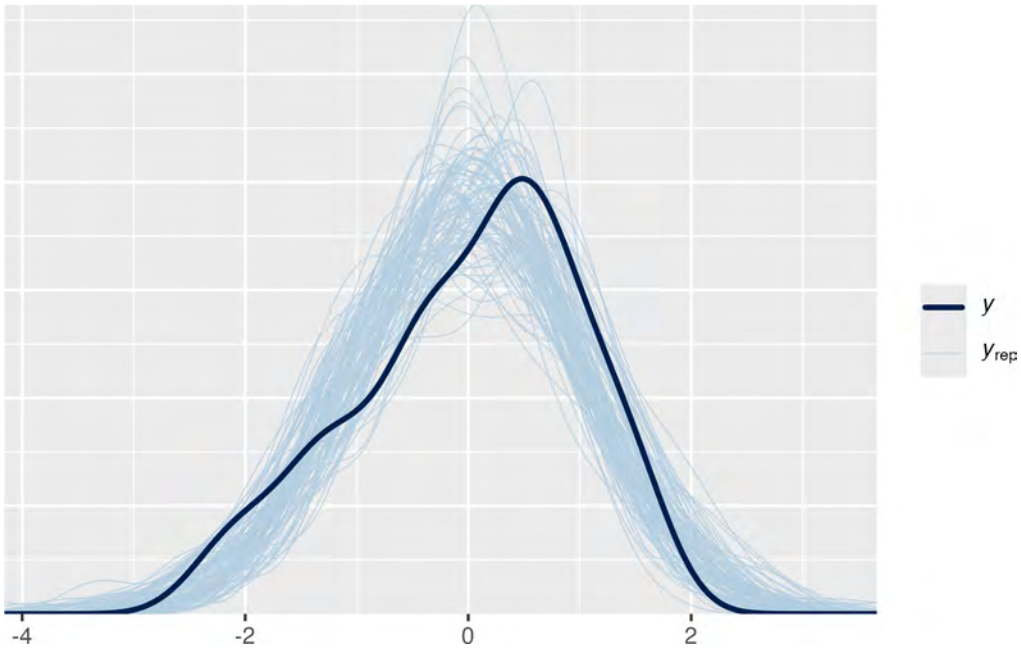





```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



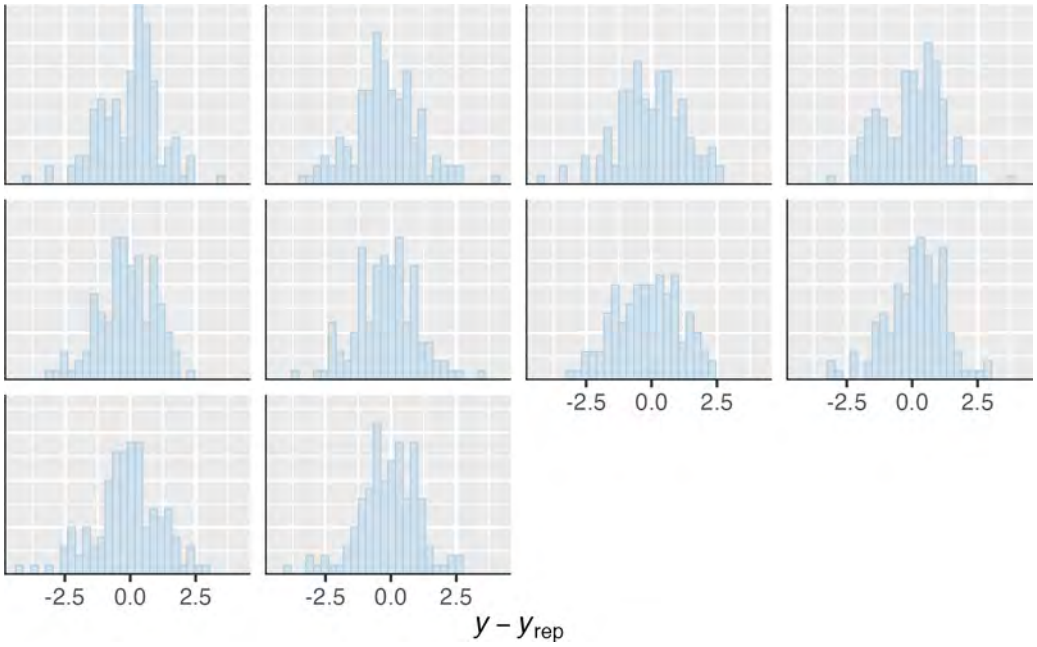

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

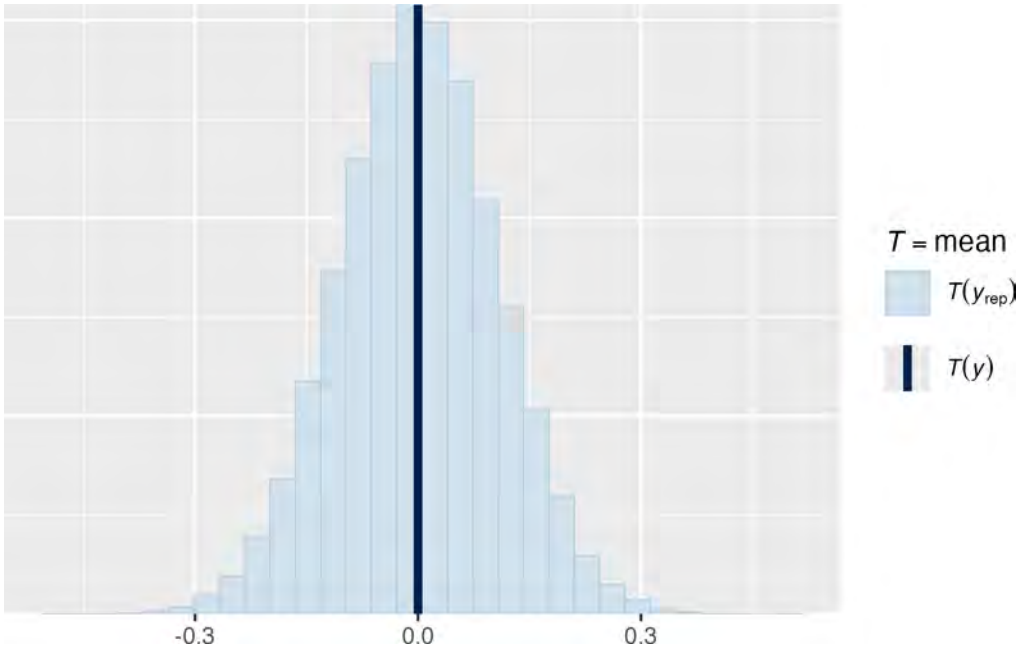
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



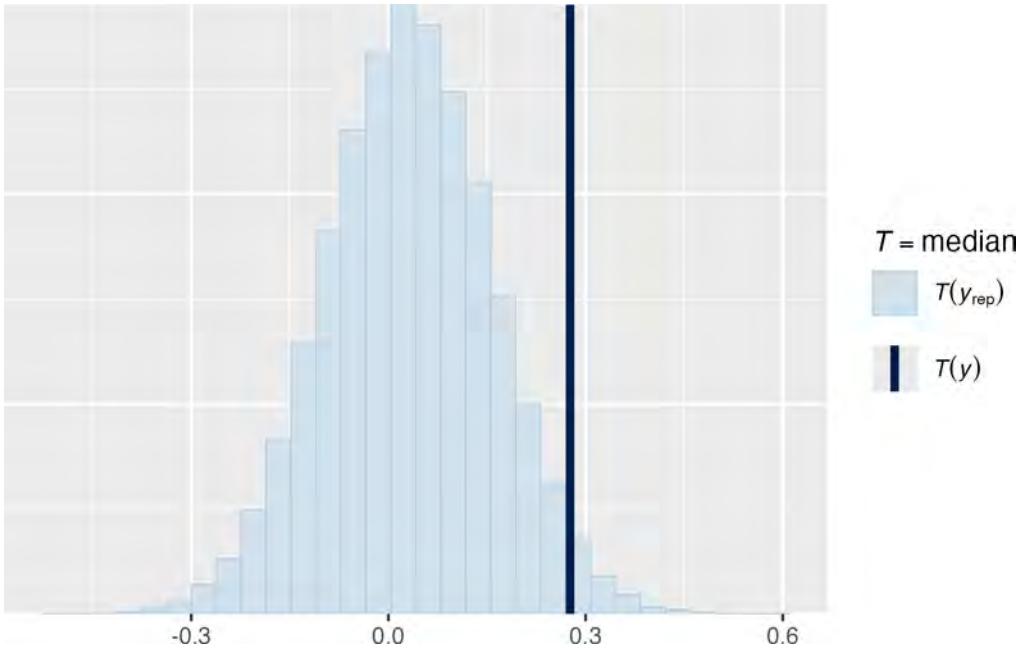

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



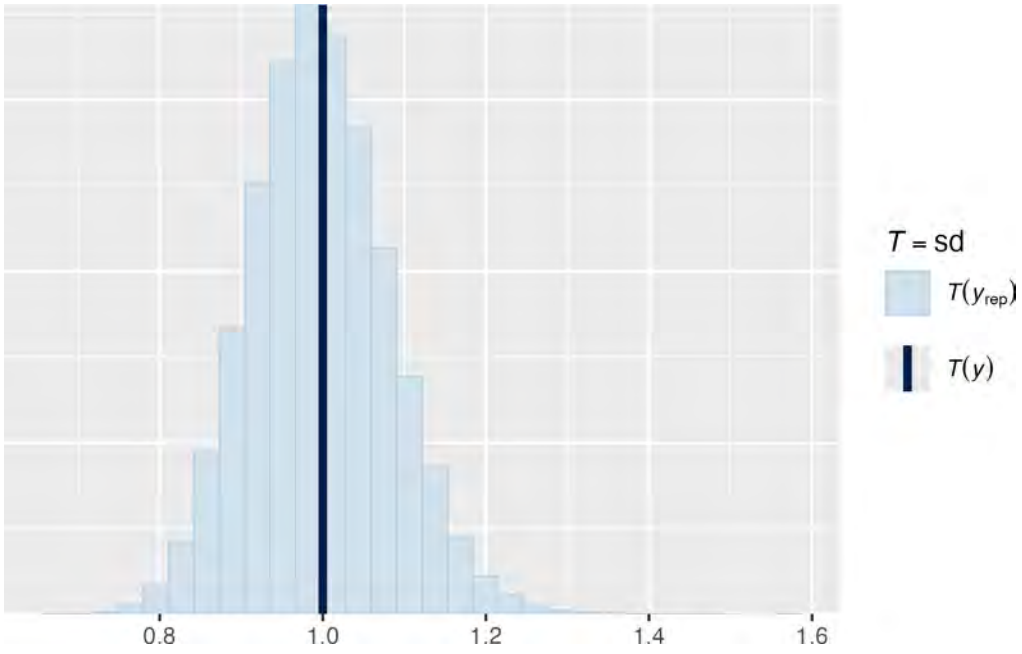

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

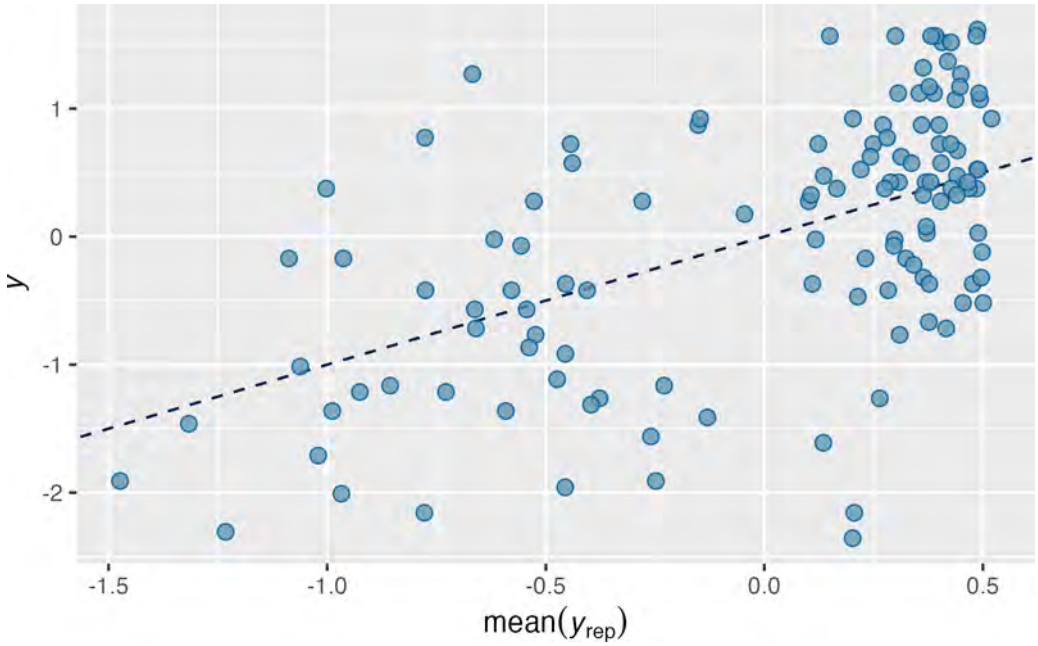



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

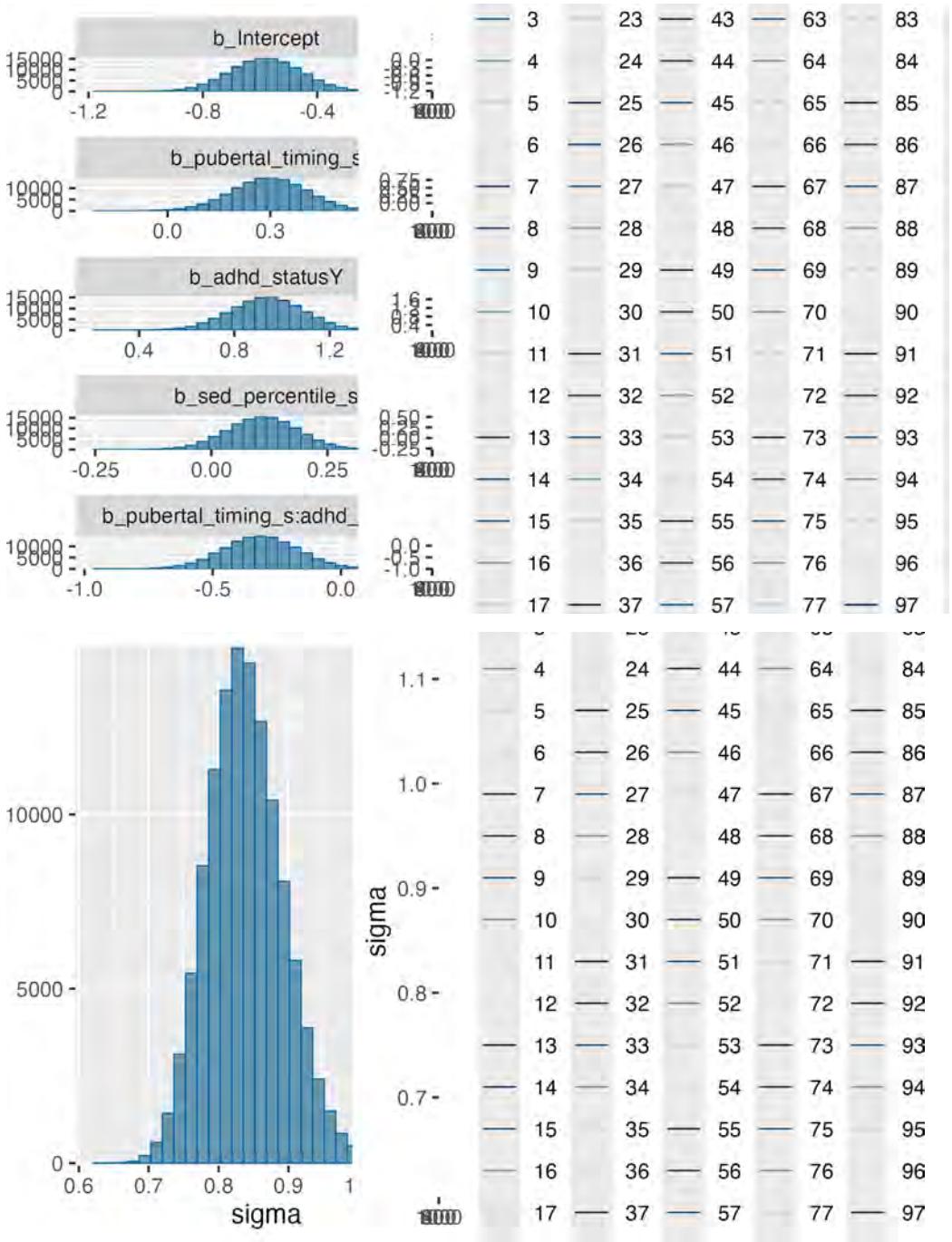
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H1 Prior 4 Model output

```
Estimate  Est.Error      Q2.5     Q97.5
```

```
R2 0.3127828 0.05544119 0.1980685 0.4146656
```

```
Summary of Posterior Distribution
```

Parameter	Mean	Median	CI	pd
b_Intercept	-0.64	-0.64	[-0.90, -0.37]	100%
b_pds_total_s	0.21	0.21	[-0.06, 0.48]	93.67%
b_adhd_statusY	1.00	1.00	[0.67, 1.33]	100%
b_age_cont_s	-3.79e-03	-3.60e-03	[-0.21, 0.20]	51.31%
b_sed_percentile_s	0.09	0.09	[-0.06, 0.25]	87.17%
b_pds_total_s:adhd_statusY	-0.06	-0.06	[-0.38, 0.26]	64.89%

Parameter	% in ROPE	meaningful
b_Intercept	4.17e-03%	TRUE
b_pds_total_s	20.19%	FALSE
b_adhd_statusY	0%	TRUE
b_age_cont_s	65.46%	FALSE
b_sed_percentile_s	54.30%	FALSE
b_pds_total_s:adhd_statusY	43.82%	FALSE

H1 Prior 4 Model diagnostics and PP checks

```
=== Rhat and ESS ===
```

```
Rhat values close to 1 indicate convergence. ESS should ideally be > 400.
```

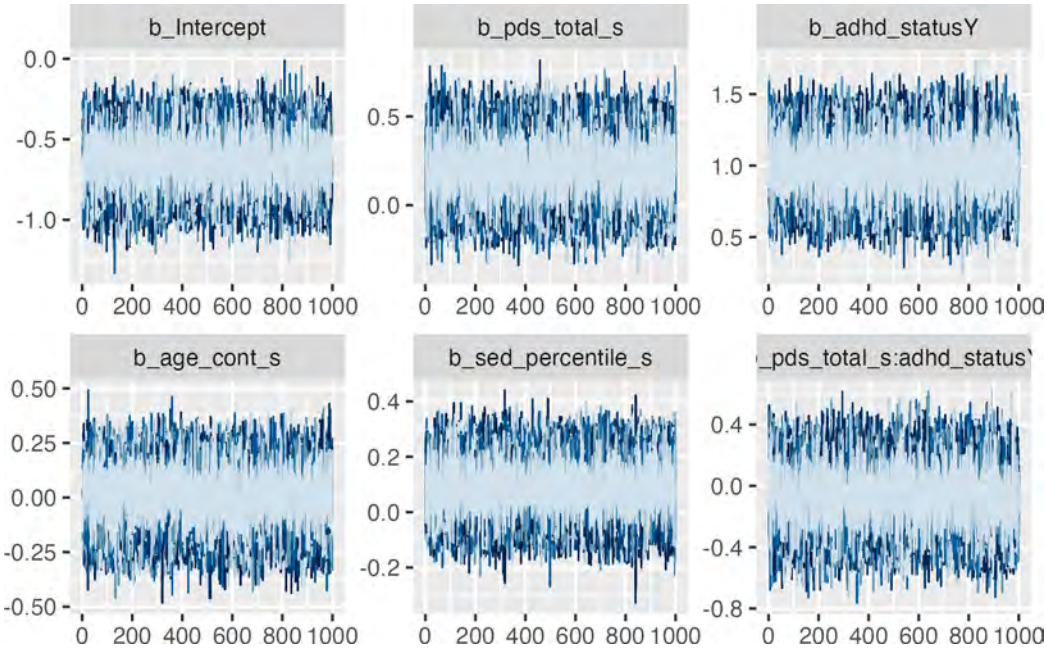
```
# A tibble: 10 x 4
   variable                    rhat ess_bulk ess_tail
   <chr>                      <dbl>    <dbl>    <dbl>
 1 b_Intercept                 1.00  126760.   91327.
 2 b_pds_total_s               1.00   79908.   84510.
 3 b_adhd_statusY              1.00  124100.   92802.
 4 b_age_cont_s                1.00   41374.   90307.
 5 b_sed_percentile_s          1.00   26227.   89590.
 6 b_pds_total_s:adhd_statusY  1.00   96750.   89047.
 7 sigma                       1.00  123151.   88610.
 8 Intercept                   1.00  137370.   90696.
 9 lprior                      1.00  120556.   88141.
10 lp__                        1.03    2646.    3325.
```

```
=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

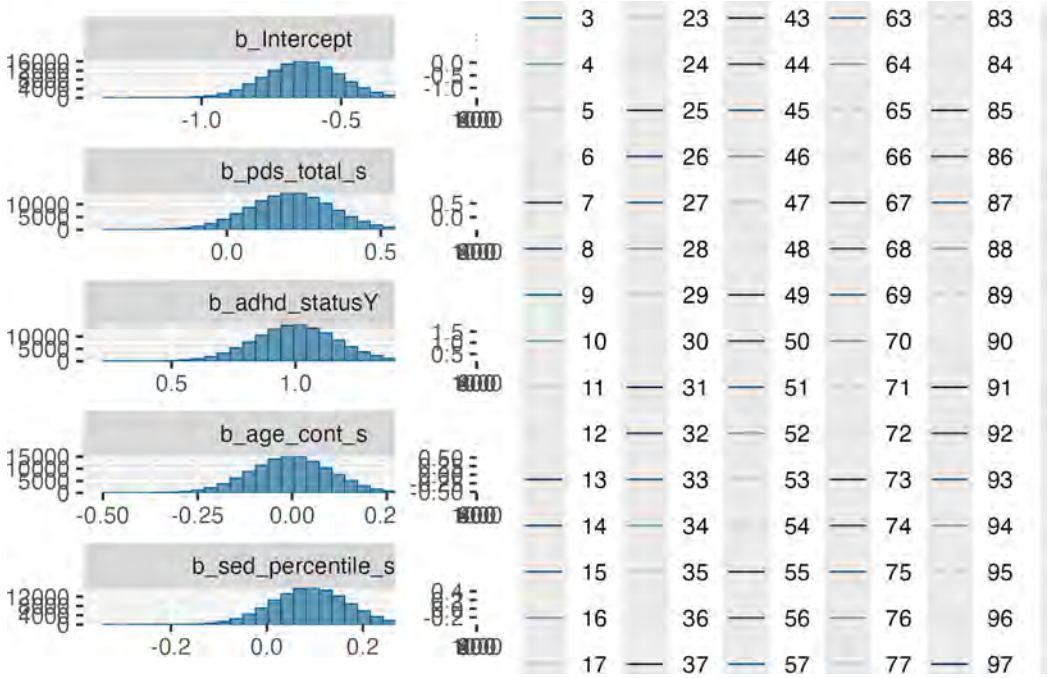
Priors specified for the model.

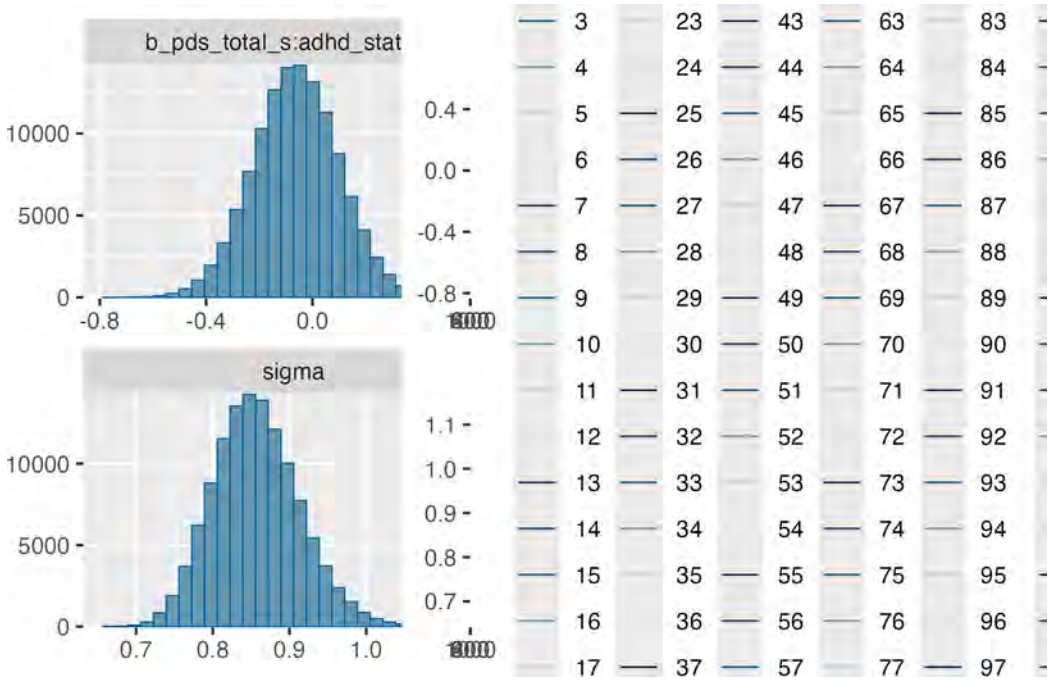
             prior     class                      coef group resp dpar nlpar lb
     normal(0, 3)         b
     normal(0, 3)         b              adhd_statusY
     normal(0, 3)         b                age_cont_s
     normal(0, 3)         b               pds_total_s
     normal(0, 3)         b pds_total_s:adhd_statusY
     normal(0, 3)         b          sed_percentile_s
```

```
    normal(0, 3) Intercept
exponential(0.5)     sigma                                                    0
ub tag       source
               user
       (vectorized)
       (vectorized)
       (vectorized)
       (vectorized)
       (vectorized)
               user
               user


=== Posterior Predictive Checks ===

Density plots for each parameter.
```

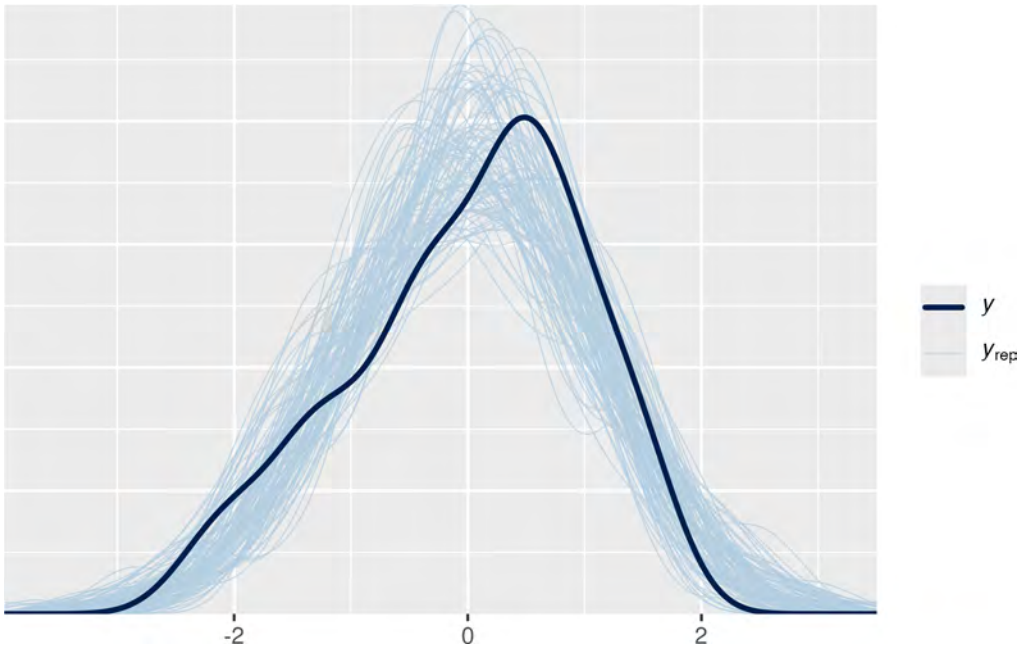





```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



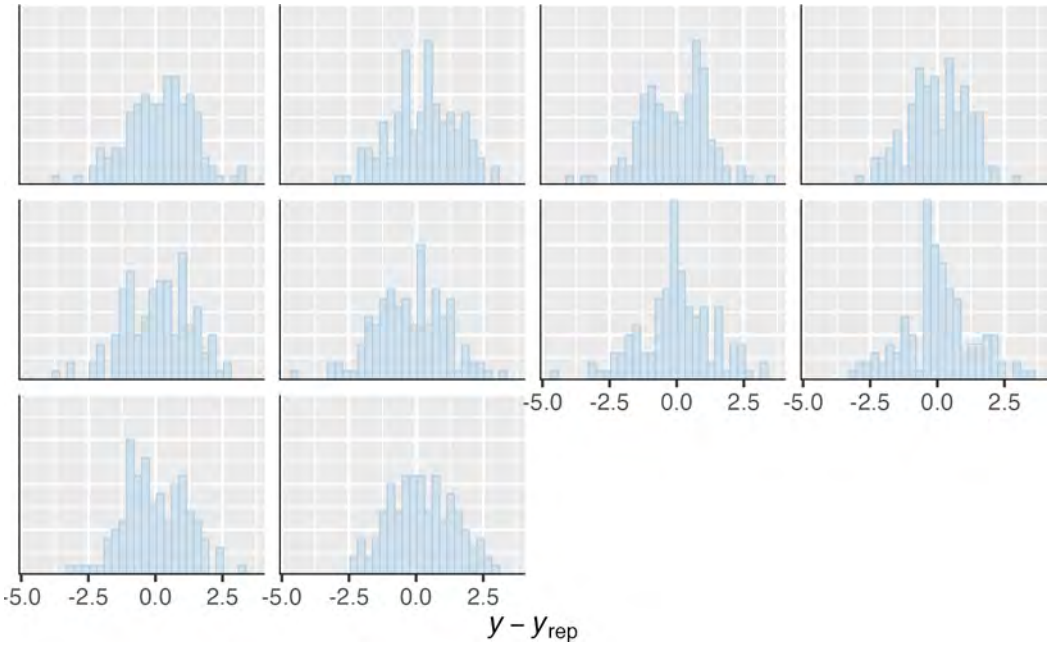

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

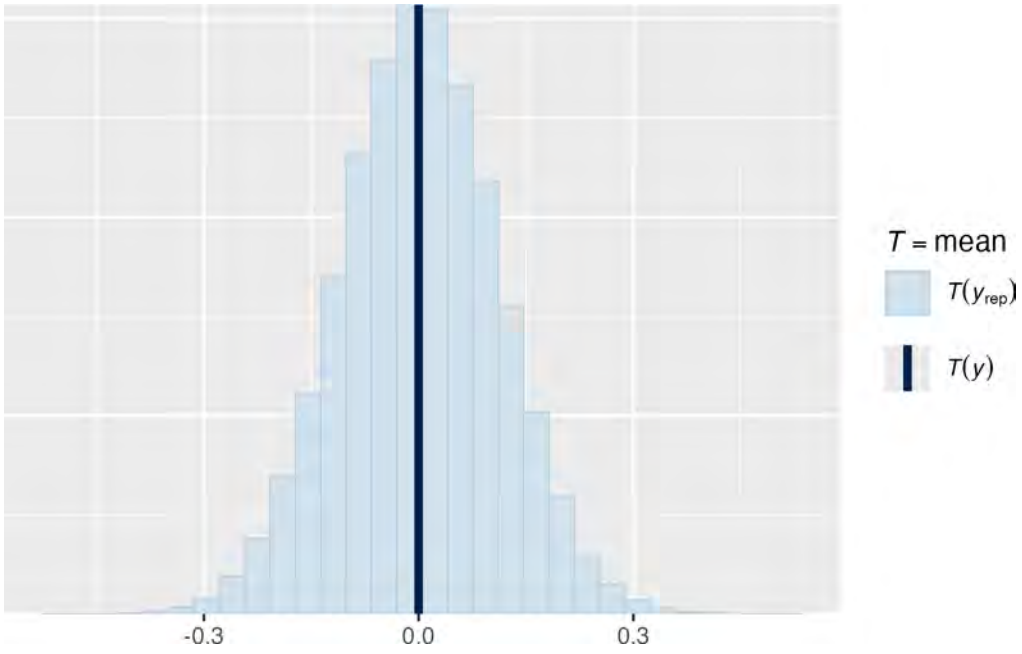
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



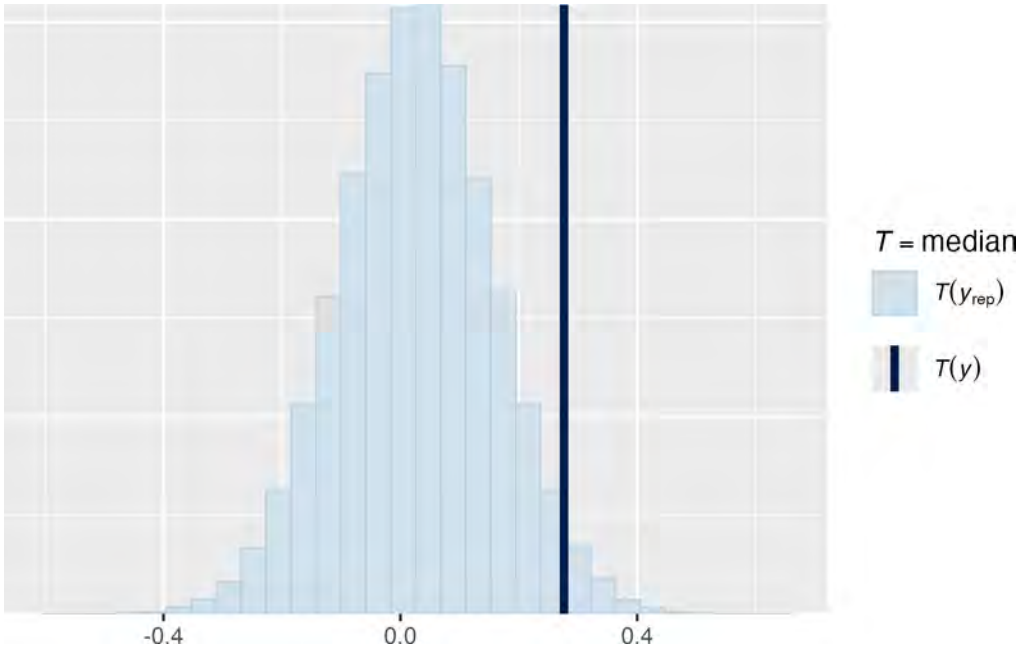

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



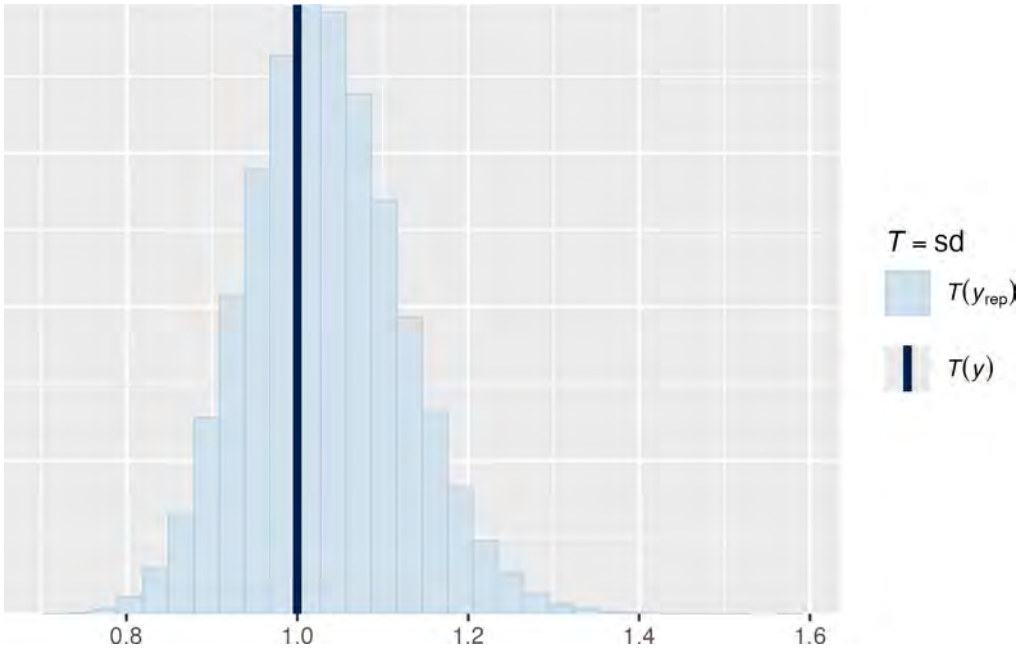

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

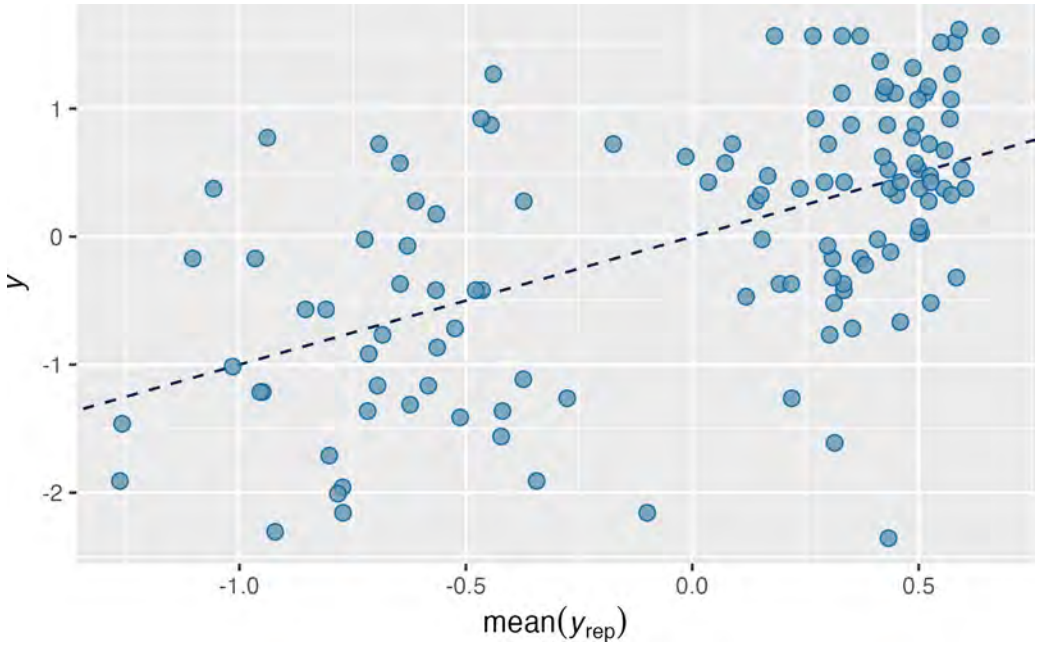



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

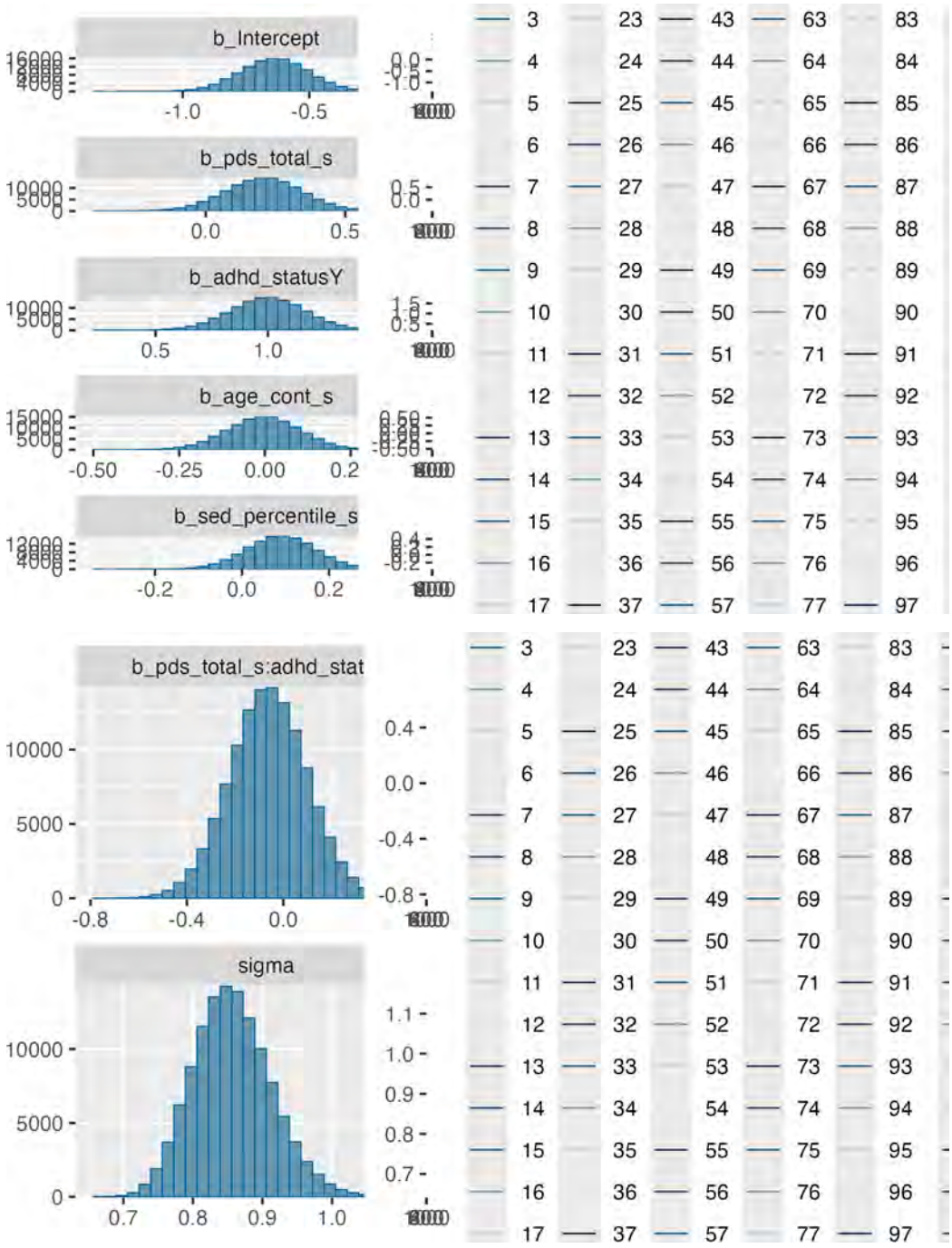
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Prior 4 Model output

```
Estimate  Est.Error      Q2.5     Q97.5
```

```
R2 0.3250011 0.05568248 0.2089701 0.4264435
```

```
Summary of Posterior Distribution
```

Parameter	Mean	Median	CI	pd
b_Intercept	-0.63	-0.63	[-0.89, -0.38]	100%
b_pubertal_timing_s	0.32	0.32	[0.09, 0.55]	99.68%
b_adhd_statusY	1.03	1.03	[0.71, 1.34]	100%
b_sed_percentile_s	0.12	0.12	[-0.03, 0.27]	93.91%
b_pubertal_timing_s:adhd_statusY	-0.35	-0.35	[-0.66, -0.04]	98.72%

Parameter	% in ROPE	meaningful
b_Intercept	5.00e-03%	TRUE
b_pubertal_timing_s	2.81%	TRUE
b_adhd_statusY	0%	TRUE
b_sed_percentile_s	40.03%	FALSE
b_pubertal_timing_s:adhd_statusY	5.01%	FALSE

H2 Prior 4 Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

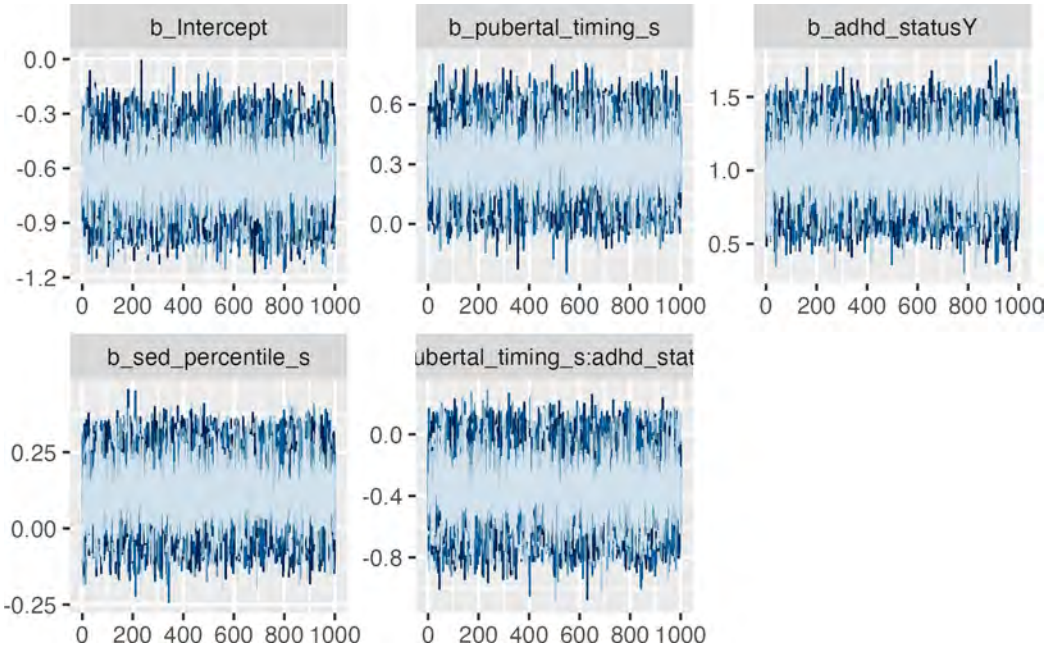
# A tibble: 9 x 4
  variable                         rhat ess_bulk ess_tail
  <chr>                           <dbl>    <dbl>    <dbl>
1 b_Intercept                      1.00  143653.   92452.
2 b_pubertal_timing_s              1.00   84582.   88239.
3 b_adhd_statusY                   1.00  143994.   91204.
4 b_sed_percentile_s               1.01   18790.   91662.
5 b_pubertal_timing_s:adhd_statusY 1.00   84126.   89158.
6 sigma                            1.00  136808.   90902.
7 Intercept                        1.00  153621.   90549.
8 lprior                           1.00  138783.   86135.
9 lp__                             1.05    1440.    1802.


=== Divergent Transitions ===
```

```
Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

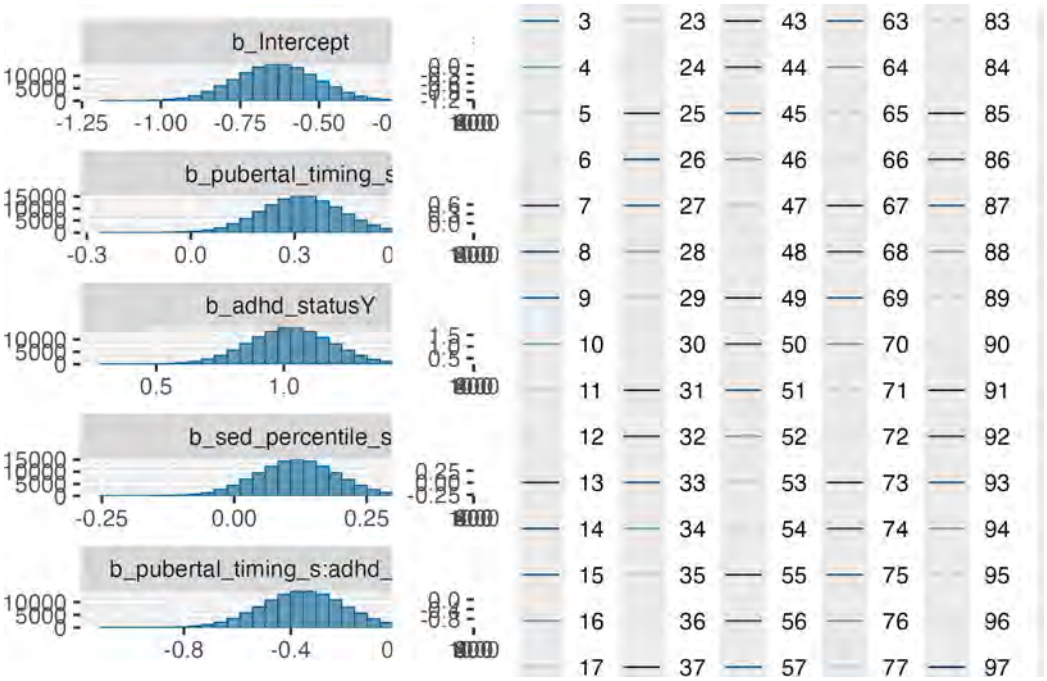
Priors specified for the model.

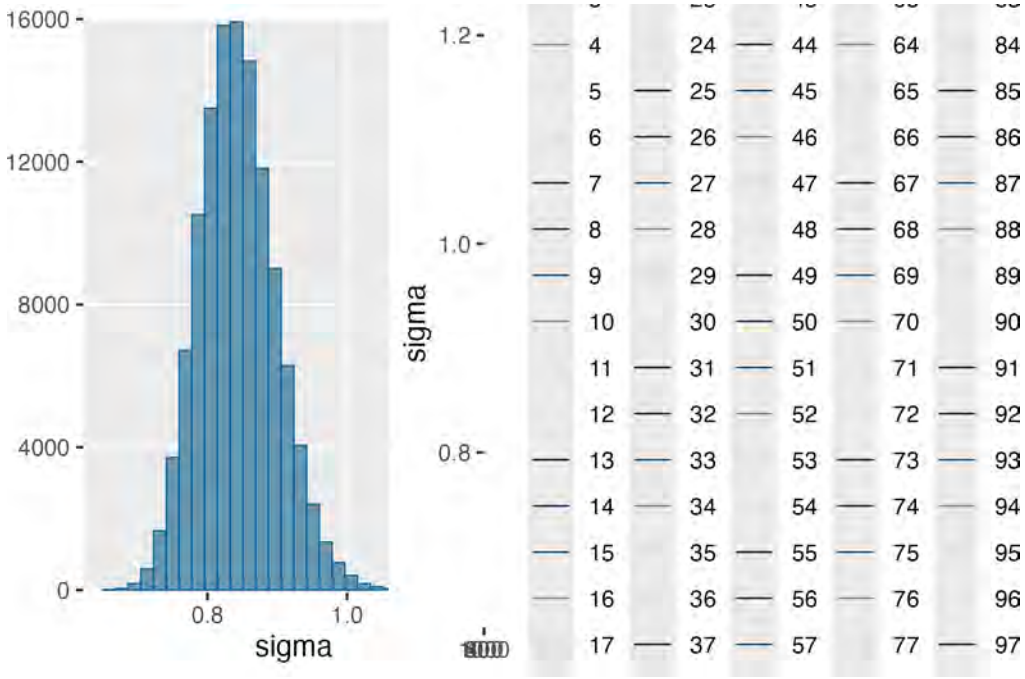
             prior     class                          coef group resp dpar
     normal(0, 3)         b
     normal(0, 3)         b                   adhd_statusY
     normal(0, 3)         b               pubertal_timing_s
     normal(0, 3)         b pubertal_timing_s:adhd_statusY
     normal(0, 3)         b                sed_percentile_s
     normal(0, 3) Intercept
 exponential(0.5)     sigma
 nlpar lb ub tag       source
                         user
```

```
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                        user
        0               user
```

```
=== Posterior Predictive Checks ===
```

Density plots for each parameter.

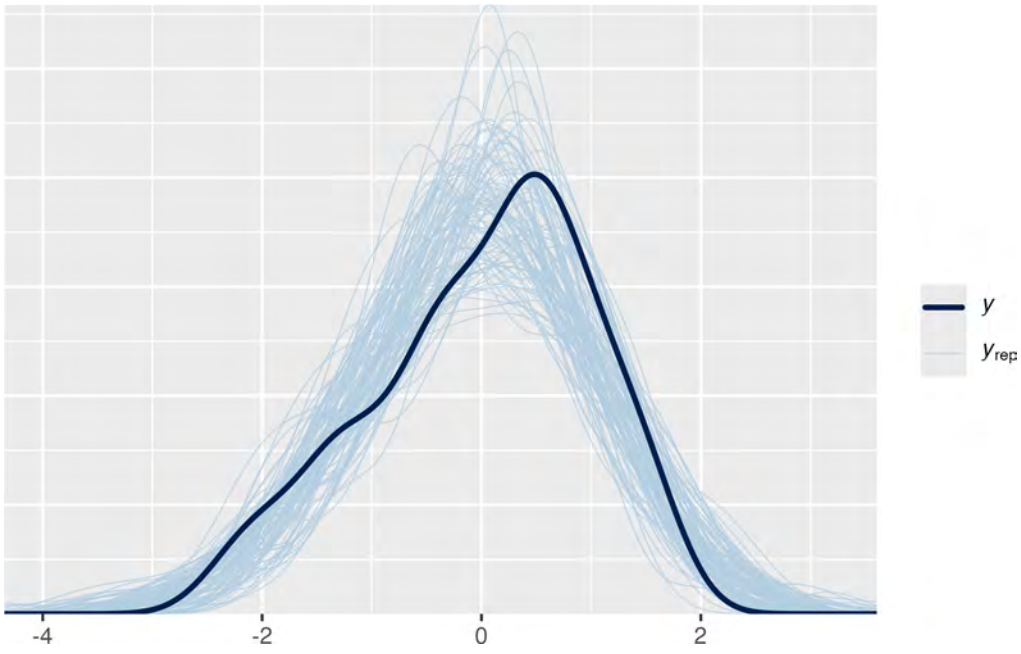





```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



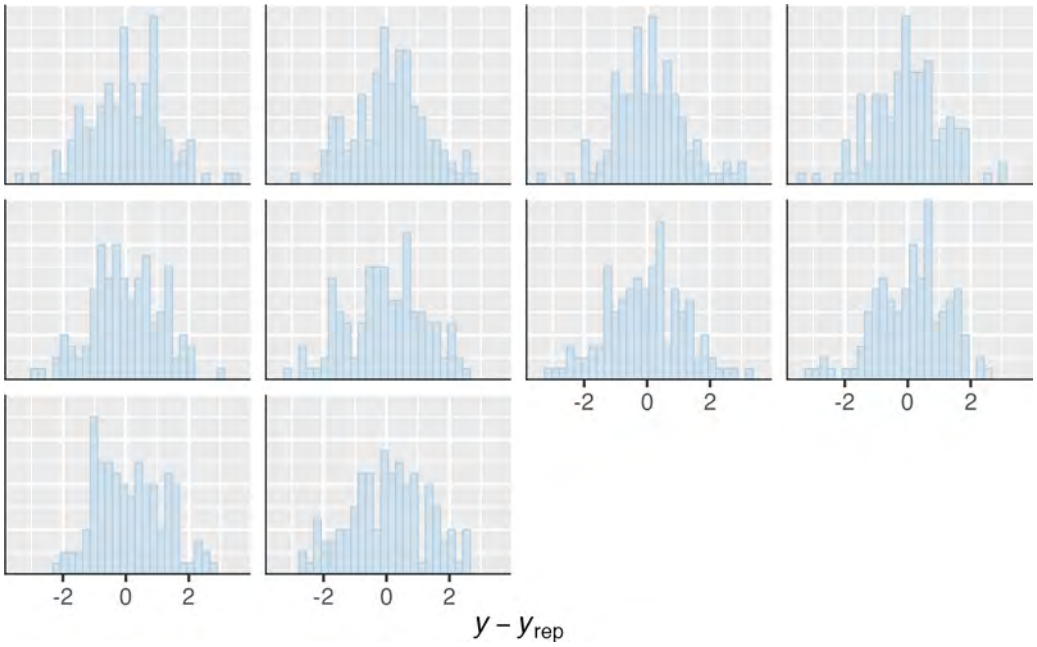

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

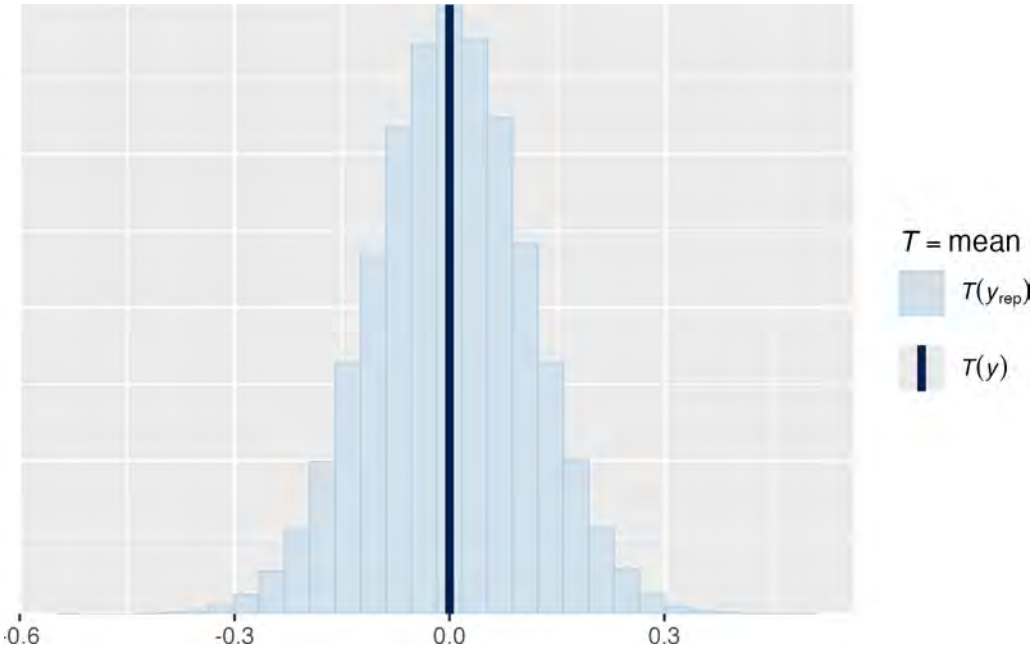
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



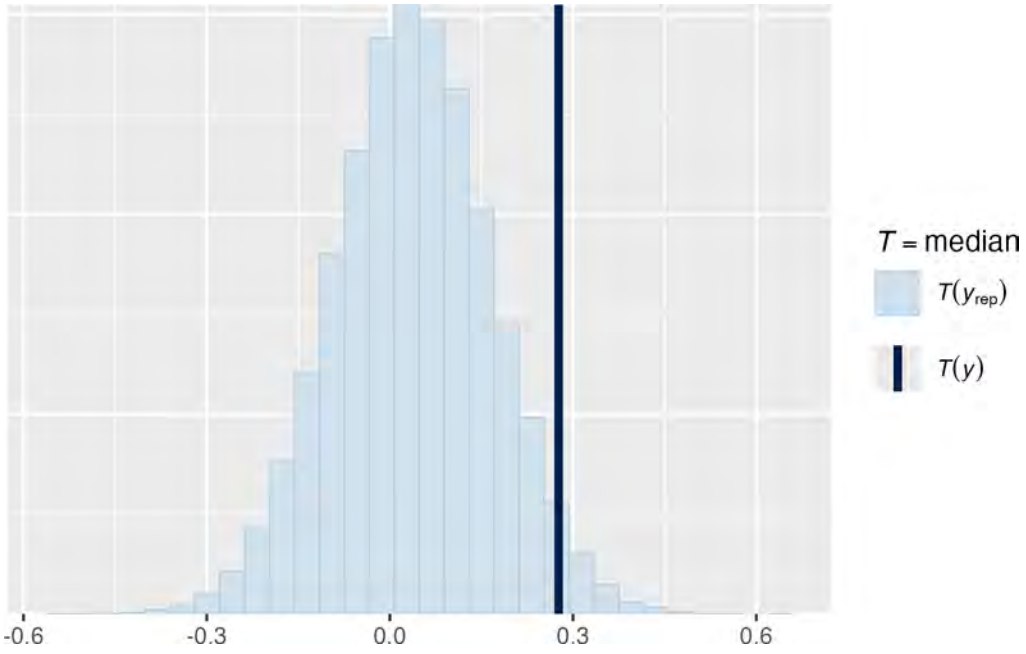

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



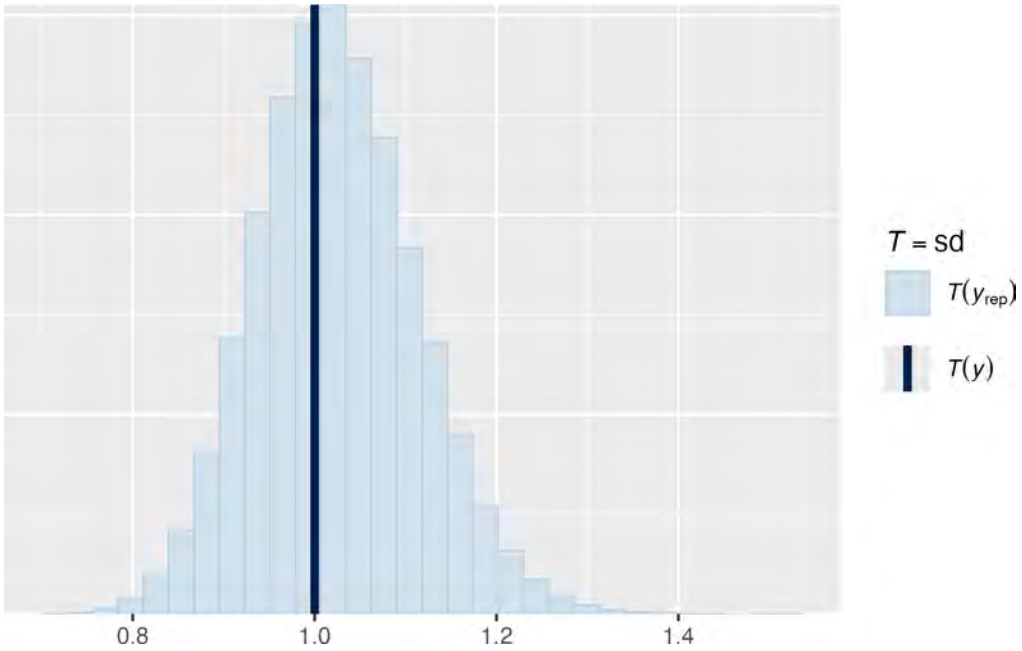

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

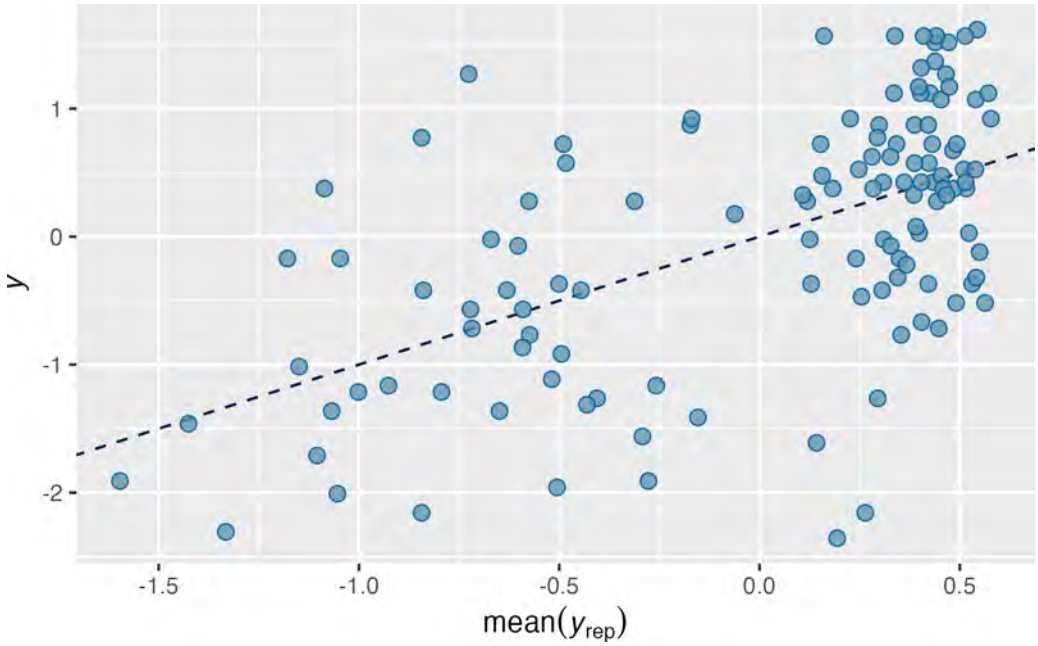



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

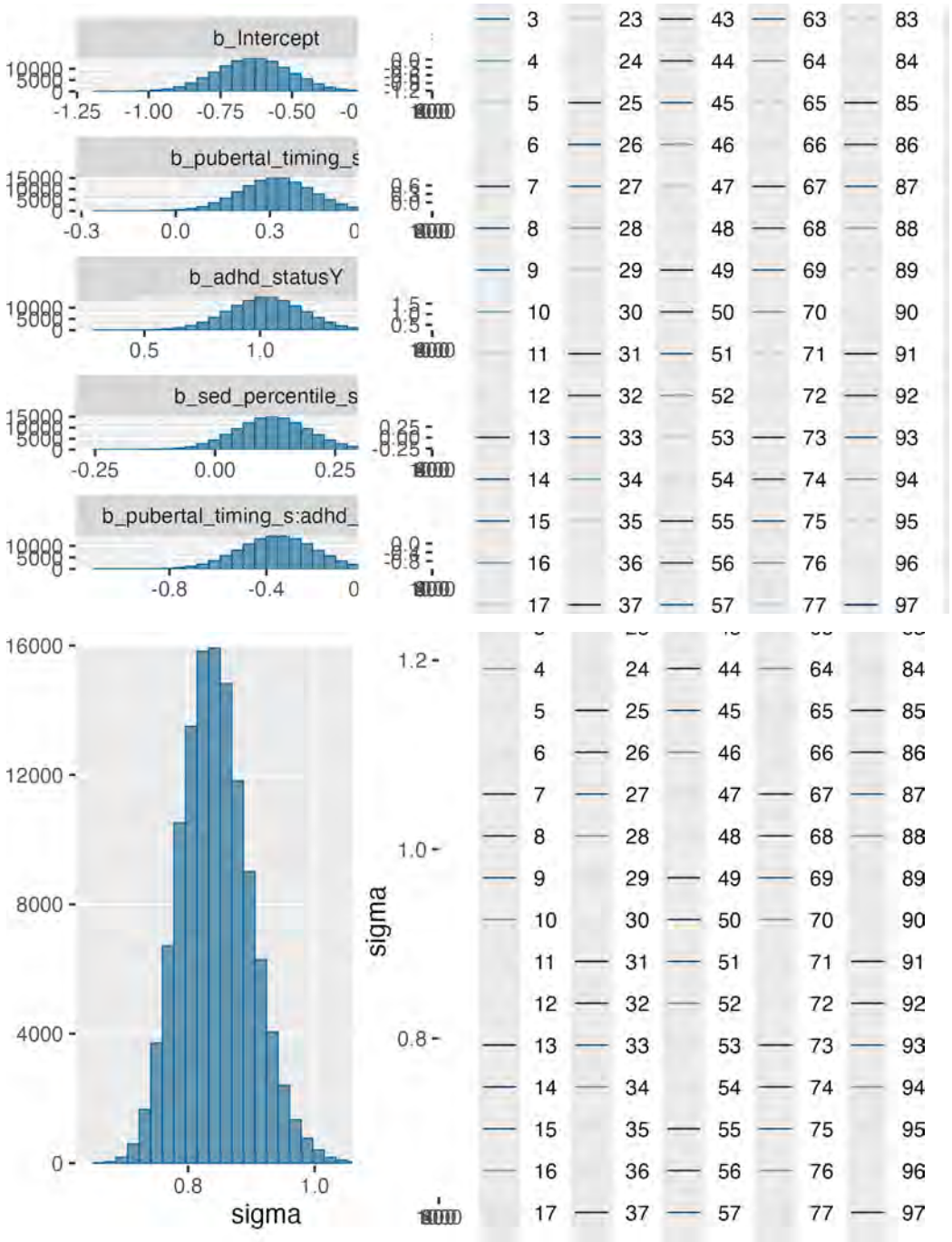
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Model excluding unconfirmed ADHD output

```
Estimate  Est.Error      Q2.5     Q97.5
```

```
R2 0.314132 0.05789314 0.1938652 0.4192914

Summary of Posterior Distribution

Parameter                         |  Mean | Median |             CI |     pd
-------------------------------------------------------------------------------
b_Intercept                       | -0.62 |  -0.62 | [-0.88, -0.37] |   100%
b_pubertal_timing_s               |  0.32 |   0.32 | [ 0.09,  0.54] | 99.67%
b_adhd_statusY                    |  0.95 |   0.95 | [ 0.63,  1.27] |   100%
b_sed_percentile_s                |  0.15 |   0.15 | [-0.02,  0.31] | 96.21%
b_pubertal_timing_s:adhd_statusY  | -0.32 |  -0.32 | [-0.63,  0.00] | 97.60%

Parameter                         | % in ROPE | meaningful
-------------------------------------------------------------
b_Intercept                       | 3.33e-03% |       TRUE
b_pubertal_timing_s               |     3.00% |       TRUE
b_adhd_statusY                    |        0% |       TRUE
b_sed_percentile_s                |    28.10% |      FALSE
b_pubertal_timing_s:adhd_statusY  |     8.03% |      FALSE
```

H2 Model excluding unconfirmed ADHD diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

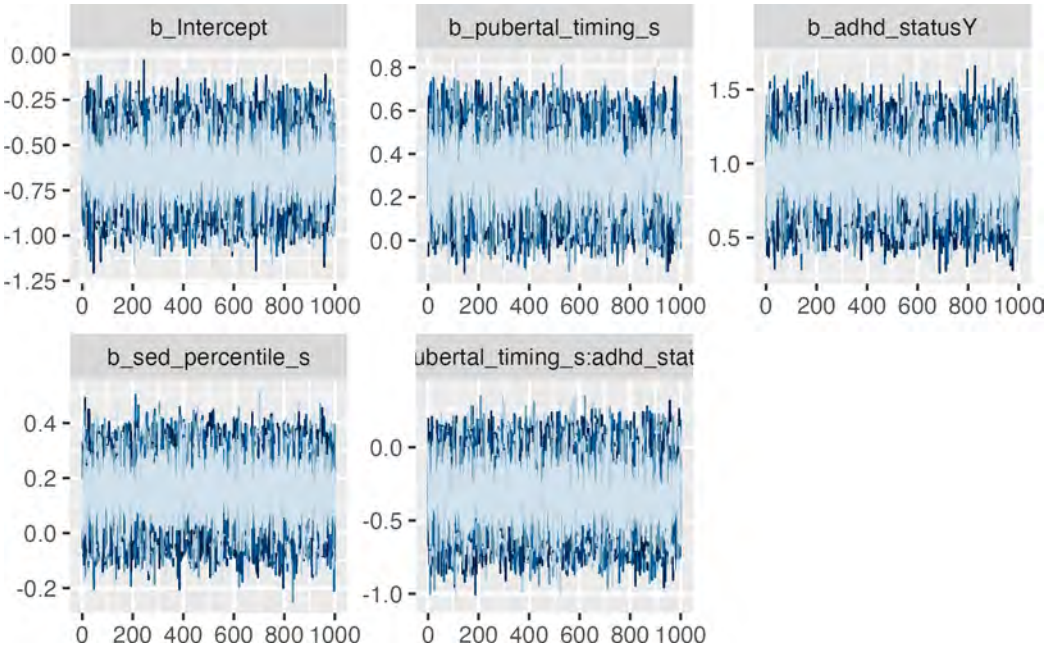
# A tibble: 9 x 4
  variable                         rhat ess_bulk ess_tail
  <chr>                           <dbl>    <dbl>    <dbl>
1 b_Intercept                      1.00  147959.   92582.
2 b_pubertal_timing_s              1.00   89841.   90609.
3 b_adhd_statusY                   1.00  148775.   91356.
4 b_sed_percentile_s               1.00   23314.   89478.
5 b_pubertal_timing_s:adhd_statusY 1.00   93705.   92328.
6 sigma                            1.00  147042.   93647.
7 Intercept                        1.00  155241.   89301.
8 lprior                           1.00  143004.   89746.
9 lp__                             1.05    1555.    1872.


=== Divergent Transitions ===
```

```
Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```




```
=== Prior Summary ===

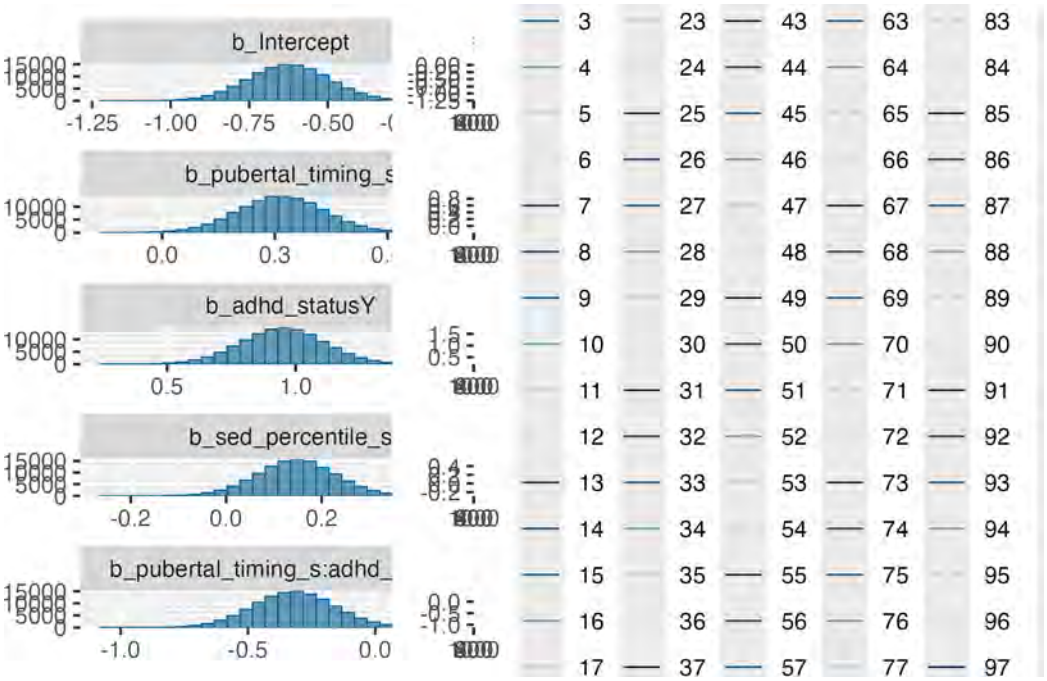
Priors specified for the model.

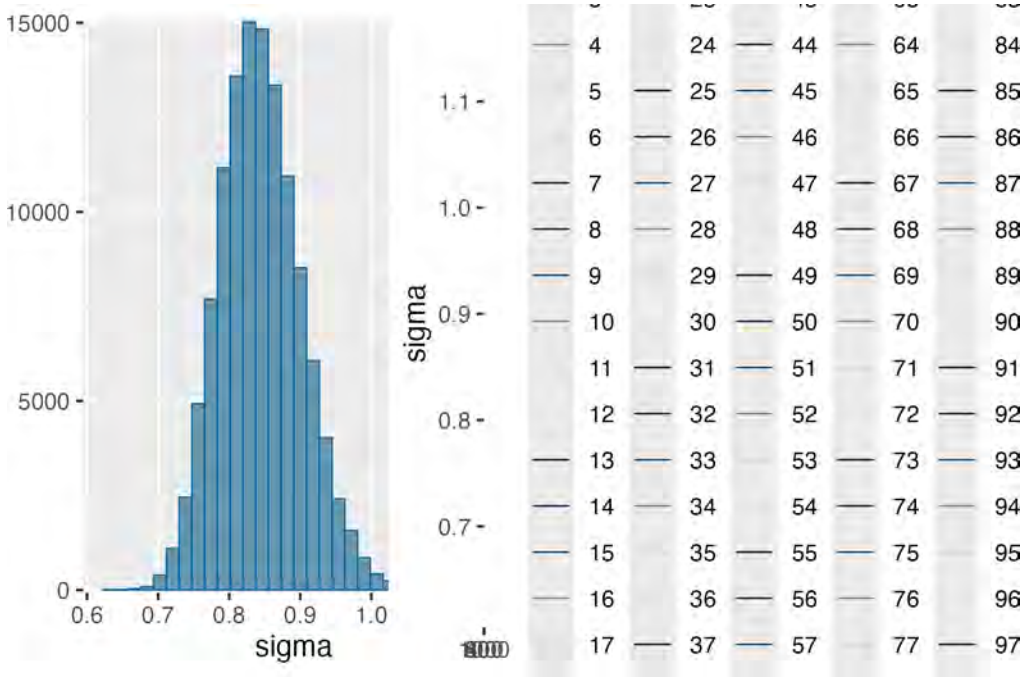
          prior     class                          coef group resp dpar nlpar
   normal(0, 1)         b
   normal(0, 1)         b                   adhd_statusY
   normal(0, 1)         b               pubertal_timing_s
   normal(0, 1)         b pubertal_timing_s:adhd_statusY
   normal(0, 1)         b                sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma
 lb ub tag       source
                   user
```

```
            (vectorized)
            (vectorized)
            (vectorized)
            (vectorized)
                    user
  0                 user


=== Posterior Predictive Checks ===

Density plots for each parameter.
```

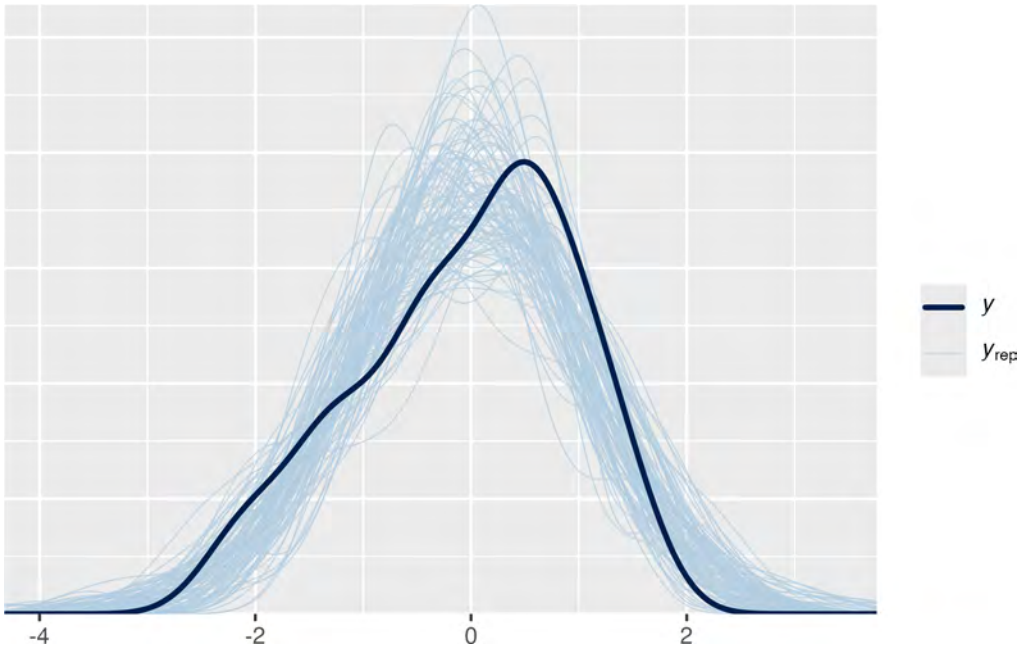





```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



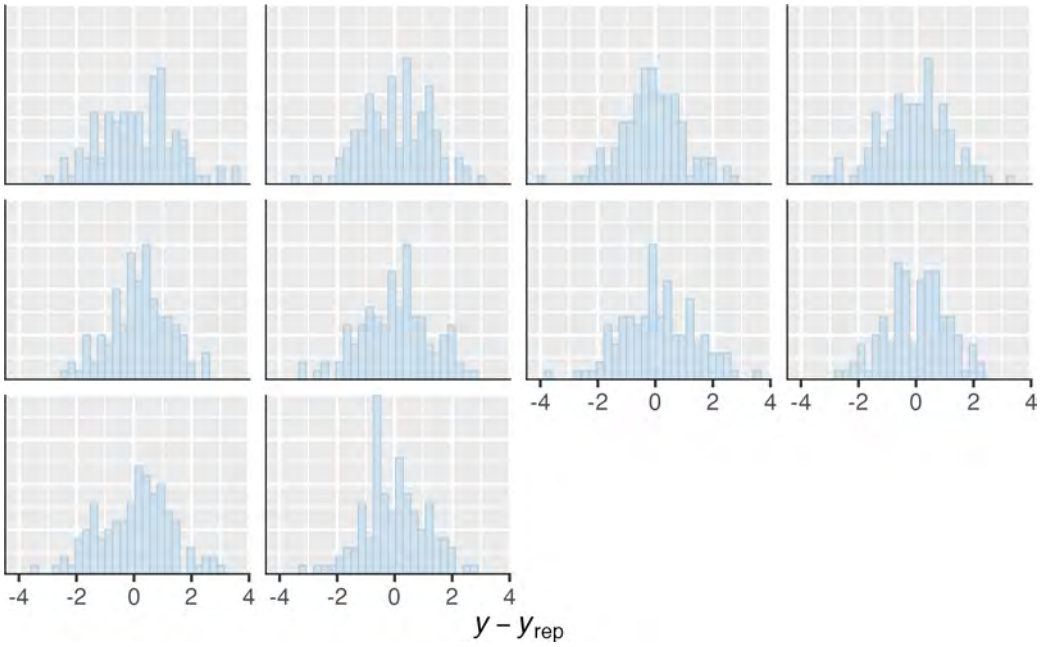

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

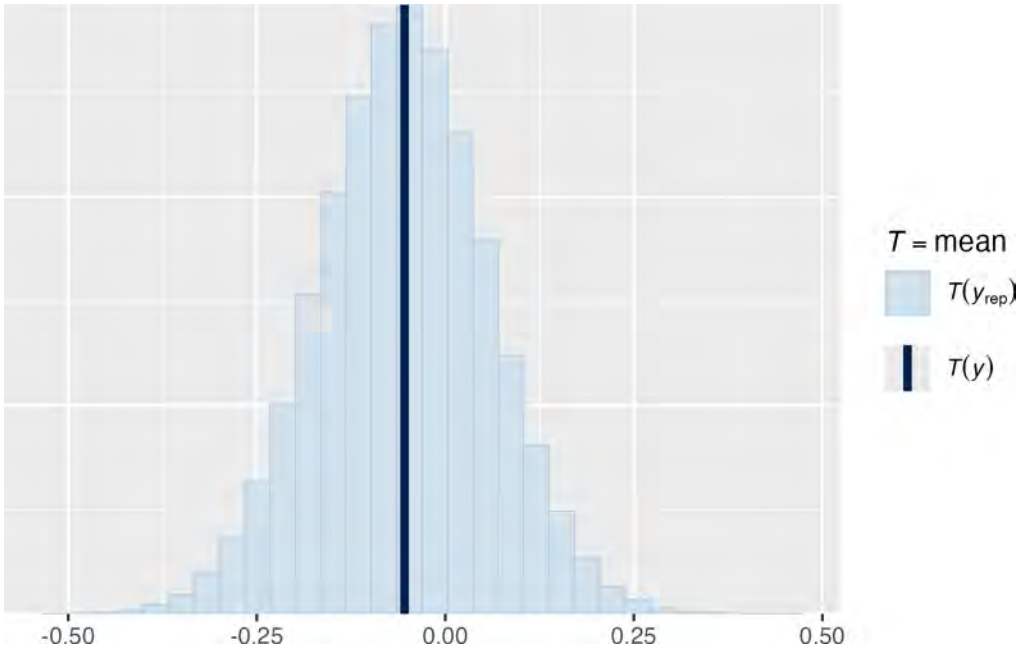
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



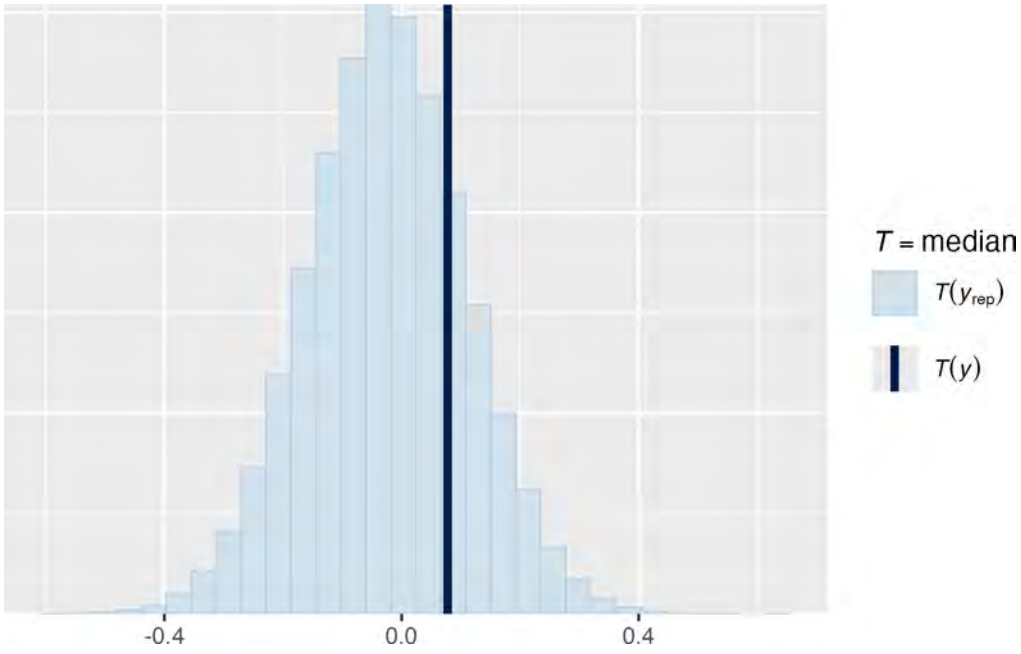

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



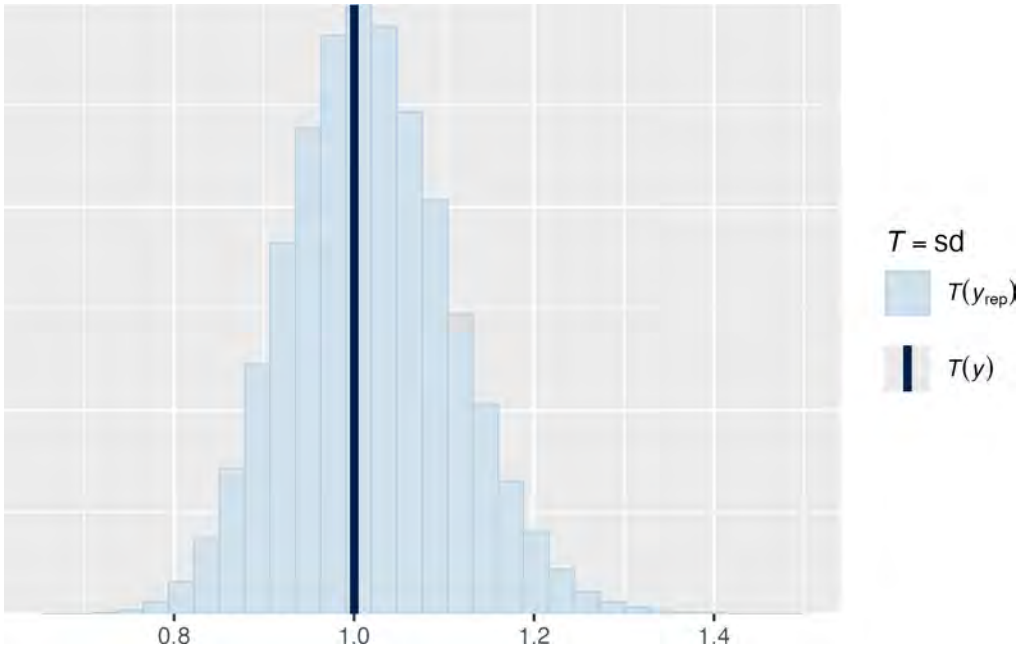

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

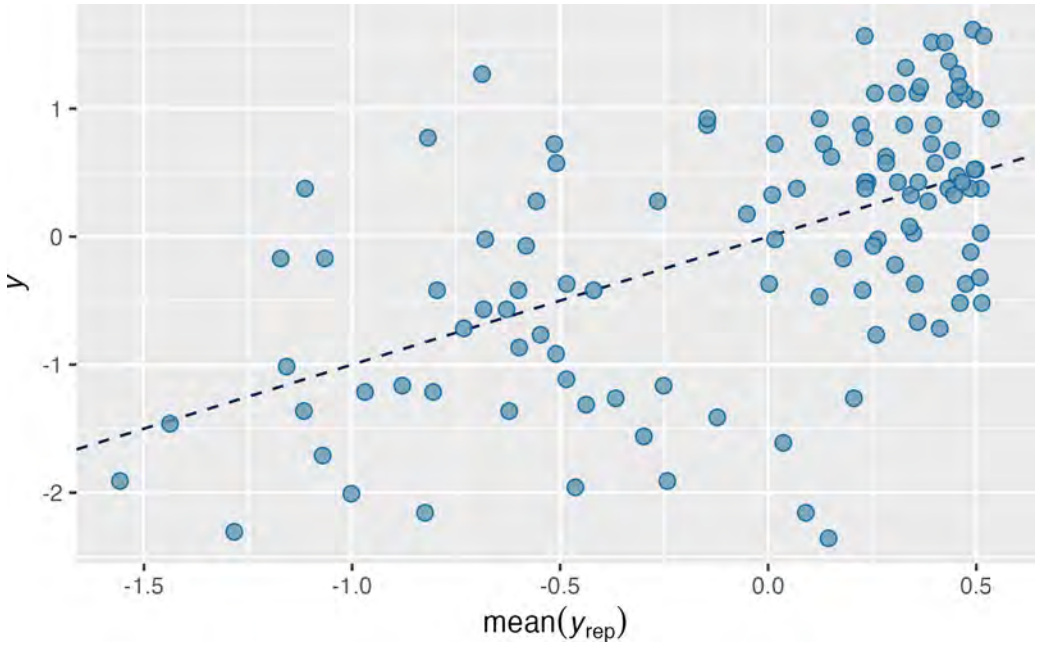



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

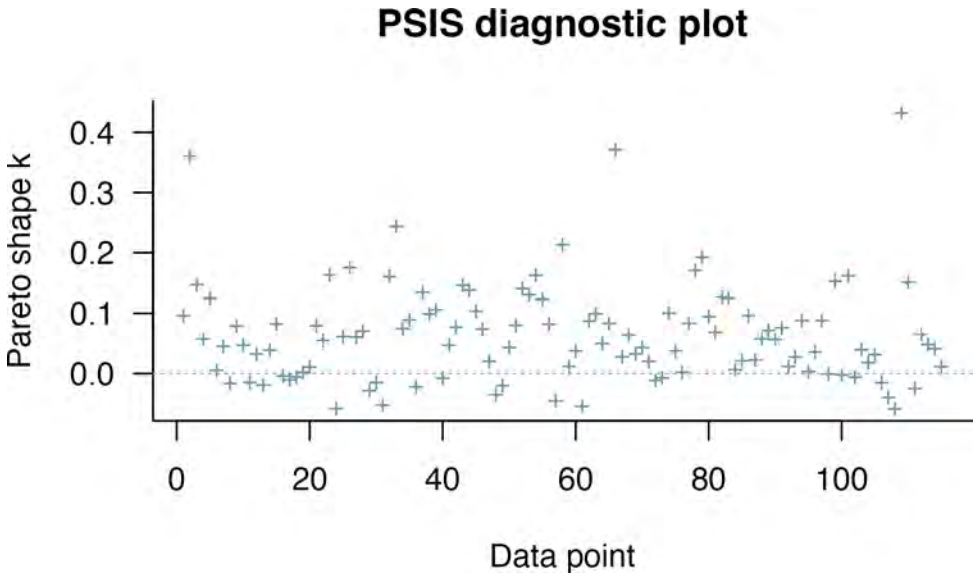
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```


PSIS diagnostic plot
Pareto shape k
0.4
0.3
0.2
0.1
0.0
0
20
40
60
80
100
Data point